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Adaptive Behavioral Graph Analysis for Android Malware Detection

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Abstract

Zero-day Android malware detection motivates detectors that can be trained without malware labels, yet many graph methods still lean on supervised family labels or treat behavioural deviation as a single scalar. This thesis asks whether an application *behaves like itself* under Adaptive Behavioural Reference Graphs (ABRG): fixed-category behavioural graphs whose reference may be read against a benign population or against the app’s own past.

The study uses two complementary corpora. On AndroCT, labelled malware evaluation is available but timestamps and repeated sessions are not. On a ContextDroid Frida corpus (v2 / v2_extended), multi-session traces and wall-clock structure are available, but malware labels are not. Representation, scoring, and temporal design choices are compared under pinned protocols rather than tuned for a single headline score.

On AndroCT the tensors carry strong malware signal under full supervision (histogram gradient boosting 0.976 AUC on the flattened feature vector; adjacency alone 0.959), yet benign-only reconstruction does not yield a reliable detector, and message passing does not improve supervised readouts on these graphs. Within-session self-reference by windowing also fails to detect: windows are largely exchangeable. On the Frida corpus, a self-reference built from prior sessions strongly anticipates the held-out session’s graph structure: Stage 2 Frobenius medians 1.28 within-app versus 38.3 cross-app, with pooled L_2 self/cross ratios of $60\times$ – $411\times$ across reference depth k . That signal is identity structure in the session stream, not malware prediction. Ordering, recency weighting, and cold-start accumulation do not beat cumulative memory, and a description-seeded reference before any session ($k=0$) does not substitute for observed behaviour.

Population deviation and self-reference therefore remain complementary halves of the title question: the former can be scored as malware detection on AndroCT; the latter shows strong identity structure without a malware evaluation arm under the same sensor. Unifying both under one labelled, multi-session corpus is left open. Reproducible artefacts for the measured claims are deposited with the thesis.

Keywords: Android malware detection; behavioural graphs; self-reference; unsupervised anomaly detection; dynamic analysis; ContextDroid.

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A.1 Framing

A.1.1 Problem and design

Zero-day Android malware detection is the design commitment: a detector that can be trained without malware labels, so that previously unseen malicious behaviour is not required to have appeared in a labelled training set. That commitment is a constraint, not a concession. The evaluation in this chapter therefore scores generalisation to unseen *behavioural groups* on AndroCT, not to unseen AVClass families: the corpus ships no family labels (§A.10.1).

Adaptive Behavioural Reference Graphs (ABRG) represent each application as a per-app reference graph over a fixed universe of behavioural categories. Static attributes seed the nodes from the APK; dynamic trace events refine activity and transition structure; detection is a *deviation* from a benign-only reference rather than a raw magnitude. Explainability is the same object as a named per-node profile on the fixed category universe (§A.4.1, §A.4.4), not a post-hoc attribution of a scalar; whether that profile localises is measured in §A.6.7. On AndroCT the recency channel and the wall-clock edge filter δ are unexercised (no timestamps). Whether the app behaves like itself is the design question: this chapter measures it within a session by windowing the same trace (Arm B; E0/E2, §A.6.8) as well as on the whole-trace graph; cross-session self-reference is Chapter C (§A.4.3, §A.10.2).

A.1.2 The exercise matrix

No single corpus can test the whole design. That is why the thesis has more than one chapter. Table A.1 states which ABRG commitments each corpus can exercise and, for v2, whether a commitment was later measured (Chapter B). Chapter A is the only chapter with malware labels; Chapters B and C recover timestamps and multi-session structure on a corpus that has no malware evaluation arm.

Table A.1. Which ABRG design elements each corpus can exercise. The v2 column records measurement where Chapters B/C have run the commitment (corpus-level; no AUC).

design element	Chapter A (AndroCT)	Chapters B/C (v2)
δ temporal edge filter	unavailable (no timestamps)	available; retention measured (§B.5)
Recency edge channel rec_v , w_{rec}	unavailable	measured; $W_{\text{REC}} \leq W_{\text{CUM}}$ (§C.4.2)
Per-app cross-session refinement	unavailable ($N=1$)	measured; within/cross separation (§C.3); ordering ORDERING_NEUTRAL (§C.4.1); $k=1$ reference / dilution (§C.4.3); $k=0$ description null (§C.5)
Malware evaluation	available	unavailable (no malware)

A.1.3 Contributions

The contributions below are ordered by defensibility. Numbers match §A.9.

1. A controlled measurement that message passing *does not help* under full supervision on these behavioural graphs. Condition-mean M1 – M2 is negative in all three Split-A poolings and under Ward mean-pool holdout. Add pooling is unanimous ($\Delta = -0.028$, sd 0.013, 0/5

reversals). Mean and max pooling do not clear paired seed noise (2/5 reversals each). A sign test on the four condition means gives $p=0.125$. Three poolings share graphs and split; Ward is a different design (§A.6.5).

2. A quantified supervision ladder on one corpus and split: on the shared T22 tensor, the cost of unseen behavioural groups is 0.127 AUC (stratified 0.976 to pooled out-of-fold 0.849); the further cost of dropping malware labels is 0.073 against the raw one-class incumbent, an API-level pooled representation rather than the T22 tensor of the first two rungs—the matched T22 input centroid (0.777) gives 0.072, so the figure does not rest on that difference. A random-group control (0.971 pooled out-of-fold, the same statistic as rung 2) shows the holdout is group structure, not fold size. The groups are a family proxy, not AVClass families (§A.6.6, §A.10.1).
3. The deviation vector carries signal the scalar destroys, but the climb is not a monotone reference ladder. The D-3 factorial (§A.6.7.1) shows per-node readout on the deviation representation is the only distinguishable benign-only margin ($D-C=+0.039$); a benign-centred scalar norm of the raw flattened input (cell A, 0.793) is statistically indistinguishable from D_1 (0.800; paired DeLong $p=0.551$) on the full split yet fails benign-group holdout (0.712 versus 0.789; §A.6.7.1). Supervised readout on the same profiles reaches 0.962, locating an upper bound. The same deviation statistic against a per-app rather than population reference does not detect at window granularity (best raw 0.700 below the 0.703 floor; E0 best size-matched 0.624; E2 0.675 with the interval spanning the floor; $|\text{PREFIX} - \text{SCATTERED}|=0.006$). A supervised diagnostic ceiling on d (0.915–0.934, not a detector) and d_{self} 0.934 versus d_{cross} 0.847 (DeLong $p\approx 3\times 10^{-6}$) show the null is not representation capacity and that self-reference is primary under supervision; cross-session evaluation is Chapter C (§A.6.8). Sparse aggregation (D-1) and higher criticism (D-2) falsify alternative readouts (§A.6.7.2, §A.6.7.3).
4. A domain counter-example to a published reconstruction remedy. Yu Cai, Hao Chen, and Kwang-Ting Cheng [5] argue that a sufficiently narrow bottleneck cannot implement an identity map and that reducing latent entropy limits unwanted anomaly reconstruction. On these graphs a genuine bottleneck ($h=2$, five-fold compression of a 10-dimensional node feature) scores 0.568, inverted and worst of the widths swept (§A.6.2, §A.3.2).
5. Evidence across seven objective families that trained and untrained graph models are indistinguishable *where paired* on this representation, including the deviation-profile (D_1 centroid) pair that carries the benign-only headline (random-init 0.812 versus trained 0.800; paired DeLong $p=0.492$). GAE reconstruction, kernels, and pooling have no untrained pair; those cells are not claimed as Wilcoxon tests (§A.6.3).
6. Separation of edge *count* from edge *pattern*: a T1K one-class WL kernel reads structure (structure-only 0.627 versus edges-removed 0.500). Message passing does not help the T22 GIN ($M1 \leq M2$ at every condition mean; §A.6.5). The two comparisons do not share a configuration.
7. A corpus-validation record: seven artefacts identified and corrected, two of which had already produced headline numbers (0.814 residualised pooling; 0.734 V2 invocation floor) (§A.7).

A.1.4 Research questions

The chapter answers eight questions. Closures are in Table A.23.

1. Does the representation carry malware signal at all?
2. Does benign-only reconstruction detect malware?
3. Is failure attributable to the objective, the architecture, or the representation?
4. Does graph topology contribute, and can message passing extract it?
5. What does the benign-only guarantee cost?
6. Where in the deviation signal does the information sit?
7. Does an application deviate from its own prior behaviour, and does that deviation detect malware?
8. Is any of it deployable?

A.2 Corpus

Chapter A evaluates ABRG on a public, year-matched pair of Android behavioural traces with both benign and malware labels. This section records why that corpus was chosen, which validation gates were applied (and which withdrawn), how the eligible population was formed, and how malware binaries were handled for static seeding.

A.2.1 Selection: sensor symmetry

Sensor symmetry is the deciding constraint: benign and malware traces must share the same instrumentation, otherwise differences in graph sparsity are confounded with the sensor itself. Reconstruction error on ABRG graphs is dominated by sparsity, so a mixed-sensor corpus produces a label-aligned structural nuisance that the autoencoder is designed to notice. That is why a mixed pairing of self-collected benign traces with a public malware corpus under a different sensor was rejected: both classes must be traced by the same instrument if class comparisons are to be attributed to behaviour rather than to the collector. Syscall-level corpora fail a second requirement: Java API identity cannot be recovered from raw syscalls without inventing a mapping, so category assignment would be fabrication rather than lookup.

AndroCT 2017 meets both constraints. Emulator traces for both classes were released together under the same DroidFax hooks (Zenodo record 4470320; Li, Fu, and Cai [19]), and Soot/Jimple callee signatures carry Java API identity directly into the category mapper (§A.4.1). Shared hooks are a design strength of this corpus choice—they remove a mixed-sensor confound by construction—not a concession recorded later as a threat. Because AndroCT also ships benign traces under those hooks, a ContextDroid-trained GAE could in principle be scored on them as a cross-sensor falsification; Chapter A does not run that check, but the availability of a matched benign arm is part of why this release was preferred.

A.2.2 Four validation gates

Four gates were applied before the eligible population was frozen.

Year overlap. Archive filenames are not treated as authoritative year labels; loaders require members under the archive-internal directories `benign2017` / `malware-2017`. AndroZoo index lookups confirmed `dex_date` year 2017 for all hex64 benign ground-truth SHAs ($n=608$) and for a malware control sample ($n=300$, seed 42).

Parser. An early inventory used a call-line regex that rejected Jimple `<init>` / `<clinit>` signatures. That bug dropped 12.2% of benign and 27.9% of malware lines—asymmetric and label-aligned—and left only 20 / 17 universe categories firing. The regex was fixed to accept those signatures. The current inventory is post-fix: dropped-line rates fall to near zero, and active universe categories rise to 21 / 18 (benign / malware), with `dynamic_code_loading` recovered on the benign side. All Chapter A tensors use the post-fix parser.

Benign naming collapse. Of 2256 benign archive files, 608 are hex64-named and 1648 are package-named. Package names were resolved against the AndroZoo index under `dex_date` year 2017 and `vt_detection=0`: 172 resolved uniquely, 145 remained ambiguous, and 1331 were unresolvable. Malware files are hex64-named throughout ($n=1742$). Ambiguous and unresolvable package names were not forced into the population; the benign fetch set is the union of clean hex64 traces and the 172 unique resolutions, after VT and SHA dedupe (§A.2.3).

Eligibility floor (tested and withdrawn). A mapped-event floor of 320 was applied as a selection rule and then removed: it excluded 58.5% of benign apps versus 27.0% of malware, and under that population the trivial floors that the GAE arms lost to inverted (total-event floor AUC 0.6055 and active-category floor AUC 0.6799, both `benign_higher_score`). The full confound account—and why those arm scores are not Chapter A results—is in §A.7. The live corpus uses only necessity filters (mapped ≥ 1 ; non-empty categories after re-parse), yielding eligible $n=2403$ (703 benign / 1700 malware), consistent with §A.4 and §A.5.

A.2.3 Population and attrition

Table A.2 summarises the path from Zenodo archives to the split used throughout the chapter. After header-only drops, the fetch stage retains 763 benign and 1742 malware apps. Zero-mapped exclusion is asymmetric: 50 / 763 benign (6.6%) versus 39 / 1742 malware (2.2%). A further 10 benign and 3 malware apps lose all categories after re-parse, leaving 703 / 1700 graph-eligible apps ($n=2403$), split under the benign-only protocol of §A.5.1 into train-benign 562, test-benign 141, and test-malware 1700.

Table A.2. AndroCT 2017 attrition from archive files to the Chapter A split (benign-only protocol, seed 42).

Stage	Benign	Malware	Total
Archive files	2256	1742	3998
Parsed non-empty	2231	1736	3967
Fetch (post VT / dedupe)	763	1742	2505
Mapped ≥ 1	713	1703	2416
Graph-eligible	703	1700	2403
Train / test-benign / test-malware	562 / 141	1700	—

Corpus-level inventory on the post-fix parse shows mapped-event rates of approximately 2.9% (benign) and 3.3% (malware). The category `sms` records zero events in both classes. Separately, among allowlisted callees sampled for mapper diagnostics, the empty-map rate is 0.965 (benign)

and 0.959 (malware). The mapper selects a security-relevant subset by design; the 96.5% unmapped rate reflects that selectivity rather than a coverage defect. This is an abstraction choice, not a claim that unmapped calls carry no signal—the API-level representation of §A.6.4, which selects nodes by frequency rather than category membership, includes callees outside the taxonomy and reaches a higher supervised ceiling.

A.2.4 Ethics and data handling

AndroCT traces are distributed via Zenodo record 4470320; licence and access terms are those of that record. Trace evaluation does not require shipping malware binaries with the logs. For static seeding only, approximately 1700 malware APKs (and the corresponding benign APKs) were retrieved from AndroZoo, parsed with Androguard, and never executed: no emulator run, no dynamic instrumentation of those APKs, and no outbound network from the parsing host as part of feature extraction. APK material is kept in a vault outside synced directories and outside git worktrees; only derived static reports enter the corpus cache used by Chapter A experiments.

A.3 Related work

Chapter A sits at the intersection of graph-level anomaly detection, reconstruction-based one-class scoring, and graph methods for Android malware. This section states what those literatures already establish, where the AndroCT measurements belong, and which neighbouring questions are closed by published evidence rather than by another run. Literature is reviewed through April 2026.

A.3.1 Graph-level anomaly detection

Deep one-class models for whole graphs train on a single class and score held-out graphs as anomalies. OCGIN maps each graph toward a frozen centre estimated from an untrained forward pass and scores by squared distance [34]; Zhao and Akoglu [46] also characterise a *performance flip*: swapping which class is treated as normal can reverse ranking quality, a diagnostic that the objective is not reading a stable class geometry. The base configuration pools layer readouts by mean and applies batch normalisation; OCGIN[†] replaces mean pooling with add pooling and batch normalisation with graph normalisation [30]. Both variants are run on AndroCT (§A.6.3, §A.8.1). OCGTL adds a transformation-learning contrast among K views to a one-class centre and supplies a hypersphere-collapse analysis [30]. GLocalKD distills a frozen random-target GNN into a predictor on normal graphs and scores by prediction mismatch [22]. OCPool reduces each graph to a pooled node-feature vector and fits a one-class head: it is the edge-free baseline against which any claim that “the graph helps” is judged.

Wang et al. [38] present UB-GOLD, a unified benchmark for unsupervised graph-level anomaly detection and out-of-distribution detection spanning 35 datasets and 18 GLAD/GLOD methods, and exclude detectors that require anomaly labels at training time. Of UB-GOLD’s six GNN-based GLAD methods, this chapter implements OCGIN, OCGTL, and GLocalKD, chosen to span the reconstruction and one-class objectives whose mechanism is under test, not to reproduce the full benchmark; SIGNET, CVTGAD, and GLADC are not implemented (SIGNET and CVTGAD are the two strongest of the six on UB-GOLD Table 3 means across the fifteen GLAD datasets: 74.13 and 73.67 AUROC). Yang et al. [43] locate normal-only training as the dominant *partially unsupervised* GLAD paradigm and argue for moving to fully unsupervised training on unfiltered data. Broader surveys of deep graph anomaly detection collect the same families under a common taxonomy [29].

Published deep one-class graph models report floor AUCs from chance to the low sixties on molecular and protein graph benchmarks; Qiu et al. [30] tabulate separate ranges for base OCGIN and OCGIN[†] on those datasets (§A.8.1). The comparison is *across problem domains*—benchmark graphs are molecular and protein structures, not mobile behaviour—so the bands set scale, not a matched baseline.

A.3.2 Reconstruction-based anomaly detection

Autoencoder anomaly detection assumes that a model trained only on normal inputs will reconstruct those inputs and fail on anomalies. Bouman and Heskes [4] treat that assumption as unreliable: unwanted anomaly reconstruction is common enough to be a research object rather than an implementation accident. Competing explanations include an *identical shortcut* in which the network learns a near-identity map [45, 21]. Bercea, Rueckert, and Schnabel [3] instead challenge the assumption that normal training data occupy a low-dimensional in-distribution manifold—an adjacent but distinct critique.

Cai, Chen, and Cheng [5] argue that the shortcut diagnosis is misplaced if the bottleneck is narrow enough. Their Proposition 1 states that an autoencoder cannot implement an identity through a linear bottleneck when the latent width d is less than half the pre-bottleneck width; Proposition 2 then requires the latent entropy to match that of normal data—enough to reconstruct normality, not enough to represent lesions. Empirically they treat simple latent-dimension reduction as a practical entropy constraint and report that extremely small widths (often $d=4$ on their medical images) can be optimal, while too-small widths eventually fail to represent normality.

That prescription does not transfer to the AndroCT dual-reconstruction GAE as a sufficient fix. Node features are 10-dimensional; hidden width $h=2$ is a genuine compression, $h=8$ a modest bottleneck, and $h=64$ an expansion. Floor AUC at $h=2$ is 0.568, worse than the tied $h=8$ / $h=64$ pair at 0.638, and every width remains inverted (§A.6.2). A real bottleneck did not prevent anomaly reconstruction, and the expansion did not hurt relative to the modest bottleneck. The result is a domain counter-example to using latent-width reduction as a remedy for unwanted reconstruction on these graphs, not a replication of Cai et al.’s medical-image protocol [5].

Memory-augmented autoencoders (MemAE; Gong et al. [11]) address out-of-bounds reconstruction by decoding from stored normal prototypes. The critique literature notes a substantial drop in reconstruction ability and added optimisation cost; the module also does not target the collinearity mechanism measured on AndroCT (§A.6.2). It is a remedy not attempted, with that reason.

A.3.3 Android malware detection with graphs

Graph methods that report strong Android detection either pre-filter to a small subgraph or compute structural features directly. None of the high-performing published pipelines runs message passing over a large graph of mostly uninformative nodes. The comparators in this subsection are static call-graph pipelines; ABRG builds its graphs from runtime traces (DroidFax on AndroCT, Frida on v2_extended). The topology argument nevertheless transfers, because it concerns the structure of the graph passed to the model—node count, occupancy, and how much of it is uninformative—not how that structure was obtained.

MsDroid [15] classifies 2-hop subgraphs around *sensitive* APIs. SeGDroid [20] and related sensitive-function-call extractors repeat the filter-first pattern. MalScan [40] scores social-network centrality on the function-call graph without learning an aggregator. GDroid [10] builds a heterogeneous App-API graph so that structure is cross-app rather than within-app topology. ANAKIN [1]

attaches explanations to API-node call graphs. HertDroid [24] pairs an influential-node filter with a heterogeneous graph transformer: filter, then attend.

Wu, Luktarhan, Tian, and Song [39] (S-FCSG subgraph) state the message-passing failure in this chapter’s own terms: nonsensical nodes cause informative node features to drift toward uninformative ones during propagation. That is the mechanism measured by the supervised GIN ablation (M1 worse than edges-removed M2; §A.6.5) and interpreted in §A.8.2.

MaMaDroid [23] is the closest published relative of the ABRG design. It abstracts call sequences to package or family Markov chains and feeds those transition statistics to a conventional classifier: transitions are *features*, never message-passing topology, and the method works. Flattened adjacency under full supervision on AndroCT reaches floor AUC 0.959 (Table A.5)—the same finding, independently rediscovered, that a transition matrix already carries the label when a non-GNN head is allowed to use it. The architectural question left for this chapter is why one-class heads and GIN stacks fail to extract that matrix, not whether the matrix exists.

The 22-node all-nodes universe is therefore the configuration those systems avoid. ABRG keeps every behavioural category present so that deviation profiles remain comparable across apps (§A.4.1); the cost of that choice is exactly the uninformative-node regime S-FCSG warns about. Observability (§A.10.1) is a second, distinct cost of the same choice: some slots are structurally empty under the acquisition protocol, not merely rarely used.

A.3.4 GNN expressiveness and over-smoothing

Repeated neighbourhood aggregation drives node embeddings toward indistinguishability [18, 26]. GIN is constructed to reach the 1-Weisfeiler–Leman upper bound on distinguishable neighbourhoods [41]; a single aggregator is already sub-1-WL when neighbourhood multisets have uncountable support [7]. Those three results are independent explanations of a non-positive mean M1 – M2: over-smoothing, hitting the 1-WL ceiling, and a readout that cannot count.

On the canonical 22-node graph a four-layer GIN saturates its receptive field in roughly two hops: with $k=5$ sequence proximity the graph is small and dense enough that further layers mix already-mixed colours. Over-smoothing that takes five or six layers on large sparse graphs happens immediately here. That is a geometric argument, not a measured Dirichlet-energy diagnostic; it is consistent with OCGIN and OCGTL also using four GIN layers and collapsing on T22 while behaving better on T1K (§A.8.2).

Graph kernels sit on the other side of the same bound: the Weisfeiler–Lehman kernel reads subtree patterns without training a smoother [35]. Chapter A uses that family as a structure probe, not as a competing deep detector (§A.5.5).

A.3.5 Evaluation methodology

Random splits inflate Android malware figures when they leak time or space. TESSERACT [28] names temporal and spatial bias; later reappraisals keep that diagnosis in force. Concept-drift detectors treat unseen families and time slices as the evaluation object, not as an afterthought (CADE [42]; Transcending TRANSCEND [2]—distinct from TRANSCEND [16]). Longitudinal Android benchmarks supply the temporal splits those methods assume (LAMDA [14]).

Tran et al. [37] quantify generalisation gaps under family and distribution shift for *supervised* graph-based Android classifiers. CADE and Transcending TRANSCEND establish drift-aware evaluation as standard practice; the supervision ladder in §A.6.6 adopts that spirit: rung 1 is a random supervised split, rung 2 holds out behavioural groups, and rung 3 drops malware labels. Ward clusters are a proxy for family holdout because AndroCT ships no family labels (§A.10.1);

the claim is alignment with drift-aware evaluation, not identity with AVClass family splits or with Tran et al.’s OOD protocol [37].

A.3.6 Questions closed by citation

This subsection records questions prior work settles for the design used here, and one it does not.

Graph transformers. Chen et al. [6] report, in the paper body rather than the abstract, hierarchical CFG/FCG models on 499K Android applications spanning 2012–2022, with a cross-anchor drift protocol on 433,488 test apps (v2 refined the corpus from 595K in v1). Hierarchical modelling outperformed both flat-graph GNNs and GraphGPS [32] under matched feature and hidden dimensions, and the discriminative signal localised to CFG-level cyclomatic complexity rather than aggregate FCG metrics. That is independent corroboration, at a scale this thesis cannot match, of the structural null in §A.6.4: aggregate edge-count and density summaries remain at chance across every node vocabulary tested. Their remedy is hierarchy—nesting control-flow inside function-call graphs—which a dynamic behavioural trace does not provide.

Attention is not an automatic escape from representation collapse. Dong, Cordonnier, and Loukas [8] prove that pure self-attention without skip connections or MLPs converges doubly exponentially toward rank-1 token uniformity; over-smoothing remains the message-passing analogue (§A.3.4). Graphormer [44] and GraphGPS [32] are representative graph-transformer architectures benchmarked by HiGraph above; neither paper analyses asymptotic collapse for the full stack, and a secondary preprint [13] that notes the absence of published collapse guarantees for those models does not settle the question on its own.

Graph rewiring. Rewiring modifies connectivity to relieve over-squashing and over-smoothing. Tori, Holst, and Ginis [36] report that curvature-based gains are highly sensitive to training and rewiring hyperparameters and that reported state-of-the-art improvements often arise from favourable configurations rather than consistent improvement over the original topology. That sensitivity is the reason rewiring is not an AndroCT axis.

Memory modules. MemAE is the reconstruction-side analogue: it reduces reconstruction capacity by design and adds a memory optimisation that does not address near-collinear malware reconstructions on this corpus (§A.3.2).

A.4 Method

This section specifies the behavioural graph representation used throughout Chapter A and the graph autoencoder (GAE) that produces the deviation scores analysed later. The representation is fixed for all experiments in this chapter unless an ablation states otherwise. Comparison detectors that consume the same graphs are listed in §A.5.5; only construction and the reference encoder are defined here.

A.4.1 Fixed category universe

Each application is represented as a directed attributed graph over a fixed node set of 22 behavioural categories (Table A.3). The universe is identical for every app. Three further hook categories collected in traces (`lifecycle`, `reflection`, `navigation`) are excluded from the graph by design.

Table A.3. Fixed behavioural category universe used as graph nodes in Chapter A (T22).

<code>accounts</code>	<code>audio</code>
<code>camera</code>	<code>clipboard</code>
<code>content_access</code>	<code>crypto</code>
<code>database</code>	<code>device_info</code>
<code>dynamic_code_loading</code>	<code>file_io</code>
<code>ipc_intents</code>	<code>location</code>
<code>media</code>	<code>native_code</code>
<code>network</code>	<code>notifications</code>
<code>package_manager</code>	<code>process</code>
<code>sms</code>	<code>storage</code>
<code>telephony</code>	<code>webview</code>

AndroCT logcat lines expose Soot/Jimple callee signatures. Each callee is mapped to zero or more categories by a deterministic rule set aligned with the Frida hook taxonomy (`categorize_callee` / `categorize_soot_callee`): exact hook-API labels where available, otherwise package/API prefixes, with a fixed priority order when several categories fire. Unmapped events and the three non-graph hook categories contribute no node outside the universe; they simply leave the corresponding graph slots inactive. Corpus-level mapped rates and parser attrition are reported in §A.2.

Because the node set does not depend on the app, graphs are directly comparable and can be stacked as tensors of constant shape. Inactive categories remain as zero-feature, zero-degree nodes rather than being dropped. Comparability across a structurally unobservable coordinate is vacuous: on AndroCT only 18 of 22 categories are environment-PRODUCIBLE under the Monkey protocol (§A.10.1). Four coordinates are excluded from that count: `sms` and `telephony` are PERMISSION_GATED (dangerous runtime permissions not granted under Monkey); `clipboard` and `dynamic_code_loading` are STIMULUS_LIMITED (APIs mapped but rarely reached).

A.4.2 Node features

Each node carries a ten-dimensional feature vector. The first seven coordinates are static (Androguard on the matching APK); the last three are dynamic (the mapped trace):

$$x_v = [s_v, \text{declared}_v, \text{gate}_v^{(0)}, \text{gate}_v^{(1)}, \text{gate}_v^{(2)}, \text{reach}_v, \text{epoch}_v, \text{act}_v, \text{sess}_v, \text{rec}_v].$$

The static slice is, in order: a sensitivity score $s_v \in [0, 1]$ from Android permission protection levels; a binary declared-capability flag; a three-dimensional gate over the protection-level buckets normal, dangerous, and signature; a count of manifest components from which the category is statically reachable; and an application-version epoch epoch_v . On AndroCT that epoch is unused and is written as zero on every node. On the graph-eligible population ($n=2403$) the static slice is never the zero vector: the minimum Euclidean norm is 3.06.

Dynamic coordinates are filled from the trace. Activity act_v is stored as a raw mapped-event count and is fed to the encoder as that node’s share of the application’s total mapped events. Session occupancy sess_v is the fraction of processing windows in which the category was active; with one whole-trace graph per app ($N=1$) it reduces to an indicator of whether the node fired. The recency slot rec_v is defined in the design but is never populated on AndroCT (no timestamps); construction asserts that it remains zero.

A.4.3 Edges

Edges are directed category-to-category transitions. On AndroCT, an edge $u \rightarrow v$ is created when an event of category v follows an event of category u within a sequence proximity of $k=5$ events. Self-transitions are not written. Edge weights use the cumulative channel w_{cum} only; the recency channel w_{rec} is never incremented.

At tensorisation, outgoing w_{cum} values on each source are converted to transition shares (outgoing probabilities); terminals with no out-edges receive zero outgoing weight. Raw counts remain stored in the graph; the GAE consumes the shares. This is the shares-not-counts normalisation used throughout Chapter A.

The reference representation is one whole-trace graph per eligible app ($N=1$). The same traces are also partitioned into windows for reconstruction and self-reference measurements (§A.6.2, §A.6.8).

Temporal proximity filter δ . In the full ABRG design, edge formation is the conjunction of sequence proximity k and a wall-clock tolerance δ : candidate pairs whose timestamps differ by more than δ are discarded. The filter can only remove edges; gap duration is never written as an edge attribute. AndroCT traces carry no timestamps, so the AndroCT builder applies k alone and leaves δ unexercised. The cost of enabling δ on a timestamped corpus is quantified in Chapter B (§B.5). This chapter reports no number for that cost; the measurement and its caveats live there.

A.4.4 Per-app tensor

The objects consumed by classical detectors and by deviation-profile extraction are built from the same graphs. Flattening the 22×10 node-feature matrix yields 220 coordinates; flattening the 22×22 adjacency yields 484; concatenation yields a 704-dimensional vector per app. Message-passing models instead receive the sparse form ($X \in \mathbb{R}^{22 \times 10}$, E , w) with optional edge weights on the encoder.

A.4.5 Graph autoencoder

The encoder is a two-layer graph convolutional network (GCN) with ReLU after the first layer, implemented as PyTorch Geometric `GCNConv` modules that accept the transition-share edge weights. A standard inner-product decoder reconstructs adjacency from the latent node embeddings Z . When the training objective includes attribute reconstruction, a linear feature decoder maps Z back to $\hat{X}$.

Chapter A trains this encoder under more than one objective on the same tensors. Construction and Runs 2–4 used hidden width $h=64$; Run 5 then swept $h \in \{2, 4, 8, 16, 64\}$ at dual-reconstruction weight $\alpha=0.2$. Widths $h=8$ and $h=64$ were near-tied under that sweep; $h=8$ was carried forward.

The configuration carried forward as the reference method is Run 5: deterministic full-adjacency weighted binary cross-entropy on the 22×22 adjacency (including the diagonal), plus node-feature mean-squared reconstruction, combined as $\alpha \cdot \mathcal{L}_{\text{struct}} + (1 - \alpha) \cdot \mathcal{L}_{\text{feat}}$ with $\alpha=0.2$, hidden width $h=8$, one whole-trace graph per app ($N=1$), 300 epochs, learning rate 0.01, weight decay 0, seed 42, and training on train-benign apps only. This is the encoder that produces the scalar deviation score D_0 and the per-node deviation profiles D_1 – D_5 reported in §A.6.7. Earlier objectives—Run 2 stochastic positive-edge reconstruction and Run 3 deterministic structure-only full-adjacency BCE—are interventions in the reconstruction sequence of §A.6.2, not alternative reference methods. Training and scoring pins for those runs are collected in Appendix A.11.

Two alternative representations—an API-level vocabulary and a caller→callee invocation edge definition—were constructed to test whether the structural findings depend on this granularity; both are described where their results are reported (§A.6.4).

A.5 Experimental protocol

This section fixes the evaluation apparatus used throughout Chapter A: how apps are split, how detectors may be fit, how scores become AUC figures, which trivial baselines any claimed detector must clear, which controls accompany those claims, and which comparison methods populate the objective-family sweep. Results appear in §A.6.

Numbered tables in this chapter are the live objects (Table A.*n* in the compiled PDF). Files `T1_*.csv`–`T8_*.csv` under `chapter_a/tables/` are generation sources; a caption or sentence that names sheet T5 or T8 points at that CSV, not at a second table in the PDF.

A.5.1 Splits and digest

Two app-level splits are used. Both are defined once here and referred to by name thereafter.

Benign-only split. Eligible apps are partitioned with seed 42 and `test_ratio=0.2` into train-benign ($n=562$), test-benign ($n=141$), and test-malware ($n=1700$). All malware apps are held in the test partition; none enter the training set. This is the split used by the reference GAE and by every one-class or reconstruction detector in this chapter (OCGIN, OCGTL, GLocalKD, OCPool, classical flat one-class, kernels). The defining property is the training regime—malware-free fit—not the architecture that consumes the tensors.

Stratified supervised split. When both labels are available for training (supervised probes and the supervision ladder), apps are partitioned by a stratified both-class 80/20 draw with the same seed 42. Benign train/test membership for the leave-one-cluster-out ladder arm remains fixed at the benign-only split counts (562/141); only the malware grouping changes.

A leave-one-cluster-out arm (Ward clustering of malware on standardized 704-dimensional vectors; cluster count selected by silhouette over {5, 10, 15, 20, 30}) is used where group holdout is the experimental axis. That arm is meaningless for methods fit on train-benign alone: the benign training set is identical across folds, so only supervised readouts vary.

Every experiment that reuses the benign-only partition asserts the SHA list digest whose prefix is `6129eb13d6a4...`,¹ thereafter referred to by that prefix alone.

A.5.2 Benign-only training commitment

For all reconstruction and one-class detectors, model parameters are fit on train-benign apps only. Malware graphs and labels never enter `fit`, vocabulary construction, PCA, kernel Gram fitting, or centre estimation. Supervised probes (HistGradientBoosting, logistic regression, end-to-end GIN) are the sole exception; they are diagnostic ceilings, not proposed detectors, and are always labelled as such (§A.5.5).

¹Full digest: `6129eb13d6a46457cd60627372b7b5479df0aa1f4efc9bbb70adc17826c64000`. Computed over the ordered train / test-benign / test-malware SHA lists in corpus-cache metadata (list order, not sorted); see `abrg/apigraph/split.py` (`_sha_digest`) and `abrg/output/androct_2017/final_validation/SUMMARY.md`.

This commitment conforms to the standard unsupervised GLAD/GLOD protocol rather than imposing a restriction unique to this chapter. Wang et al. [38] state that in their target scenarios, “all the samples in the training set are normal/ID, while the anomaly/OOD samples only occur in the testing set,” with no validation set containing anomalies during training (§3.3 of that benchmark). The absence of a labelled validation set here is therefore protocol alignment, not a design omission. Mixing anomaly/OOD graphs into the training set is a separate robustness experiment in that benchmark (§4.3), not the default setting. Yang et al. [43] term this setting *partially unsupervised*: training on normal samples only, with anomalies appearing solely at test. The chapter’s benign-only commitment is that protocol. The guarantee removes malware labels from training but does not make training label-free: it assumes a benign pool verified clean of anomalies, which in practice requires expert filtering. The 562 train-benign applications are benign by AndroCT’s labelling; that cleanliness has not been independently verified here. That is a scope condition on the guarantee, not a defect in the protocol.

A.5.3 Metric and direction

Primary discrimination is reported as a floor AUC $\text{AUC}_{\text{floor}} = \max(\text{AUC}, 1 - \text{AUC})$ together with an explicit score direction (**malware_higher_score** or **benign_higher_score**). The floor makes inverted detectors numerically comparable to non-inverted ones; the direction string records which class ranks higher on the raw score. Where a method is intended to flag malware by large deviation, a benign-higher direction is a failure mode, not a re-labelable success.

Averaging per-fold floor AUCs can hide direction flips: a fold with raw AUC 0.4 contributes 0.6 to a mean floor and looks like weak discrimination rather than inversion. When folds are the unit of analysis, this chapter therefore reports pooled out-of-fold raw AUC (or an explicit flip count) alongside any floor summary; the quantitative inflation is measured in §A.6.6.

Bootstrap percentile intervals on scores use $B=2000$ resamples unless a nested procedure is required (§A.5.4).

A.5.4 Trivial floors, controls, and nested uncertainty

Trivial floors. Any claimed detector on the canonical T22 representation is interpreted against scalar baselines that use no learned parameters: counts and simple graph summaries scored on the benign-only split (Table A.4). The mapped-event count (0.7025) is the size bar used as the primary trivial gate; total event count sits just below it (0.6912). Structural summaries (active-node / category counts near 0.52; edge count and density at 0.5267) are near chance. Floors on API-level and invocation-edge representations are reported with those representations (§A.6.4), not here.

Table A.4. Trivial floors on the canonical T22 representation (benign-only split). Floor AUC = $\max(\text{AUC}, 1 - \text{AUC})$; 95% bootstrap intervals ($B=2000$).

Metric	Direction	Floor	CI low	CI high
mapped_event_count	malware higher	0.7025	0.6434	0.7545
total_event_count	malware higher	0.6912	0.6294	0.7477
edge_count	malware higher	0.5267	0.5013	0.5781
graph_density	malware higher	0.5267	0.5013	0.5781
distinct_active_categories	benign higher	0.5165	0.5007	0.5699
active_nodes	benign higher	0.5164	0.5007	0.5704

Controls. Claims in later sections are paired with at least one of the following seven controls, chosen for the claim being tested: (i) random-init / untrained encoders of identical architecture (including frozen-teacher baselines); (ii) shuffled-label readouts on the same scores; (iii) raw-input centroid or other flat one-class detectors on the same tensors; (iv) random-group holdout of matching cluster sizes when leave-one-cluster-out is the axis; (v) shuffled-support controls on deviation / support-derived scores; (vi) volume and density diagnostics—Spearman correlations of each configuration’s scores against the six trivial scalars of Table A.4 (mapped/total events, active nodes and categories, edge count, density)—used to reject size-coupled candidates; (vii) the trivial-floor comparison itself (Table A.4), treated as a named control rather than background: a detector that does not clear the mapped-event bar is not reported as a win. Which control applies is stated beside the result; the control inventory does not itself assert AUCs as findings.

Nested bootstrap. When vocabulary construction, feature selection, or another data-dependent preprocessing step is part of the fitted pipeline, score bootstrap alone understates uncertainty. In those cases a nested bootstrap refits the inner pipeline on resampled train-benign apps and rescores a fixed evaluation set, typically with outer $B=200$. Abandonments of that procedure are recorded as process artefacts (§A.7), not as silent omissions.

A.5.5 Comparison methods

The detectors below are not a convenience sample. The central claim of this chapter—that failure on AndroCT sits in the objective and the readout, not in the representation or the message-passing architecture—requires that each major objective family in graph-level anomaly detection be represented under the same tensors and the benign-only split. Selection required a published method (or classical baseline), available code or an unambiguous reimplementing target, and compatibility with benign-only training. Hyperparameter grids are collected in Appendix A.12; only the objective and the one configuration detail that matters for interpretation are stated here.

Reconstruction. The reference method is the dual-reconstruction GAE of §A.4.5 (structure BCE plus feature MSE, $\alpha=0.2$, $h=8$). Reconstruction is the standard deep formulation of graph anomaly scoring and the ABRG design commitment; all later interventions and deviation profiles are readouts of this encoder (or of named objective variants of it).

One-class embedding. OCGIN maps graphs toward a frozen centre θ estimated from an untrained forward pass and scores by squared distance [46]; OCGTL adds a transformation-learning contrast among K views to a one-class centre [30]. Both ask whether a hypersphere / transformation objective succeeds where reconstruction fails. Configurations carried forward use four-layer GIN encoders with hierarchical readout (embedding dimension 128); OCGTL follows the paper learning rate 0.001 and temperature $\tau=1$. The sweep uses $K \in \{4, 6\}$: $K=6$ matches the paper’s six learnable transformations; $K=4$ is an additional ablation. Batch size is 64 (reference 128), training runs 300 epochs (reference 500), with no learning-rate scheduler or early stopping.

Distillation. GLocalKD distills a frozen random-target GNN into a predictor on benign-only data and scores by prediction mismatch [22]. It is included because the chapter already observes untrained encoders outperforming trained ones; a published random-teacher method tests whether that pattern can be turned into a detector. Primary score is the graph-level mismatch; mean/add/max pooling are reported as variants.

Kernel and shallow embedding. Weisfeiler–Lehman, propagation, shortest-path, FGSD, NetLSD, Graph2Vec, and GL2Vec embeddings (with degree-label variants) supply structure-aware features without message-passing training. They test whether topology carries usable signal that deep GLAD fails to extract. Detectors on the resulting vectors or Gram matrices are classical one-class heads fit on train-benign only; anything not clearing the mapped-event floor is not treated as a result.

Classical one-class on flat tensors. Centroid (Euclidean / cosine), Isolation Forest, k -NN, Ledoit–Wolf Mahalanobis, and RBF one-class SVM are fit on the same flattened T22 tensors (704 / 220 / 484 coordinates). They bound what a graph model must beat without using edges as message-passing structure.

Pooling without edges. OCPool reduces each graph to a pooled node-feature vector and fits a one-class SVM. It is the edge-free baseline against which any claim that “the graph helps” is judged.

Supervised probes. HistGradientBoosting, logistic regression, and a supervised GIN are fit under the stratified supervised split (and, where relevant, the Ward holdout arm). They establish representation ceilings and price the cost of removing malware labels; they are not proposed detectors and are never scored under the benign-only training commitment.

A.6 Results

Results are organised by research question. The benign-only headline is D_1 centroid Euclidean scoring, floor AUC 0.800 (nested interval [0.757, 0.815]); at a 1% false-positive target that read-out reaches only 0.24% recall and 0.33% wild precision (§A.6.9). Supervised probes establish that the T22 representation carries malware signal (RQ1; §A.6.1). Reconstruction and embedding interventions under benign-only training fail to convert that signal into a detector that clears the mapped-event floor (RQ2; §A.6.2). Seven objective families and classical one-class heads localise the failure to label-free scoring rather than to a missing adjacency signal (RQ3; §A.6.3). Granularity and message-passing tests ask whether finer nodes or true call edges move the structural null (RQ4; §A.6.4, §A.6.5). The supervision ladder prices unseen malware groups (RQ5; §A.6.6). Deviation profiles versus the scalar score (RQ6; §A.6.7), within-session self-reference (RQ7; §A.6.8), and operating points at deployment-like prevalence (RQ8; §A.6.9) close the section.

A.6.1 The representation carries the signal

Under the stratified supervised split, flat classifiers read strong discrimination from the same tensors later scored by one-class methods (Table A.5). HistGradientBoosting on the full 704-dimensional vector reaches floor AUC 0.976; node features alone reach 0.957, and the flattened 22×22 adjacency alone reaches 0.959. Logistic regression is lower but still far above the mapped-event floor (0.908 / 0.931 / 0.855). These probes are diagnostic ceilings, not proposed detectors: both labels enter training.

Table A.5. Supervised probes on T22 tensors (stratified supervised split, seed 42). Floor AUC; direction malware-higher throughout.

features	HGB	LR
full (704)	0.9762	0.9085
node only (220)	0.9565	0.9308
adjacency only (484)	0.9593	0.8554

Zeroing crypto or file_io activity, alone or together, changes HGB by at most 0.014 on the full vector; zeroing the notification or process controls moves it by at most 0.003. The ceiling is therefore not an artefact of a single ad-library-heavy category: discriminative mass is distributed across the representation.

A.6.2 Benign-only reconstruction fails, with a measured mechanism

Table A.6 lists seven interventions on the benign-only split. Runs 2–4 use hidden width 64; Run 5 and the later rows use the canonical width 8. Every reconstruction arm is inverted (benign median error higher than malware) and below the mapped-event floor 0.703. Run 2’s floor AUC is 0.538 on the full 1841-app test set after NaN repair; an earlier figure computed on 1794 of those apps is withdrawn.

Table A.6. Reconstruction and embedding interventions (benign-only split). Floor AUC. Mapped-event floor = 0.7025. Runs 2–4: $h=64$; Run 5 onward: $h=8$ unless noted. Run 8 reports best trained embedding score and the matched random-init mean/centroid score.

row	h	objective / score	AUC _{floor}	direction	clears floor
Run 2	64	stochastic recon	0.5381	benign higher	no
Run 3	64	det. full-adj. BCE	0.6161	benign higher	no
Run 4	64	dual-recon $\alpha=0.2$	0.6379*	benign higher	no
Run 5	8	dual-recon $\alpha=0.2$	0.6379 [†]	benign higher	no
whiten	8	Run 5 + Ledoit–Wolf whiten	0.5600	benign higher	no
Arm B	8	dual-recon, $N=8$ windows (mean)	0.6012	benign higher	no
Run 8	8	emb. max/cosine (trained)	0.6834	malware higher	no
		emb. mean/centroid (random-init)	0.7591	malware higher	yes

*Run 4 = 0.637939; [†]Run 5 = 0.637893 (five decimals).

Geometry supplies a mechanism. Pairwise cosine distance among malware graphs (mean 0.462) exceeds that among test-benign graphs (mean 0.325); PCA needs one component to reach 90% variance on malware versus two to three on benign partitions. Malware mass is more collinear in the flattened space, so a benign-fitted reconstructor can assign *lower* error to malware without ever seeing a malware label.

Embedding-space scoring on the frozen Run 5 encoder (Run 8) is the first arm that is not inverted: best trained floor AUC 0.683, still below the size floor. An untrained encoder of identical architecture scores 0.759 under mean pooling and Euclidean centroid distance, clearing the size floor but remaining below the input-space centroid (0.777). Embedding scores are also *more* density-coupled than reconstruction error (mean $|\rho|$ 0.444 vs. 0.257 over mapped/total/edges/density). Training therefore does not merely fail to help: the trained encoder is worse than random initialization on the same scoring head, and more entangled with volume.

Three negatives close the reconstruction case. Multi-window Arm B ($N=8$) yields 0.601 (mean) and 0.621 (max), still inverted—the adaptive windowing claim fails on this corpus. Whitening

train-benign node rows before the canonical dual-recon fit drops the floor AUC to 0.560: targeting the measured collinearity makes the detector worse, so collinearity is a correlate of the inversion, not a remedy. Bottleneck width $h=2$ under the same dual-recon objective—the latent-dimension reduction of Yu Cai, Hao Chen, and Kwang-Ting Cheng [5]—scores 0.568, still inverted and the worst of the five-width dual-recon sweep at $\alpha=0.2$ ($h=2$: 0.568; $h=4$: 0.629; $h=8$: 0.638; $h=16$: 0.626; $h=64$: 0.638; all inverted). Cai, Chen, and Cheng [5] argue that a sufficiently narrow bottleneck cannot implement an identity map and that reducing latent entropy toward that of normal data limits unwanted anomaly reconstruction; on these graphs a genuine bottleneck ($h=2$, five-fold compression of a 10-dimensional node feature) did not prevent inversion (Table A.6 omits this row; it is a domain counter-example to that remedy as a sufficient fix, not a further intervention on the main sequence; §A.3.2).

A.6.3 Seven method families

Table A.7 summarises trained versus untrained (or random-init) floor AUCs for the seven objective families of §A.5.5, plus the deviation-profile (D_1 centroid) pair that carries the chapter headline. Paired Wilcoxon signed-rank tests at $n=5$ seeds cannot reach $p<0.05$ by construction (smallest attainable two-sided $p=0.0625$); per-family *indistinguishable* verdicts therefore report test resolution, not absence of effect. GLocalKD is the clearest instance: all five seeds favour the untrained predictor ($\Delta_{\text{floor}} \approx 0.106$) yet Wilcoxon $p=0.0625$. The aggregate direction is nevertheless consistent: five of six families in the table favour the untrained or no-training arm (GAE reconstruction has no paired untrained cell); among the five trainable objectives with an explicit untrained comparison (OCGIN, GLocalKD, OCGTL, GAE embedding distance, deviation profile D_1 centroid), all five favour untrained—a sign test gives $p=0.031$. No single family reaches significance; the claim is cross-architecture consistency, not per-family proof. The GAE-reconstruction cell is the T22 Run 5 point only: the earlier Wilcoxon pairing against invocation-V2 reconstruction is withdrawn as a representation artefact, and no T22 untrained reconstructor was fit for a replacement pair. Embedding distance uses the Run 8 mean/centroid deterministic pair ($n=1$; Wilcoxon not applicable)—0.669 trained, not the best trained max/cosine 0.683 in Table A.6. OCGIN, GLocalKD, and OCGTL retain their five-seed pairs; kernels and pooling have no untrained column.

Table A.7. Seven method families against the mapped-event size floor (0.7025), plus D_1 centroid trained versus random-init (paired DeLong $p=0.492$, not distinguishable). Floor AUC. Pairing sources: GAE recon = T22 Run 5 $h=8$ $\alpha=0.2$ (trained only); GAE embedding = Run 8 mean/centroid_euclidean; OCGIN = OCGIN_plus Variant A vs. random-init (5 seeds); GLocalKD = T22 pool=mean, s_{graph} , full loss (5 seeds); OCGTL = T1K $K=4$ vs. untrained $K=4$ (5 seeds); kernels = WL $h=3$; pooling = OCPool_mean raw

(validated residualisation R0).

family	trained	untrained	clears floor
GAE reconstruction	0.6379	—	no
Deviation profile (D_1 centroid)	0.8004	0.8118 [†]	yes / yes
GAE embedding distance	0.6691	0.7591	no / yes
OCGIN	0.5660	0.6538	no
GLocalKD	0.5594	0.6650	no
OCGTL	0.5503	0.5614	no
graph kernels	0.6726	—	no
pooling / one-class	0.7765	—	yes

[†]Random-init D_1 centroid: paired DeLong $p=0.492$; bootstrap CI on Δ_{floor} includes zero (§A.6.7).

For GLocalKD specifically, training *reduces* floor AUC relative to an untrained predictor (Table A.7: 0.5594 versus 0.6650; 5/5 seeds favour untrained; $\Delta \approx 0.106$). Because the method’s mechanism is distillation from a frozen random teacher into a trained predictor, an untrained predictor beating the distilled one means the mismatch between two independently random networks exceeds the mismatch after optimisation—distillation contributes negatively on this corpus. A paper-configuration validation arm (`glocalkd_papercfg`; lr 10^{-4} , 512/256, 150 epochs, dropout, batch normalisation, and StepLR as in Ma et al. [22]; §A.10.4) preserves the same direction (trained 0.5955 versus untrained 0.6633 on s_{graph} ; 5/5 seeds), so the brief profile’s $100\times$ learning rate alone does not explain the reversal.

The pooling incumbent 0.776 is mean-pooled node features concatenated with the static global vector, scored by a one-class SVM. Score-resample CI [0.727, 0.821]; nested bootstrap on the residualised readout is [0.699, 0.800] (source sheet T8). A related OCPool configuration that pools node features alone (no static global) scores 0.752 under the same SVM head; it is a different representation, not a competing estimate of the same quantity, and is not carried as the chapter incumbent.

Classical one-class detectors on the same 704-dimensional tensors, same split, provide a flat baseline ladder (Table A.8). The Euclidean input centroid (0.777) matches the pooling incumbent; every other head sits between 0.603 and 0.650, all below or near the size floor and well below supervised adjacency readout.

Table A.8. Classical one-class baselines on flattened T22 tensors (704-dim, benign-only train). Floor AUC; direction malware-higher.

method	AUC _{floor}
centroid Euclidean	0.7769
Isolation Forest	0.6500
k NN ($k=1$ / 5 / 20)	0.6402 / 0.6332 / 0.6069
Mahalanobis (Ledoit–Wolf)	0.6134
one-class SVM (RBF)	0.6033

The sharpest contrast in the chapter is structural and label-dependent. Supervised HGB reads 0.959 from the flattened adjacency alone. On that same raw 484-dimensional matrix, eight unsupervised one-class detectors cap at 0.545 (k NN, $k=5$). PCA reduction of the adjacency lifts the best unsupervised score to 0.683 (pca8, one-class SVM), recovering part of the gap but nowhere near closing it. Discriminative transitions occupy a minority of the 484 cells; unsupervised scoring has no mechanism to localise them, and linear dimensionality reduction only partially substitutes for that localisation.

Two collapse findings separate architecture failure from objective failure. OCGIN_{orig} hypersphere collapse occurs on all five seeds. OCGTL on T22 at $K=4$ and $K=6$ collapses by encoder agreement (mean pairwise cosine ≈ 0.9999) with train score-floor fraction 0.0—not the OCGIN near-zero-score signature—while the same OCGTL configurations on T1K do not collapse at all, consistent with a 22-node graph saturating a four-layer receptive field. These are reimplementations (AGPL reference not vendored; GTL interpreted as within-graph); the collapses are reported as process findings under that caveat.

Per-node distillation mismatch on the T1K family-sweep winner was once read as a null under the brief training profile (Table A.7): largest class-mean gaps $+0.000961$ and -0.000922 with alternating signs (`glocalkd/`). That reading is withdrawn. A paper-configuration arm on the same T1K representation, one seed, produces gaps of order 10^0 rather than 10^{-3} and reverses the graph-level score direction (§A.10.4). The 22-category localisation result rests on the D_1 per-node

ablation (§A.6.7), not on GLocalKD.

Performance-flip diagnostics on OCGIN (Variant A + Variant B) provide independent corroboration of the published pathology: a fully inverting scorer gives arm A + arm B ≈ 1.0 . `OCGIN_orig` sums to 1.083—near-complete flip, reproducing on AndroCT the instability Zhao and Akoglu [46] name on benchmark data; Qiu et al. [30] Table 2 reports the same pattern for base OCGIN (e.g. DD 26.3 versus 75.2, AIDS 26.0 versus 95.5 depending on which class is treated as normal). `OCGIN_plus` sums to 1.335—improved over the base arm but far short of 2.0. Random-init `OCGIN_plus` (0.654; Table A.7) exceeds trained `OCGIN_plus` (0.566) by 0.088 and falls inside the published *base* OCGIN band (≈ 0.507 – 0.624 ; Table A.22); this is another instance of the untrained-beats-trained pattern in Table A.7, not a separate finding.

A.6.4 Granularity and edge definition

The structural null of §A.6.3 is a property of one representation. This subsection asks whether a finer node vocabulary or a true call edge changes the trivial floors.

API-level vocabulary. The corpus yields 50,331 distinct callees. Node vocabularies are built from train-benign apps only: ranking over both classes would leak labels. Three selection rules (TF-IDF, document frequency, raw frequency) are crossed with four sizes ($K \in \{100, 300, 500, 1000\}$).

At $K=1000$ under TF-IDF, in-vocabulary coverage is 0.719 (train benign), 0.688 (test benign), and 0.444 (test malware). Because the vocabulary is benign-derived, the out-of-vocabulary (OOV) rate is itself a floor: 0.727 under TF-IDF, 0.636 under raw frequency, and 0.625 under document frequency. Document frequency is selected as the vocabulary whose OOV floor sits closest to chance; subsequent T1K detectors use that control unless a caption states otherwise.

Abstraction cost. On the TF-IDF $K=1000$ tensors, supervised HistGradientBoosting reaches floor AUC 0.996 (logistic regression 0.970), against 0.976 on the taxonomic T22 full vector (§A.6.1). A frequency-selected vocabulary that ignores the 22-category taxonomy therefore raises the supervised ceiling. That is the measurement behind §A.2’s claim that mapper selectivity is an abstraction choice, not a coverage defect: unmapped callees still carry signal once they are allowed to be nodes.

Structural floors do not move. Edge count and density remain near chance across K (0.512 / 0.509 / 0.507 / 0.513), slightly *below* the 22-node value 0.527 (Table A.9). Active-node count rises to 0.606 at $K=1000$, but that shift tracks the OOV effect rather than a new topological separator. Under the selected document-frequency vocabulary the edge floor is 0.501.

Table A.9. Trivial floors on API-level proximity graphs (TF-IDF vocabulary unless noted). T22 baselines: edge/density 0.5267, active nodes 0.5164. OOV is a score, not a graph summary.

K / vocab	edge	density	active nodes	OOV rate
100 TF-IDF	0.5116	0.5116	0.5416	0.7113
300 TF-IDF	0.5092	0.5092	0.5960	0.7223
500 TF-IDF	0.5074	0.5074	0.5933	0.7065
1000 TF-IDF	0.5126	0.5126	0.6055	0.7269
1000 docfreq	0.5013	0.5013	0.5659	0.6249

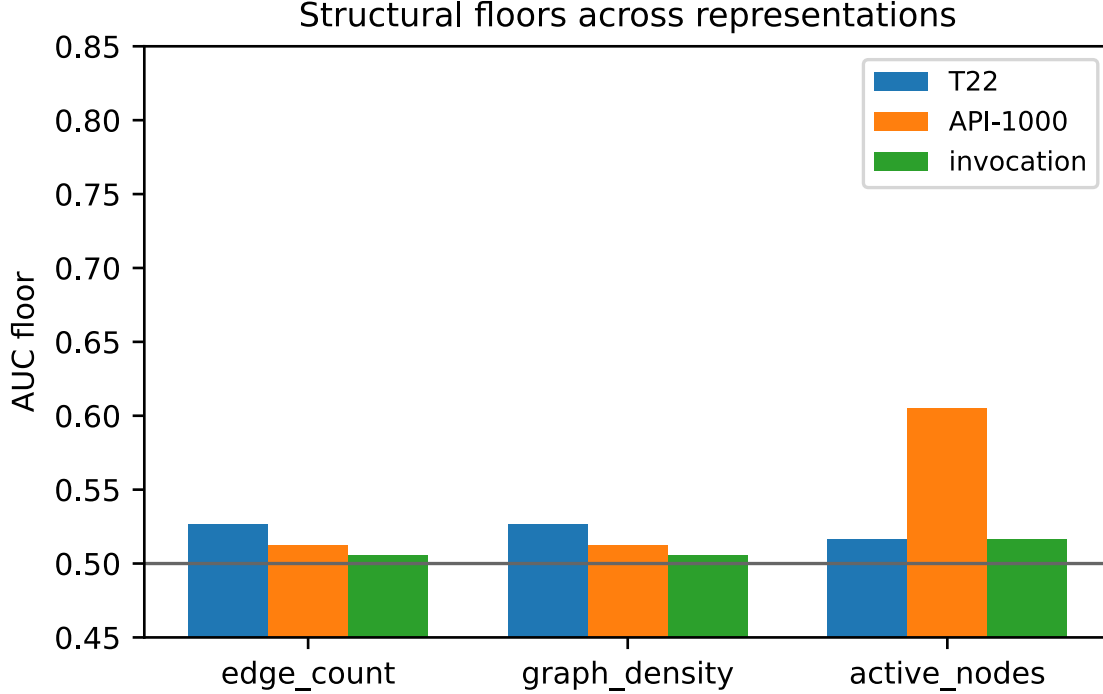


Figure A.1. Structural floors across 22-node proximity, API-level $K=1000$, and 22-category invocation (no self-loops; V2 withdrawn). Source: `chapter_a/figures/F6_granularity.svg`.

Invocation edges. AndroCT lines are `caller` $\rightarrow$ `callee`, so true call structure is available and was tested. API-vocabulary invocation (V2) produced an edge floor of 0.734 and was withdrawn: median drop rates were 0.949 (train benign) versus 0.9999 (test malware), because 68.7% of callers are app-package code absent from a callee-derived vocabulary. Ancestor projection (V3) recovered edges and the floor fell to 0.507. The process record belongs in A.7; 0.734 is not a live structural result.

Twenty-two-category invocation, with both endpoints passed through the mapper, is the representation that can be compared to T22 proximity without that vocabulary artefact. The either-unmapped rate is approximately 0.999 in both classes, with a class gap of 0.0012, so the edge construction discards a comparable share of both. This says nothing about coverage being good; it says the loss is unbiased—unlike V2, where the drop favoured malware. Edge and density floors are 0.506 (no self-loops) and 0.508 (with self-loops): real call structure, unbiased loss, still chance (Figure A.1).

A.6.5 Topology versus message passing

If adjacency carries supervised signal that one-class heads cannot localise (§A.6.3), the remaining architectural question is whether message passing extracts it. A supervised GIN is fit under the stratified split (M1: full graph; M2: edges removed; M3: node features replaced by a constant) and under Ward holdout (Table A.10). Split-A numbers are seed means over $n=5$ (seeds 42–46). Mean pooling: M1 0.884, M2 0.900, M3 0.590. Message passing does not help: M1 – M2 is negative at every condition mean. Add pooling is the one unanimous seed-level result ($\Delta = -0.028$, sample sd 0.013, 0/5 reversals). Mean pooling ($\Delta = -0.016$, sd 0.028, 2/5 reversals) and max pooling ($\Delta = -0.012$, sd 0.020, 2/5 reversals) sit inside paired seed noise. Ward holdout on mean pool

is a fold-weighted aggregate, not a matched five-seed pair: 0.790 versus 0.805 ($\Delta = -0.015$). A two-sided exact sign test on the four condition-mean signs gives $p=0.125$. The four conditions are not independent samples: three poolings of one model on the same graphs and split, plus one Ward aggregate. Pooled OOF M1 on Split-B is 0.762. Flip inflation is a ladder property and is measured in §A.6.6, not here.

Table A.10. Supervised GIN ablations on T22 (seed-mean floor AUC). Split-A: stratified supervised split, $n=5$ seeds. Split-B: Ward $k=30$ holdout, weighted mean versus pooled out-of-fold. Paired M1 – M2 seed statistics in Table A.11. Full grid: source sheet T5 (`chapter_a/tables/T5_message_passing`).

pool	M1	M2	M3	M1–M2	B wtd M1	B pooled M1
mean	0.8838	0.9000	0.5903	−0.0162	0.7904	0.7617
add	0.8690	0.8975	0.5789	−0.0285	0.7774	0.6772
max	0.8765	0.8886	0.6788	−0.0121	0.7718	0.7413

Table A.11. Split-A paired M1 – M2 over seeds 42–46 (`splitA_results.json`). Positive cells are seeds where message passing helps. Ward is fold-weighted, not this design.

pool	42	43	44	45	46	mean Δ	sd	M1>M2
mean	+0.002	+0.019	−0.053	−0.018	−0.031	−0.016	0.028	2/5
add	−0.032	−0.013	−0.048	−0.019	−0.029	−0.028	0.013	0/5
max	−0.016	+0.010	−0.013	+0.001	−0.042	−0.012	0.020	2/5

A Weisfeiler–Lehman kernel on T1K ($h=3$), fit as a one-class SVM on the benign-only split, is the family figure 0.673 (source sheets T3 / T5). Under the ablation configuration that produced that winner, as-built, structure-only (constant features), and edges-removed scores are 0.6730, 0.6268, and 0.5000 (Table A.12; Figure A.2). Those numbers show that a kernel can read T1K topology without this GIN stack. The ablation is benign-only and on T1K; it shares no split, supervision, or vocabulary with M1 versus M2 on T22, so it does not test whether the T22 GIN extracts edges.

Table A.12. WL $h=3$ ablation on T1K (one configuration). Family number carried elsewhere is the grid winner 0.6726; this row is the matched as-built / structure-only / edges-removed triple.

	as-built	structure only	edges removed
$\text{AUC}_{\text{floor}}$	0.6730	0.6268	0.5000

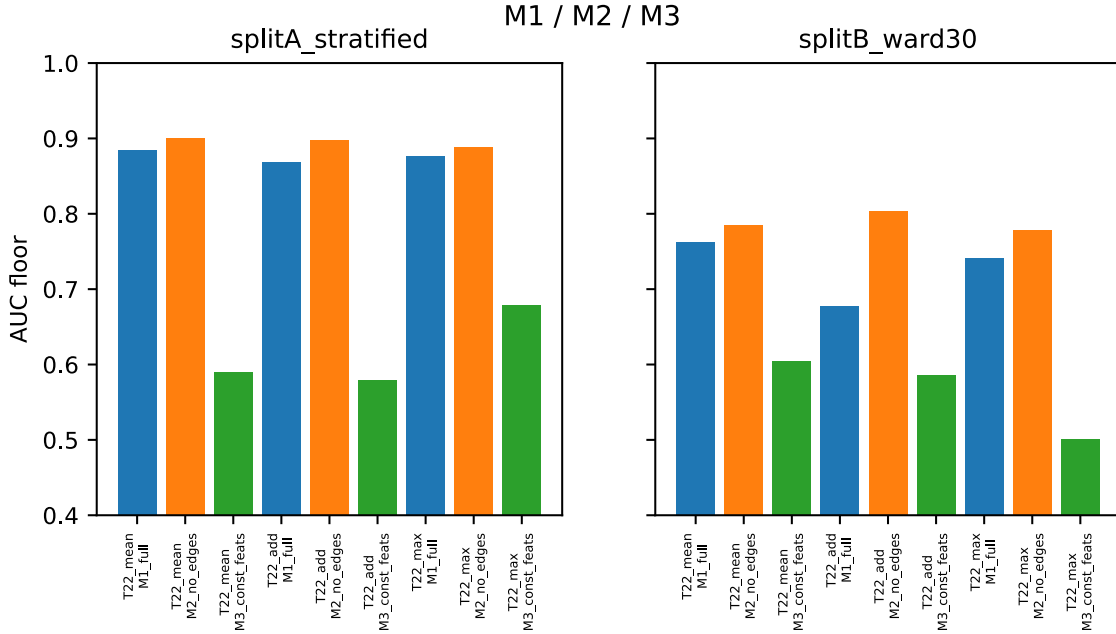


Figure A.2. Message-passing and kernel ablations. Source: `chapter_a/figures/F5_message_passing.svg`.

A.6.6 The supervision ladder

Three rungs price the cost of removing malware labels (Table A.13; Figure A.3). Rung 1 is the stratified supervised probe already reported (0.976). Rung 2 holds out Ward clusters of malware ($k=30$). The carried figure is pooled out-of-fold raw AUC 0.8489, not the superseded per-fold floor mean 0.8492 (withdrawn as a headline: three of thirty folds have raw AUC below 0.5, and averaging floors inflates the mean by 0.0118 relative to the mean raw 0.8374; Table A.13).

Table A.13. Supervision ladder (HGB full unless noted). Rung 2 and the random-group control are both carried as pooled out-of-fold raw AUC, not as per-fold floor means, and are computed over the same 5930 scored rows; the withheld floor means are shown beneath each. Full sheet: source sheet T4.

row	AUC	CI	note
rung 1 (stratified)	0.9762	[0.9642, 0.9859]	seed 42
rung 2 pooled OOF	0.8489	[0.8347, 0.8619]	carried
rung 2 mean raw	0.8374	—	3/30 raw <0.5
rung 2 mean floor (superseded)	0.8492	—	inflation 0.0118
random-group control	0.9713	[0.9659, 0.9766]	pooled OOF; 0/30 flips
random-group mean floor (withheld)	0.9864	—	12/30 folds = 1.0
rung 3 OCPool raw	0.7765	[0.7273, 0.8215]	score bootstrap
rung 3 OCPool R2	0.7544	[0.7019, 0.8013]	score bootstrap
Nested bootstrap on R2: [0.6994, 0.7995] (source sheet T8).			

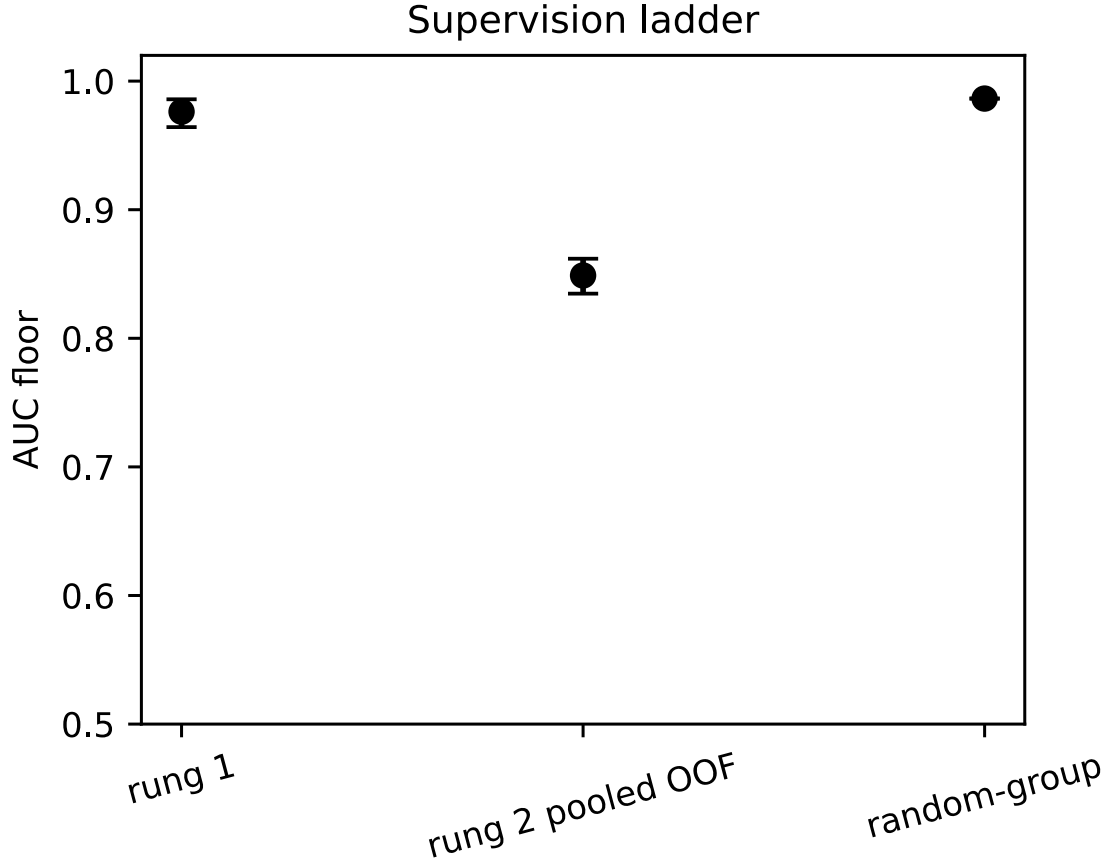


Figure A.3. Supervision ladder. Source: `chapter_a/figures/F1_ladder.svg`.

Logistic regression on the same behavioural holdout is not reported as a ladder result: nine of thirty folds flip, with inflation 0.195.

A random-group control that preserves fold sizes but destroys behavioural clustering scores 0.971 [0.966, 0.977] with 0/30 flips, so the rung-2 drop is group structure rather than fold-size artefact. The control is read as pooled out-of-fold raw AUC, the same statistic as the carried rung-2 figure, over the same 5930 scored rows (1700 malware and 141 test-benign replicated across thirty folds) and the same fold-size vector; 0.8489 and 0.9713 are therefore like-for-like. Its per-fold floor mean (0.9864) is withheld for the reason rung 2's own floor mean is: eleven of the thirty clusters hold a single malware app, and a fold with one positive can only score 0 or 1, so twelve of thirty control folds are exactly 1.0. That inflates any per-fold mean and leaves a pooled ranking unaffected.

Volume does not explain the clusters. The between/within variance ratio is 1.90 on the non-volume tensor against 0.026 on mapped-event count; fold AUC versus fold median volume is $\rho=0.058$ ($p=0.76$).

Rung 3 is the one-class incumbent already carried (0.776 raw / 0.754 residualised). The gap from rung 1 to pooled rung 2 is the price of unseen malware groups under supervision; the further drop to rung 3 is the price of dropping malware labels. Rungs 1 and 2 are T22 tensors while the incumbent is an API-level pooled representation (§A.6.3), so the rung-2-to-rung-3 gap is read against the matched T22 input centroid on the same split (0.777; Table A.8), which gives 0.072 against the 0.073 above. The residualised figure 0.754 is not used as a supervision cost: its OOV regressor exists only for an open API vocabulary and has no T22 analogue, so the gap to it absorbs

a representation change and a vocabulary correction alongside the loss of labels.

Split-B is meaningless as a generalisation test for methods fit on train-benign alone: that training set is identical across all thirty folds, so only the evaluation slice changes. The divergence is quantitative. On one benign-only head (D4 Mahalanobis) the protocol-weighted mean of per-fold floor AUCs is 0.808, while the single pooled out-of-fold ranking is 0.518. Weighted fold means are optimistic relative to one global ranking when fold malware counts range from 1 to 381.

A.6.7 Deviation vector versus scalar

This subsection has three parts: (i) the scalar reconstruction score fails while per-node deviation profiles still carry malware signal under supervision; (ii) the carried benign-only headline is D_1 centroid Euclidean scoring, with the controls that bound that claim; (iii) follow-up experiments (D-3, D-1, D-2) falsify alternative aggregations and a monotone “reference ladder” reading. Naming: D_0 – D_5 are deviation *scores* / *profiles* from the Run 5 encoder; D-1, D-2, and D-3 are *follow-up experiments* on those profiles (sparse aggregation, higher-criticism ladder, and a 2×2 factorial), not additional profile indices.

Scalar versus vector. The scalar reconstruction score is the failure already measured in §A.6.2: Run 5 D_0 floor AUC 0.638, inverted, below the mapped-event floor. The same encoder yields per-node deviation profiles D_1 – D_5 . Under a stratified supervised readout those profiles recover almost the raw-input ceiling (Table A.17). HistGradientBoosting on D_3 scores seed-mean floor AUC 0.962 (seeds 42–46); the seed-42 fit used for ROC and operating points is 0.961. The raw-input control on the same split is 0.975. A shuffled-label control on D_3 (logistic regression with ℓ_1) scores 0.504. The vector therefore contains the signal the scalar discarded; converting it at that level still requires malware labels.

Benign-only headline. The carried benign-only result is D_1 centroid Euclidean scoring: floor AUC 0.800 (Table A.17). Nearby one-class numbers are controls, not co-headlines: one-class scoring of $|D_0 - \mu_{\text{benign}}|$ (Euclidean distance from the train-benign mean of the scalar score) reaches 0.762, above the scalar but still a reconstruction-family result; the RAW input-tensor centroid on the same split scores 0.777.

Nested uncertainty on the D_1 headline uses outer $B=200$: percentile interval [0.757, 0.815]. The full-sample point lies inside that interval (bootstrap bias -0.003).

Volume and residualisation controls on the same D_1 scores: among the six Table A.4 volume scalars, Spearman $|\rho|$ reaches 0.2789 (`total_event_count`, $\rho=+0.2789$); static-feature norm reaches $\rho=+0.3301$ (legacy check 3, not a Table A.4 scalar—the chapter’s earlier “ $|\rho| \leq 0.33$ ” bound cited that covariate). Residualising on mapped-event count (OLS fit on train-benign) leaves $R^2=0.000029$ and floor AUC 0.805. The ranking holds in each test-set volume tercile (0.785 / 0.796 / 0.854). The strongest volume-family association is with *capability breadth* (`distinct_active_categories`, $\rho=+0.085$ among Table A.4 six; static norm $\rho=+0.330$), not raw trace volume—consistent with the architectural reading of `ipc_intents` and with the α -sweep finding that $\rho(\text{static norm})$ rises from 0.55 to 0.60 as feature weight increases.

Benign-group holdout (Ward clustering of train-benign, $k=5$) gives pooled out-of-fold AUC 0.789 [0.764, 0.814] with 0/5 folds inverted (source sheet T8). Mahalanobis on the same holdout inverts on 5/5 folds (pooled OOF 0.242): covariance estimated on a cluster-reduced benign set is unstable; the centroid is not.

The subsections below falsify alternative readings of that headline: a monotone reference ladder

(D-3), sparser aggregators (D-1), and higher-criticism / Fisher combination on per-coordinate p -values (D-2).

A.6.7.1 Referencing versus localisation (D-3)

This experiment falsifies a monotone ladder in which each step adds reference geometry. Table A.17 lists one-class rows; it does *not* support that ladder. A 2×2 factorial separates representation (RAW input tensor versus DEV deviation profile) from readout (scalar collapse versus per-node L_2 centroid), all scored benign-only on split A (562 train-benign / 141 test-benign / 1700 test-malware).

Table A.14. D-3 factorial: referencing $\times$ localisation (floor AUC). Cell A has no persisted artifact: $s_i = \|X_i^{\text{raw}}\|_2$ (704-d flattened input), score $|s_i - \mu_{\text{train-benign}}|$. Cells B–D: **results/D3_factorial_2x2.csv**.

cell	repr. $\times$ readout	AUC _{floor}	direction
A	RAW $\times$ SCALAR	0.7927	malware higher
B	RAW $\times$ PER-NODE	0.7769	malware higher
C	DEV $\times$ SCALAR	0.7617	malware higher
D	DEV $\times$ PER-NODE (D_1)	0.8004	malware higher

Margins (floor AUC differences): referencing at SCALAR readout $C - A = -0.031$ (referencing *hurts*); referencing at PER-NODE $D - B = +0.024$ (not distinguishable under paired DeLong/bootstrap); readout on RAW $B - A = -0.016$; readout on DEV $D - C = +0.039$ (the only distinguishable margin, DeLong $p = 0.019$); interaction $(D - B) - (C - A) = +0.055$ (larger than any main effect). The axes are strongly non-additive: referencing pays only under a per-node readout; the per-node readout pays only on the deviation representation. Neither operation helps alone. Shuffled labels on cell D score 0.5049 (noise).

Cell A is scored as a control beside the D_1 headline, not as a competing carried detector: a new baseline, not previously computed—a scalar norm of the raw flattened input, benign-centred, scores 0.7927—above the RAW input centroid (0.7769), above the mapped-event floor (0.7025), and above every one-class row in Table A.17 except D_1 . Paired against D_1 on the same test apps ($\Delta_{\text{floor}} = +0.0078$, inside the nested interval $[0.757, 0.815]$): DeLong $p = 0.551$; bootstrap CI on Δ_{floor} $[-0.017, +0.033]$ contains zero; Spearman $\rho = 0.779$. Headline AUC alone therefore does not separate the two readouts.

A full volume battery (**results/cell_a_volume_battery.md**) scores all seven Table A.4 covariates plus **static_feature_norm**. Cell A is *less* volume-coupled than D_1 on every one (max Table A.4 $|\rho| = 0.094$ versus 0.279; static-feature norm $+0.167$ versus $+0.330$; OLS residualisation $R^2 = 0.0016$ versus ≈ 0 ; residualised floor AUC 0.789 versus 0.805). Volume terciles are 0.743 / 0.819 / 0.842 (versus D_1 0.785 / 0.796 / 0.854). The readouts separate only under benign-group holdout (Ward $k = 5$): pooled OOF 0.712 $[0.683, 0.740]$ versus D_1 0.789 ($\Delta = -0.076$; 0/5 folds inverted in both). Nested bootstrap gives a tighter interval for the unfitted statistic— $[0.792, 0.807]$, bias $+0.002$ (versus D_1 $[0.757, 0.815]$, bias -0.003)—as expected when nothing is fitted beyond a scalar centre.

That holdout gap is the same axis the supervision ladder prices at 0.127 (stratified 0.976 to pooled rung 2 0.849; Table A.13): unseen behavioural groups. Cell A versus D_1 measures that cost in the one-class setting without malware labels.

Verdict: **CELL_A_FAILS_GROUP_GENERALISATION**. Cell A is a new trivial baseline above the 0.7025 floor, not a competitor; D_1 's contribution over cell A is robustness to unseen benign behavioural groups, not volume independence.

The D_1 result is not a univariate volume detector, nor a raw-share detector in disguise. The input-tensor centroid on the same split scores 0.777; zeroing the **ipc_intents** block of that 704-

dimensional vector drops the floor to 0.568 ($\Delta=0.209$), yet the raw IPC activity share (`act_v_frac` on the `ipc_intents` node, scored directly without train-benign centring) scores 0.575 and is *benign*-higher. On the D_1 profile the same node dominates: zeroing its deviation coordinate drops 0.800 to 0.591 ($\Delta=0.209$). Of the remaining 21 non-`ipc_intents` coordinates, four are structurally zero on AndroCT (`sms`, `clipboard`, `dynamic_code_loading`, `telephony`; §A.10.1): ablating an already-zero slot cannot change the score by construction, so their near-zero Δ is guaranteed rather than measured. The measured claim is over the 17 PRODUCIBLE non-`ipc_intents` coordinates; among those, the next node (`network`) contributes $\Delta=0.0025$ (Figure A.4). Univariate centroid distance on that D_1 coordinate scores 0.793 *malware*-higher.

Across 18 PRODUCIBLE coordinates, comparing train-benign-centred $|x_j - \mu_j|$ on the raw `act_v_frac` feature (D-3 RAW column) to the same scoring on the D_1 deviation coordinate: 13/18 show a positive referencing Δ ; 10/18 *reverse direction* (Table A.15). The direction reversal is a general property of the deviation operation on this corpus, not specific to `ipc_intents`. Reversing coordinates: `ipc_intents`, `network`, `content_access`, `crypto`, `audio`, `package_manager`, `notifications`, `storage`, `webview`, `file_io`. `ipc_intents` remains the largest instance ($\Delta_{\text{floor}}=+0.231$ with flip; uncentered raw share 0.575 versus deviation coordinate 0.793). An earlier reading of D_1 as “a univariate detector in disguise” is withdrawn; random-init L_∞ reaches 0.816 but is not distinguishable from D_1 (§A.6.7.2).

Table A.15. Per-coordinate referencing (D-3): RAW `act_v_frac` versus D_1 deviation coordinate, each scored as $|x_j - \mu_j|$ (train-benign μ_j). Four rows N/A: structurally zero on AndroCT (§A.10.1). Uncentered raw IPC share (0.575) is in the prose above; this table holds readout constant. Sorted by Δ_{floor} . Source:

results/D3_per_coordinate.csv.

category	RAW _{floor}	DEV _{floor}	Δ	flip?
<code>ipc_intents</code>	0.5625	0.7932	+0.2307	yes
<code>network</code>	0.5003	0.6327	+0.1324	yes
<code>camera</code>	0.5000	0.6207	+0.1207	no
<code>content_access</code>	0.5234	0.6436	+0.1202	yes
<code>crypto</code>	0.5017	0.6037	+0.1020	yes
<code>media</code>	0.5281	0.6272	+0.0991	no
<code>location</code>	0.5066	0.5666	+0.0600	no
<code>audio</code>	0.5038	0.5630	+0.0592	yes
<code>native_code</code>	0.5297	0.5664	+0.0367	no
<code>package_manager</code>	0.5606	0.5947	+0.0342	yes
<code>accounts</code>	0.5062	0.5393	+0.0330	no
<code>notifications</code>	0.5063	0.5182	+0.0120	yes
<code>storage</code>	0.5337	0.5389	+0.0052	yes
<code>process</code>	0.5496	0.5388	−0.0108	no
<code>device_info</code>	0.5413	0.5186	−0.0226	no
<code>database</code>	0.5654	0.5095	−0.0558	no
<code>webview</code>	0.6109	0.5464	−0.0645	yes
<code>file_io</code>	0.6014	0.5013	−0.1001	yes
<code>sms</code>	N/A	N/A	N/A	N/A
<code>telephony</code>	N/A	N/A	N/A	N/A
<code>clipboard</code>	N/A	N/A	N/A	N/A
<code>dynamic_code_loading</code>	N/A	N/A	N/A	N/A

A.6.7.2 Sparse aggregation (D-1)

This experiment falsifies the remedy that a sparser aggregator on D_1 profiles would close the readout gap left by the scalar score (§A.8.3). Twelve aggregators $\times$ two preprocessings (RAW, Z-standardised per coordinate) $\times$ two encoder arms (trained, random-init) were scored on D_1 profiles (no retrain). None beat trained D_1 ’s L_2 on trained weights (0.8004). Best trained non- L_2 : L_∞/MAX 0.7945 ($\Delta=-0.0059$); TOPK_2 0.7900; Z-standardised L_2 0.5089 ($\Delta=-0.2915$).

Per-coordinate SD ratio max/min on train-benign D_1 is 45419.7 (**ipc_intents** SD 49.86 versus **telephony** 0.001); **ipc_intents** is both the highest-variance coordinate and the one carrying the signal, so L_2 's implicit variance weighting is accidentally correct. Z-scoring flattens the informative coordinate to the level of the uninformative ones and the score collapses toward chance. $\rho(\text{coordinate SD, mapped share})=0.9122$: per-coordinate scaling is largely volume normalisation, not behavioural-volatility correction. PCA $n_{90}=1$ of 22: the profile is close to one-dimensional; there was no sparse-signal problem to solve.

§A.8.3's diagnosis of the readout gap is correct; its implied remedy (sparser aggregation) is falsified. D_1 's Euclidean norm is empirically optimal on trained profiles, demonstrated rather than assumed. MAX beat MEAN in Arm B (0.621 vs. 0.601) and in E0 (best cell SCALAR MAX): that pattern holds where signal is distributed and fails where it is concentrated on one coordinate—a scoped rule, not a general one.

Random-init L_∞ as negative control. Random-init RAW L_∞ reaches 0.8161 versus trained D_1 0.8004 ($\Delta_{\text{floor}}=+0.0157$). Not distinguishable: DeLong $p=0.340$; paired bootstrap CI on Δ_{floor} $[-0.047, +0.016]$ contains zero; Spearman $\rho=0.842$ between scores. It *is* the univariate detector: **ipc_intents** is the argmax coordinate for 1824/1841 apps (99.08%); univariate IPC deviation scores 0.793. It *loses* where it matters: benign-group holdout 0.779 versus D_1 0.789; high-volume tercile 0.774 versus D_1 0.854. The non-**ipc_intents** coordinates contribute +0.006 pooled but prevent an 0.08 degradation in the highest-volume tercile—what keeps **SIGNAL_IS_UNIVARIATE** (§A.6.7.3) from being over-read as “the other coordinates are noise.”

A.6.7.3 Sparse-signal ladder (D-2)

This experiment falsifies the claim that higher-criticism or Fisher combination on per-coordinate p -values recovers the D_1 centroid readout. Higher-criticism and Fisher-combination readouts on per-coordinate empirical p -values were measured on the same D_1 profiles (scoring pass only; no retrain). Two-sided p compares $|x_j - \mu_j|$ on each test app to the train-benign distribution per coordinate (562 apps). Verdict: **SIGNAL_IS_UNIVARIATE**—sparse p -value aggregation does not recover the D_1 centroid readout; signal remains on **ipc_intents**. The non-**ipc_intents** coordinates still matter for volume robustness (§A.6.7.2): they contribute +0.006 pooled but prevent an 0.08 degradation in the highest-volume tercile.

Uniformity on test-benign (KS versus $U(0,1)$) fails for two independent reasons (§A.10.1). Environmental gating produces degenerate nulls by construction (**sms** KS=0.936, **clipboard** KS=0.851, **telephony** KS=0.729). Sparse occupancy breaks uniformity even where the coordinate is PRODUCIBLE: **audio** KS=0.801, **content_access** KS=0.714. Mean KS by bucket: 0.83 (PERMISSION_GATED), 0.77 (STIMULUS_LIMITED), 0.45 (PRODUCIBLE); Spearman(occupancy, KS)=-0.91.

Table A.16. Sparse-signal ladder on D_1 (floor AUC, split-A digest pinned). Full 22 coordinates versus 18 PRODUCIBLE only (§A.10.1). $n=141$ test-benign / 1700 test-malware.

coordinate set	statistic	AUC _{floor}	note
22 (full)	FISHER	0.6498	best full-22 ladder
22 (full)	HC ($\alpha_0=1$)	0.5015	
22 (full)	BERK_JONES	0.5863	
22 (full)	MIN_P	0.5410	
18 (PRODUCIBLE)	FISHER	0.6730	restricted arm
18 (PRODUCIBLE)	HC ($\alpha_0=1$)	0.6471	
21 (ipc excluded)	FISHER	0.5929	vs. D_1 ipc-zeroed 0.591
reference	D_1 centroid L_2	0.8004	headline readout

Removing four coordinates that could not carry evidence lifts HC by 0.146 (0.6471 versus 0.5015; Table A.16) yet the restricted ladder still trails D_1 by 0.127 on a null it can satisfy. Excluding `ipc_intents` entirely: best ladder FISHER 0.5929 versus D_1 with ipc zeroed 0.591—two independent readouts agreeing to 0.002 that the residual coordinates carry ≈ 0.59 , below the 0.7025 floor. That is a *stronger* closure of the localisation question in §A.8.5 than the full-22 result alone: the ladder was given a fair per-coordinate null and still lost.

FISHER holdout (Ward $k=5$ benign-group, pooled OOF) scores 0.587 with 5/5 folds inverted—a third data point alongside D_4 Mahalanobis (0.242, 5/5 inverted) and input whitening (0.638→0.560): any benign-fitted *distributional* estimate is unstable under group holdout on this corpus, while a centroid is not (D_1 holdout 0.789, 0/5 inverted). Split-conformal calibration at nominal FPR 0.01 achieves 0.035 (non-uniform p -values break exchangeability of nonconformity scores)—a failed arm, reported with its mechanism.

D_1 centroid L_2 restricted to PRODUCIBLE coordinates scores 0.8004 versus 0.8004 on all 22 ($\Delta \approx 4 \times 10^{-6}$): a structurally-zero coordinate contributes nothing to a Euclidean norm—a mechanical reason the simplest readout was resilient to a problem that degraded the p -value ladder by 0.146. Self-reference readouts show the same insensitivity: E0 (0.6997→0.6998) and E2 (0.7124→0.7128) under the same mask (§A.6.8).

Support-novelty scoring is reported with its control, not as a headline. On the API-level document-frequency vocabulary, $S_{1,\text{norm}}$ reaches raw floor AUC 0.823, but the nested bootstrap mean is 0.787 [0.755, 0.818] (bias -0.036 ; the full-sample point lies *outside* its nested interval). Shuffling the support matrix still returns 0.659 ± 0.005 across twenty seeds. On the 22-node proximity graph the same score is 0.503: a universal node set makes “unseen” nearly empty, so support novelty is granularity-dependent and at chance where every category is always in vocabulary. The converse also holds: readouts that require a well-conditioned per-coordinate null fail when some nodes are structurally empty (permission-gated or hardware-absent), because zeros encode instrumentation artefacts as well as behaviour (§A.10.1). One fixed-universe design choice therefore admits two failure modes at opposite ends of the occupancy spectrum—support novelty dies when every app has every node; higher-criticism nulls die when some nodes never fire. The T22 adjacency one-class cap remains 0.545 (§A.6.3).

Table A.17. Deviation profiles versus scalar reconstruction and raw-input controls (floor AUC). D_3 HGB headline is the seed-mean; seed 42 is the ROC source. Full sheet: source sheet T6.

row	form	AUC _{floor}	note
D_0 GAE recon	scalar	0.6379	§A.6.2
D_0 scalar distance	benign-only	0.7617	$ D_0 - \mu $ from train-benign
D_1 centroid	benign-only	0.8004	headline; nested CI in T8
RAW centroid	input control	0.7769	T6
D_3 HGB (mean 42–46)	supervised	0.9624	headline readout
D_3 HGB (seed 42)	supervised	0.9614	ROC / source sheet T7
RAW HGB full	supervised	0.9746	split-A control
D_3 shuffled LR $_{\ell_1}$	control	0.5035	labels destroyed

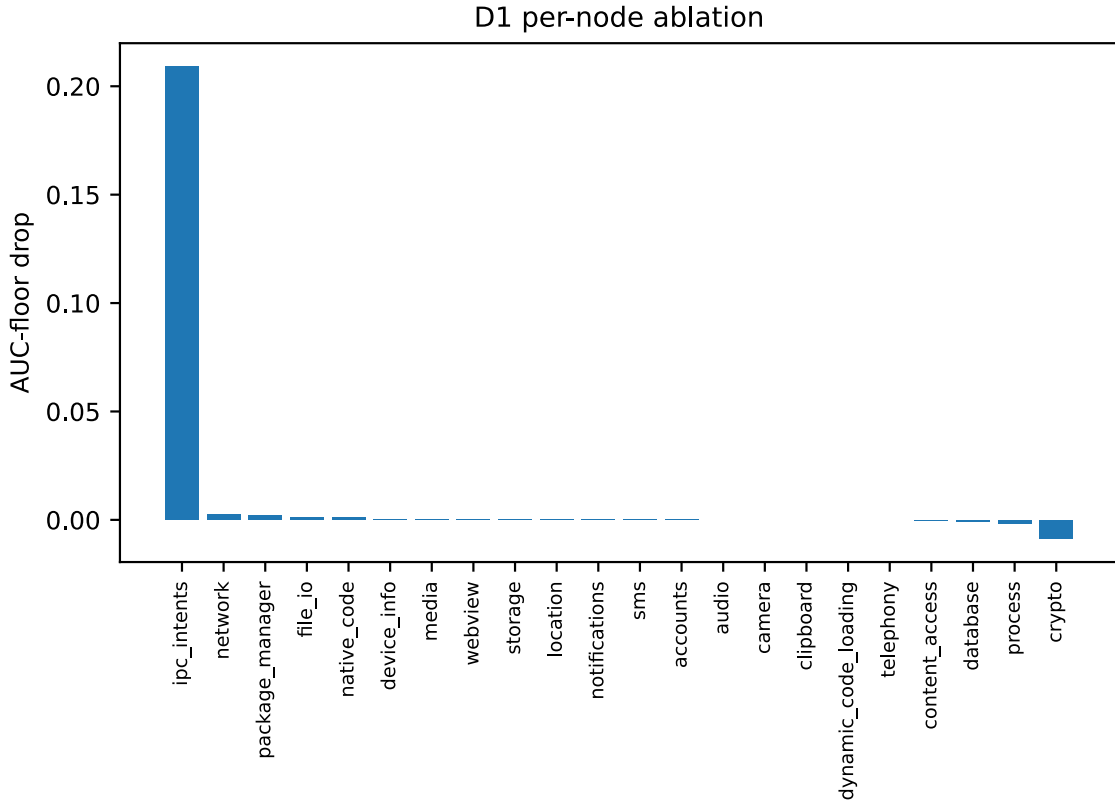


Figure A.4. D_1 per-node ablation and univariate centroid floors. Four coordinates (sms, clipboard, dynamic_code_loading, telephony) are structurally zero on AndroCT; their near-zero Δ is guaranteed, not measured (§A.10.1). Measured near-zero drops are over the 17 remaining PRODUCIBLE non-ipc_intents slots. Source: chapter_a/figures/F4_d1_node_ablation.svg.

A.6.8 Self-reference windowed detection

Chapter A’s per-app reference is built from other apps’ benign behaviour (§A.6.7). A complementary design holds out windows *within* the same session and scores deviation from a reference built from the app’s other windows. That path was specified before multi-session v2 data existed; this subsection reports the AndroCT measurement (E0, E2) and the window construction that precedes

it. All floor AUCs use $\text{AUC}_{\text{floor}} = \max(\text{AUC}, 1 - \text{AUC})$ with direction stated; the mapped-event trivial floor is 0.7025 (§A.5.4).

A.6.8.1 Window construction and fixed- W rejection

Windows are eight contiguous disjoint slots per app ($N=8$), each with equal mapped-event count within the app, built with `update_graph_sequence` ($k_{\text{burst}}=5$, cumulative weights, shares-not-counts). On the Arm B tensor cache, benign windows have median 21.0 mapped events and malware 102.0; zero-edge fractions are 0.2939 (benign) and 0.1964 (malware).

Fixed wall-clock width W was measured on a grid $\{10, 15, 20, 25, 30, 40\}$ s and rejected: no W clears a 0.15 zero-edge floor on either corpus. At $W=10$ the benign zero-edge fraction is 0.7954 on AndroCT and 0.4727 on v2 sessions-only. Recommended $W=10$ entered only via `COINCIDENT_FALLBACK`; the formal `OVERLAP` branch was empty.

The mono-category zero-edge proxy was validated, not assumed: verdict `PROXY_VALID`. At $W=20$ on a seed-42 sample, benign proxy mono-category rate 0.8340 equals builder zero-edge 0.8340 (off-diagonal of the 2×2 table zero). Sample $N=8$ benign zero-edge 0.2875 reconciles with Arm B 0.2939. The decisive bucket is benign windows with 6–10 mapped events: zero-edge 0.3729 under $N=8$ versus 0.8567 under fixed- W ($n=9711$). Same slot count, different rates: $N=8$ is app-balanced; fixed- W is mass-weighted, so high-volume apps with long mono-category runs dominate the fixed- W pool. An app-balanced fixed- W variant was measured in the W -selection branch but not evaluated for detection.

A.6.8.2 E0: isotropic self-reference readouts

Design. For each app, reference R_i is the elementwise mean of six reference windows; the test set is the remaining two. Per-node deviation is $d=|X-R_i|$ in node-feature and adjacency space separately. `PREFIX` (earliest six) and `SCATTERED` (per-app random six, seeded) split modes; `SCALAR` (L_2 norm) and `CENTROID` (distance from μ_{benign} fit on train-benign) scorers; `MEAN`, `MAX`, and `FRACTION` verdict aggregators over the two test windows (24 cells per matrix).

Results. Best raw floor AUC is 0.6997 (`PREFIX`, `SCALAR`, `MAX`, node space, direction benign-higher) — below the 0.7025 floor (Table A.18). Best size-matched floor AUC is 0.6235 on overlap window [173.0, 3567.0] mapped events ($n=53$ benign, $n=1313$ malware after filtering). With only two test windows, `FRACTION` scores take values in $\{0, 0.5, 1\}$ only.

A stratified supervised readout on d (HistGradientBoosting, seed 42) reaches a *diagnostic ceiling* of 0.9154–0.9340 floor AUC (§A.5): labels are used and this is not a benign-only detector. A hygiene control scores the same test windows against a *cross-app* reference R_j ($j \neq i$, class-blind): `PREFIX` node HGB d_{self} 0.9340 versus d_{cross} 0.8470 ($\Delta_{\text{floor}}=+0.087$, DeLong $p \approx 3 \times 10^{-6}$; `SCATTERED` node 0.9274 vs. 0.8488). The window tensor alone carries substantial supervised signal ($d_{\text{cross}}=0.847$); the margin is the claim that self-reference adds beyond cross-reference, not either value alone. One-class arms sharpen the contrast: d_{cross} collapses to ≈ 0.50 – 0.53 while d_{self} holds ≈ 0.70 (`PREFIX` node `SCALAR MAX`). Chapter C reports within-app versus cross-app session medians of 1.28 versus 38.3 under Stage 2 Frobenius combined distance, not HGB floor AUC (§C.3), and L_2 self/cross ratios of $72 \times$ – $411 \times$ at session granularity under a different builder (§C.4.3)—neither is directly comparable to the $+0.087$ window-level HGB margin here. Shuffled-label control mean floor AUC is 0.5172. Density trend versus mapped-event count is flat; volume-coupling ρ columns and the size-matched matrix are recorded in the artefact bundle. Mean $|\text{PREFIX} - \text{SCATTERED}| \Delta=0.005545$:

within-session windows are exchangeable at this granularity. CENTROID loses to SCALAR here, the reverse of the D_0/D_1 ordering in §A.6.7; that reversal is reported, not explained.

Table A.18. E0 self-reference headline cells (floor AUC, split-A digest pinned). Size-matched overlap [173.0, 3567.0]; $n=53$ benign / 1313 malware. Full 24-row matrix: `selfref/matrix_24.csv`.

tag	cell	AUC _{floor}	clears 0.7025?	direction
raw	PREFIX SCALAR MAX node	0.6997	no	benign higher
size-matched	PREFIX SCALAR MAX node	0.6229	no	benign higher
size-matched	SCATTERED CENTROID MAX adj	0.6235	no	benign higher
Diagnostic ceiling on d (supervised HGB): 0.9154–0.9340				

A.6.8.3 E2: variance-aware readouts

E2 re-scores the E0 deviation vectors with Mahalanobis (Ledoit–Wolf on train-benign), per-node standardisation (NODE_STD), and a conformal wrapper on NODE_STDPREFIXnodeMAX.

Anisotropy diagnostic. Phase 1 verdict is LOW_VARIANCE_SEPARATION on projection ratios, but the strict MaRS condition is false: 90% of variance requires $n_{90}=9$ of 22 components, not under one third. Both facts are reported. ρ (per-node variance, node mapped-event share)=0.948: per-node variance is almost entirely density. NODE_STD is therefore substantially a volume normalisation, not a behavioural-volatility correction; this caveat applies wherever E2 gains are cited.

Results. Best raw floor AUC is 0.7124 (NODE_STD, MAX, node, PREFIX), clearing the floor on four raw cells. Best size-matched floor AUC is 0.6750 ($n=53$ benign, $n=1313$ malware): the nested interval spans the floor and the cell is underpowered, not decisive. Mahalanobis best raw is ≈ 0.709 (PREFIXMAX node) and fails size-matching (0.6706). Conformal `min_p` raw floor AUC is 0.5385 — a failed arm, reported not omitted.

Per-node ablation at the E2 headline: `ipc_intents` drop -0.0012 (does not dominate); top gain is `file_io` at $+0.014$. Contrast Chapter A D_1 ablation where `ipc_intents` drop is $+0.209$ (§A.6.7). Shuffled-label mean 0.5171; mean $|\text{PREFIX} - \text{SCATTERED}| \Delta=0.006216$.

Variance-aware scoring recovers measurable signal that isotropic readouts suppress ($+0.0127$ raw and $+0.0515$ size-matched on the best NODE_STD/SCALAR pair; Table A.19), consistent with the anisotropy diagnostic, but the margin does not survive size-matching: the E2 stop rule treats the one-class null as standing after volume control.

Table A.19. E0 versus E2 on matched cells (floor AUC delta E2–E0, raw tag). Size-matched: $n=53$ benign, $n=1313$ malware; CI spans 0.7025.

verdict	space	raw Δ	size-matched Δ
MAX	node	$+0.0127$	$+0.0515$
MAX	adj	$+0.0159$	$+0.0139$

A.6.8.4 Interpretation

Two mechanisms were measured. First, within-session windows are exchangeable ($|\Delta|=0.005545$), so there is no temporal structure for a hold-out window to violate. Second, 2017 AndroCT malware largely repackages legitimate applications with payload behaviour live from launch, so a self-reference built from the same session has already absorbed the malicious pattern. AndroCT

(18/22 PRODUCIBLE) and v2_extended (15/22) do not share the same effective coordinate system despite the common 22-node label set (§A.10.1); cross-corpus comparisons of per-node d vectors are over differing effective dimensionality. Prediction-based (E1) and sequence-based (E3) variants were not pursued: the exchangeability result removes their premise.

A.6.9 Deployability

AUC ranks apps; a detector must also pick a threshold. Table A.20 reports true-positive rate at a 1% false-positive target, with precision recomputed at a wild malware prevalence $\pi=0.01$ (Bayes). The evaluation set used for benign-only heads is 141 benign / 1700 malware ($\approx 92\%$ malware). That balance inverts the wild base rate, so test-set precision is not a deployment number.

At FPR 0.01, D_1 centroid recall is 0.24% (TPR=0.0024) and wild precision is 0.33%. OCPool mean-pool reaches TPR 0.018 / precision 0.025; $S_{1,\text{norm}}$ 0.074 / 0.095; the mapped-event floor 0.001 / 0.002. None of the benign-only heads is usable at this operating point. Self-reference readouts (E0/E2; §A.6.8) likewise carry no reportable operating point: the best raw floor AUC (0.7124) does not survive size-matching and no threshold sweep was carried to ROC form. The supervised D_3 readout (seed-42 ROC; stratified 141/340) reaches TPR 0.794 and wild precision 0.531. HistGradientBoosting on the raw input achieves observed FPR 0 at the 1% target on 141 benign negatives (TPR 0.832; wild precision reported as 1.000, but with zero observed false positives the upper 95% bound on true FPR is $\approx 2\%$)—a label-using ceiling, not a deployment recipe. Figure A.5 marks the 1% FPR line.

Table A.20. Operating points at FPR target 0.01 (wild precision at $\pi=0.01$). Benign-only rows: 141/1700. Supervised rows: stratified 141/340. Full sheet (four FPR targets): source sheet T7.

config	FPR ach.	TPR	prec. wild	split
D_1 centroid	0.0071	0.0024	0.0033	GAE
OCPool mean	0.0071	0.0176	0.0245	GAE
$S_{1,\text{norm}}$ T1K	0.0071	0.0735	0.0948	GAE
D_3 HGB readout	0.0071	0.7941	0.5307	stratified
HGB full	0.0000	0.8324	1.0000 [‡]	stratified
mapped-event floor	0.0071	0.0012	0.0017	GAE

[‡]Observed FPR 0 on $n=141$ benign; upper 95% FPR bound $\approx 2\%$.

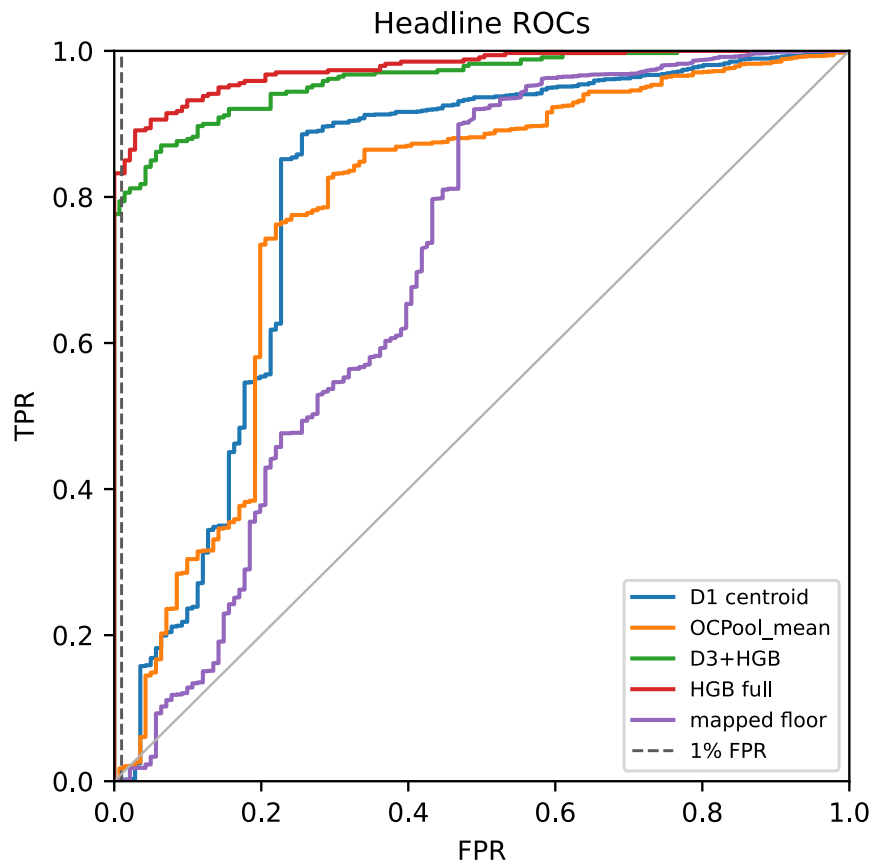


Figure A.5. Headline ROC curves with the 1% FPR line. Source: `chapter_a/figures/F3_roc_headlines.svg`.

A.7 Artefacts caught before they became results

Chapter A reports only numbers that survived corpus validation on the frozen $n=2403$ population (§A.2, §A.5). Table A.21 records seven cases where an artefact would have entered the thesis as a result had it not been caught. Two readings *did* enter briefly as headline numbers—residualised pooling at 0.814 (row 3) and the V2 invocation edge floor at 0.734 (row 5)—before withdrawal. The table is contribution 7 of the chapter framing; it is not a footnote to individual results.

Five of the seven surfaced while auditing a *denominator* or *population*—drop rates, exclusion rates, class coverage, or app counts—rather than by inspecting an AUC. Rows 3 and 4 required protocol checks on residualisation fit and bootstrap design. That pattern is the transferable lesson: on imbalanced corpora, silent subsetting and label-aligned loss often move a floor or score before the detector does anything interesting.

Table A.21. Seven artefacts identified and corrected before Chapter A results were frozen. “Had headline” marks readings briefly carried as primary numbers. Floor AUC unless noted; direction named when it flips.

#	failure	symptom	spurious read	correction
1	<code>_CALL_RE</code> / <code><init></code>	Asymmetric line loss (12.2% benign vs. 27.9% malware); 20/17 categories active; found by grepping raw traces for well-formed calls the regex rejected (§A.2.2)	—	Regex accepts <code><init></code> / <code><clinit></code> ; post-fix loss ≈ 0 ; categories 21/18; <code>dynamic_code_loading</code> recovered (DexClassLoader <code><init></code> , 27 benign / 57 malware)
2	Eligibility ≥ 320	Excluded 58.5% benign vs. 27.0% malware; trivial floors inverted on the eligible subset—mapped events $0.703 \rightarrow 0.506$ (malware-higher both); total events $0.685 \rightarrow 0.605$ (malware-higher $\rightarrow$ benign-higher); active categories $0.501 \rightarrow 0.680$ (malware-higher $\rightarrow$ benign-higher); <code>arm_b_n8</code> mean floor ≈ 0.517 , below those inverted floors	Early <code>arm_a</code> / <code>arm_b</code> runs	Gate removed; live eligibility mapped ≥ 1 only ($n=2403$); those arm scores not cited as results
3	OOV residual (eval fit)	OLS residualisation fit on the <i>eval</i> split (R1); <code>coef_oov</code> sign reversed (-2.236 vs. $+0.449$ train-benign)	Residual floor 0.8141 (<i>had headline</i>)	Train-benign-only fit (R2) floor 0.7544; raw incumbent 0.7765 unchanged
4	Vocabulary selection	<code>B_docfreq</code> chosen by <code> oov_floor - 0.5 </code> ; naive score-resample CI width 0.080 [0.771, 0.851]	Naive CI [0.771, 0.851] (width 0.080)	Nested bootstrap with per-draw vocab refit: width 0.100 [0.699, 0.800]; naive is 20% narrower than nested (nested 25% wider); selection is part of the model (§A.10)
5	V2 invocation graph	Callee-vocabulary edges; median drop 0.9487 benign vs. 0.9999 malware (68.7% app-package callers)	Edge floor 0.7338 (<i>had headline</i>)	Withdrawn; 22-category mapped invocation 0.5056 (unbiased loss, chance)
6	Run 2 NaN filter	47/1841 test apps silently excluded (empty <code>edge_index</code>); class-aligned asymmetry (7/141 benign = 5.0% vs. 40/1700 malware = 2.4%)	NaN-filtered AUC 0.5222 (benign-higher)	Finite score for every app; pre-repair subset figure withdrawn
7	Stale <code>corpus_cache</code> $\rightarrow$ T1	Population row in source sheet T1 taken from a stale graph-eligible cache (wrong n and medians) instead of post-fix inventory	Wrong corpus table	T1 population from <code>inventory_summary.json</code> ; eligible 703/1700/2403 from <code>POSTMORTEM.json</code> reconciliation

Related items that are *not* table rows: $S_{1,\text{norm}}$ ’s shuffled-support control (0.6594 ± 0.0052) is a reported diagnostic in §A.6.7, not a withdrawn headline. Ladder per-fold floor mean (0.8492 superseded by pooled OOF 0.8489) and Split-B weighted Mahalanobis (0.8083 vs. deduplicated pooled OOF 0.5178) are protocol choices already resolved in §A.6.6 and §A.6.7.

Two further process records are abandonments, not corrected artefacts: an OCGTL performance-flip Split-B run was attempted and stopped with no checkpoints retained, and a GLocalKD nested bootstrap on T1K was stopped at outer $B=20$ for cost ($B=100$ infeasible). No reported chapter number depends on either stopped run; both are disclosed here so they are not silent omissions (chapter_a/GAPS.md; chapter_a/MANIFEST.csv).

A.7.1 Verification scope and reproducibility

Reproducibility gaps. Every experiment directory that carries a SUMMARY.md now also carries a reproduce.md (16/16 previously missing files written from on-disk provenance; several mark CLI argv or library versions UNDETERMINED where evidence is absent; results/reproducibility_remediation.md). Of twenty-one reproduce_config.json files that lacked library version pins, none have evidence-backed run-time pins on disk: 0 were filled from a contemporaneous freeze/lockfile, and all 21 are marked version_provenance unverified against the post-hoc lockfile reproducibility/environment-2026-08-26.txt (captured after the runs; site-packages mtimes post-date August 2026 experiment artefacts). Chapter tables and figures still regenerate from pinned scripts and fail rather than invent numbers. These facts bound audit depth and environment claimability, not the arithmetic consistency of the reported catalogue.

Catalogue verification covers 104/104 rows: six notebooks execute, and verify_all.py exits 0. Agreement is to six decimal places between the chapter catalogue and stored JSON scalars. That check is catalogue-internal—it does not claim coverage of every number in §A.6 (§A.7). Per-app score vectors sufficient to recompute ROC AUC without re-inference are persisted for 61/104 catalogue rows today (chiefly ocdev/devread/ocgtl/supgmn predictions.csv tables); the remaining 43/104 detector artefacts store quantile summaries or aggregate AUCs only. Independent verification therefore does not re-infer trained torch models; GPU nondeterminism would matter only if those models were re-trained.

The early androct_2017/SUMMARY.md GAE arm table predates eligibility removal and Run 2 repair. It remains in the output tree as process history only. Headline claims in §A.6 trace to MASTER_RESULTS.csv rows marked headline or to explicitly labelled controls. The reconstruction intervention sequence (Table A.6) and the remaining diagnostic scalars are not catalogued there; they trace to the named run artefacts under abrg/output/androct_2017/, cited per number in the source. The 104/104 agreement reported above is therefore catalogue-internal—it checks each catalogued scalar against its stored JSON—and is not a coverage claim over every number in this chapter. Superseded scalars appear only alongside the figure that supersedes them.

These incidents changed the protocol, not only the numbers: trivial floors are recomputed whenever the eligible population changes; nested bootstrap (with vocabulary refit where selection is fitted) replaces naive score-resample intervals on those configurations; and every alternative representation carries explicit drop-rate and exclusion accounting before any floor or AUC is quoted.

A.8 Discussion

The results section reports measurements under fixed pins (§A.5). Supervision exposes strong signal; benign-only readouts recover only part of it. Message passing, scalar reconstruction, and

trained encoders do not deliver here; mechanisms are named only where measured. The benign-only headline is not deployable, and that cost is measured.

A.8.1 Positioning against published work

The D_1 centroid readout scores 0.800 floor AUC on the benign-only split. Published one-class graph detectors make that figure look strong. The operative references are the 0.7025 mapped-event floor and the operating point, where the same readout recalls 0.24% of malware at 1% FPR (§A.8.6).

The controlled comparison is on-corpus. On identical tensors, split, and protocol, D_1 at 0.800 exceeds the trained deep one-class family: OCGIN_plus 0.566, GLocalKD 0.559, OCGTL 0.550 (Table A.7; GLocalKD there is scored by s_{graph} on the brief training profile). Those trained arms are the weaker configuration of each family. GLocalKD is positioned here on Ma et al.’s [22] published test score (Eq. 13: graph error plus mean node error) under the paper-configuration training profile (`glocalkd_papercfg`; five seeds). Under Eq. 13 the trained arm is 0.513 and the untrained arm is 0.680, a gap of 0.167 and the largest trained-versus-untrained margin measured in the chapter.² Against the best measured configuration of those families the margin is 0.800 versus 0.680 (GLocalKD untrained, Eq. 13, paper-config). The supervised ceiling of 0.976 on the full 704-vector records how much signal the tensors carry once malware labels enter training.

Table A.22. Positioning on AndroCT 2017 (floor AUC, split-A GAE unless noted). Untrained arms are listed because they outperform the trained arms of the same families on this corpus. GLocalKD Eq. 13 rows: paper-configuration profile (`glocalkd_papercfg`, five seeds); brief-profile s_{graph} in Table A.7. Published OCGIN ranges: Qiu et al. [30], Tables 1 and 3, molecular/protein datasets only. [‡]OCGIN[†] range excludes AIDS (outlier: OCGIN[†] 0.975, OCPool 0.976). Full range including AIDS is 0.572–0.975.

reference point	AUC _{floor}
Published OCGIN, base config (mean pool + BatchNorm), molecular/protein range	$\approx 0.507\text{--}0.624$
This work: OCGIN_orig (mean pool + BatchNorm, 5 seeds) 0.591 (hypersphere collapse, 5/5 seeds)	
Published OCGIN [†] (add pool + GraphNorm), molecular/protein range excl. AIDS [‡]	$\approx 0.572\text{--}0.635$
This work: OCGIN_plus (add pool + GraphNorm, 5 seeds)	0.5660
This work: random-init OCGIN_plus (add pool + GraphNorm, 5 seeds)	0.654
This work: GLocalKD trained, Eq. 13, paper-config (5 seeds)	0.5133
This work: GLocalKD untrained, Eq. 13, paper-config (5 seeds)	0.6803
This work: best benign-only (D_1 centroid)	0.8004
This work: supervised ceiling (HGB full, stratified)	0.9762

Deep one-class methods cluster tightly relative to the gap between that family and other approaches. On AndroCT the trained spread is 0.016 at floor AUC (Table A.7). Wang et al. [38] report per-method means of 67.72–74.13 AUROC across six GNN-based GLAD methods on fifteen

²Table A.7 retains s_{graph} on the *brief* profile (lr 0.01, hidden 128): trained 0.5594, untrained 0.6650. The Eq. 13 pair here is from `glocalkd_papercfg` (Ma et al.’s [22] reference hyperparameters; five seeds). On that arm, s_{graph} is trained 0.5955, untrained 0.6633. Eq. 13 under the brief profile was not computed.

GLAD datasets, a 6.4-point spread on a 0–100 scale that is several times the AndroCT width.³ In both settings the family clusters under a common protocol, and Wang et al. describe their Table 3 means as performance upper bounds from test-set hyperparameter search (§3.3). These floor AUCs use no test-set access at any stage. A level gap against published figures near 0.70 is therefore a protocol asymmetry, not evidence that AndroCT would exceed published benchmarks under matched tuning.

The published OCGIN bands in Table A.22 answer a different question. Qiu et al. [30] report one-class graph anomaly detection on molecular and protein graphs for base OCGIN (mean pooling, batch normalisation) and OCGIN[†] (add pooling, graph normalisation). Those benchmarks differ from Android behavioural traces in corpus, class balance, graph size, and protocol. The bands set scale. Trained OCGIN_{plus} at 0.566 is consistent with the published OCGIN[†] range (≈ 0.572 – 0.635 excluding AIDS). That consistency is a sanity check on the reimplementations whose deviations §A.10.4 records as a threat. Qiu et al.’s [30] architectural change raises OCGIN on eight of nine published datasets. On AndroCT the matched configuration scores 0.566 against 0.591 for OCGIN_{orig}, but the base arm hypersphere-collapses on all five seeds (§A.6.3), so 0.591 cannot be read as a detector score. The stronger published implementation does not recover the objective family on this representation. Random-init OCGIN_{plus} scores 0.654, above both published OCGIN bands. An untrained encoder sitting above those trained ranges is the more striking table entry, and it previews the chapter’s central finding on a corpus those bands do not cover.

A.8.2 Why message passing does not help here

Topology is not the bottleneck under full supervision: flattened adjacency alone reaches floor AUC 0.959 (Table A.5). That figure is HistGradientBoosting on the 484-dimensional vectorised adjacency. Supervised GIN (M1) scores 0.884 on the same split. It is a different model on a different feature set: node attributes plus a four-layer graph. So 0.959 versus 0.884 is not a message-passing penalty. The controlled contrast is M1 versus M2: identical GIN, identical node features, edges present or removed (§A.6.5).

On Split-A mean pooling, seed-mean M1 is 0.884 and M2 is 0.900 ($n=5$, seeds 42–46). Per-seed deltas, pooling-wise means, sample sds, and reversal counts are in Table A.11. Ward holdout on mean pool is fold-weighted, not a matched five-seed pair: 0.790 versus 0.805 ($\Delta = -0.015$). All four condition-level means have $M2 \geq M1$. A two-sided exact sign test on those four signs gives $p=0.125$. The four conditions share the same graphs and split, except Ward, so they are not independent samples. No condition mean favours message passing, and only add pooling supports the stronger reading that it hurts.

Three explanations are consistent with a non-positive mean $M1 - M2$, and they reinforce one another on a 22-node fixed graph. Four GIN layers with $k=5$ proximity on at most 22 nodes can saturate the receptive field, so removing edges reduces over-mixing rather than stripping useful signal. That saturation bites harder because every app shares the same 22 category nodes under the 1-WL bound with identical initial colouring, leaving message passing a nearly indistinguishable palette unless dynamic shares break symmetry. Even where mixing might help, mean, add, and max readouts collapse the graph to one vector and can dilute sparse transitions that a flat supervised head still uses (§A.8.3). None of these three was isolated here.

OCGTL supplies a measured mechanism rather than a hypothesis. On T22 it collapses by encoder agreement at $K=4$ and $K=6$ (5/5 seeds). Matched T1K runs do not collapse (0/5).

³Type I intrinsic anomaly: p53, HSE, MMP, PPAR-gamma. Type II class-based anomaly: COLLAB, IMDB-BINARY, REDDIT-BINARY, ENZYMES, PROTEINS, DD, BZR, AIDS, COX2, NCI1, DHFR. Table 3’s “Avg. AUROC” row averages all 35 datasets, including GLOD scenarios, and is not the figure used here.

Centre θ is trainable, as in the reference, so this is outside Prop 2’s fixed-centre regime. $\mathcal{L}_{\text{GTL}}$ holds on both representations at 59–75% of the ceiling and does not separate the arms. Encoder agreement does: T22 cosine ≈ 0.9999 with $\mathcal{L}_{\text{OCC}} \approx 0.3\text{--}0.5$, against T1K cosine $\approx 0.96\text{--}0.98$ with $\mathcal{L}_{\text{OCC}} \approx 5.6\text{--}8.8$. Views become identical and sit on θ , a failure Prop 2 does not cover.

The same four-layer GIN family is the wrong extraction bet on T22. `OCGIN_orig` hypersphere-collapses on all five seeds (§A.6.3). Supervised GIN, OCGIN, and OCGTL all degrade on 22 nodes and behave better on 1000 API nodes, which is consistent with receptive-field saturation. T1K also opens the vocabulary and changes occupancy. §A.8.4 uses that contrast for support novelty: $S_{1,\text{norm}}$ is 0.787 nested on T1K and 0.503 on T22. This chapter does not separate node count from vocabulary. No Dirichlet-energy diagnostic was run. The same family is the one whose trained and untrained one-class encoders are indistinguishable (§A.6.3).

A.8.3 Why unsupervised scoring does not reach the structural signal

Labels identify which coordinates matter. On the same split, supervised HGB on the 484-dimensional adjacency alone reaches floor AUC 0.959. The best unsupervised adjacency readout in the granularity sweep stalls at 0.545 (kNN on flattened adjacency, $k=5$). PCA on that adjacency lifts the best one-class score to 0.683. That is partial recovery. It does not close the gap.

Density-style benign-only scores aggregate over hundreds of dimensions. Most coordinates are near zero for most apps; a handful of behavioural categories carry the class shift. A scalar reconstruction loss or a Euclidean centroid on the full vector averages signal with noise.

That diagnosis implies a remedy: reduce the weight of the empty coordinates. Three interventions test it, and all three fail. Input whitening centres and whitens graph tensors against train-benign statistics before the encoder, the direct intervention on low effective rank. Run 5 floor AUC drops from 0.638 to 0.560 and remains inverted (Table A.6). Replacing L_2 with a sparser aggregator on trained D_1 profiles is falsified (D-1; §A.6.7.2). Higher-criticism readouts fail even on the restricted 18-coordinate null, confirming univariate signal rather than rescuing localisation (D-2; §A.6.7.3). The mechanism claim survives. What fails is the inference from mechanism to remedy.

D_1 centroid scoring (0.800) is the $\text{DEV} \times \text{PER-NODE}$ cell of the D-3 factorial (Table A.14). It is the only distinguishable margin over scalar deviation (D_0 , 0.762; DeLong $p=0.019$, bootstrap CI $[+0.007, +0.072]$; §A.6.7.1). Referencing at scalar readout hurts ($C-A=-0.031$). Cell A is a benign-centred L_2 norm of the raw flattened input. It scores 0.793 against D_1 ’s 0.800 on the full split, statistically indistinguishable. Under benign-group holdout the same cell drops to 0.712 against D_1 ’s 0.789 (§A.6.7.1). The two are equal on the easy evaluation. They separate on the hard one.

Node-level ablation shows most of the D_1 gain on `ipc_intents` ($\Delta=0.209$ AUC when that coordinate is zeroed), a concentration whose reading is given in §A.8.4. That readout still fails deployment thresholds (§A.8.6) and still clears the size floor only narrowly.

A.8.4 What the tested configuration establishes

The δ temporal filter, the recency edge channel, the recency node slot, and per-app cross-session refinement were structurally unexercisable on AndroCT (Table A.1; §A.10.2). The measurements below concern a cross-app, single-session, timestamp-free reference.

What held up. The fixed 22-node universe (§A.4.1) makes cross-app comparisons and deviation profiles definable. It is also what puts support novelty at chance on T22 ($S_{1,\text{norm}}$ 0.503; §A.6.7).

Shares-not-counts normalisation (§A.4.3) enforces length invariance on dynamic activity under that universe.

The D-3 factorial (§A.6.7.1) is the primary readout result. Deviation-plus-per-node scoring (cell D, 0.800) is the only distinguishable benign-only margin over scalar deviation. Referencing without per-node structure does not carry the climb.

The deviation profile concentrates the D_1 gain on `ipc_intents` ($\Delta=0.209$); the next node contributes $\Delta=0.0025$ (§A.6.7). That is the mechanical localisation the commitment asked for. §A.8.5 shows the named coordinate is architectural. Localisation of this kind is discharged in form but not in the sense the design intended: the profile identifies which coordinate carries the class shift, and that coordinate is not a behavioural deviation.

What did not. A trained encoder was the design’s extraction bet. Untrained or random-initialised arms match or beat trained arms on OCGIN (0.654 versus 0.566), on GLocalKD under Eq. 13 (0.680 versus 0.513), and on the GAE embedding readout (§A.6.3). The bet does not pay on this corpus.

The scalar GAE reconstruction objective and its single-number readout invert under benign-only training (§A.6.2). Message passing was treated as the extraction mechanism; supervised ablations show it does not help on these graphs (§A.8.2).

Support-based novelty is granularity-dependent. On the open API vocabulary (T1K), $S_{1,\text{norm}}$ reaches floor AUC 0.787 nested (0.823 raw). On the fixed 22-category universe, where every app shares the same node set, the same score is at chance (0.503). Per-coordinate null readouts fail when some nodes are structurally empty (§A.10.1). Corpus validation (§A.7) caught several ways those mistakes would have been reported as results.

A.8.5 Mechanisms and synthesis

AndroCT and v2_extended share the same nominal 22-node label set but not the same effective coordinate system in practice: the observability audit classifies 18/22 categories as PRODUCIBLE on AndroCT and 15/22 on v2 (342 reference-tier sessions). Per-node deviation profiles from the two corpora are therefore vectors of differing effective dimensionality even when tensor shapes match (§A.10.1).

The reference-needs-variance principle. A benign reference is informative when it encodes variation the test observation does not share. Three measurements converge on that reading. Population benign (562 apps, cross-app) drives D_1 centroid floor AUC 0.800 (§A.6.7). Per-app cross-session on v2: within-cross medians 1.28 versus 38.3 (Chapter C, §C.3; k -curve §C.4.3). On AndroCT the same windows and split modes detect reference identity (d_{self} versus $d_{\text{cross}} + 0.087$ HGB PREFIX node, DeLong $p=2.7\times 10^{-6}$; +0.184 on adjacency).

Windows are exchangeable. Within-session windowing is the only form of self-reference AndroCT permits: one trace per app, no timestamps (§A.10.2). PREFIX versus SCATTERED moves at most 0.017 raw. On SCALAR, competitive near 0.70, every $|\Delta|$ is ≤ 0.0018 ; the larger gaps are CENTROID cells at 0.50–0.62, where the readout fails to detect (§A.6.8.4). The null has a positive control: those same windows detect reference identity (above) and label signal (shuffled 0.517 versus HGB ceiling 0.915–0.934). The design could have seen an effect an order of magnitude larger than anything ordering produced. No E0 or E2 readout reached a reportable operating point (§A.6.8, §A.6.9). Cross-session self-reference, which AndroCT cannot support, is exercised in Chapter C (§C.1, §C.3). Within-session windows do not encode temporal structure on AndroCT.

The covariance rule. Full covariance fails on this representation in four places. D_4 Mahalanobis inverts on 5/5 benign-group holdout folds (pooled OOF 0.242; §A.6.7). Input whitening drops Run 5 floor AUC from 0.638 to 0.560 (Table A.6). E2 Mahalanobis fails size-matching (0.6706 best node cell). FISHER combination inverts on 5/5 holdout folds (0.587 pooled OOF; §A.6.7.3). E2 diagonal NODE_STD is the variant that helped. At $n \approx 562$ with 22 coordinates, per-coordinate variance is estimable; off-diagonal covariance and benign-fitted joint nulls are not. Centroid readouts (D_1 , 0/5 inverted) remain stable.

Sparse graphs recur as four observations consistent with one candidate mechanism: aggregation can destroy the structure a comparison depends on. That mechanism is a reading of the measurements, not a finding isolated from alternatives. D_0 reconstruction inversion under benign-only training (Table A.6) and benign windows scoring above malware in E0 isotropic readouts are one phenomenon: benign scoring higher under sparse conditions on two datasets. Fixed- W mono-category collapse (0.8340 at $W=20$ benign; §A.6.8.1) concerns window construction. REFERENCE_DILUTION on the v2 k -curve (§C.4.3) belongs to Chapter C.

The conclusion in §A.8.4 is that the D_1 profile localises the class shift to a coordinate that is architectural. The supporting measurements follow. Chapter A D_1 ablation: zeroing `ipc_intents` drops floor AUC by 0.209 (§A.6.7). E2 self-reference ablation at the same node drops -0.0012 ; the top within-app gain is `file_io` at $+0.014$ (§A.6.8.3); on v2 under nav-first guided stimulation the same coordinate fires on far fewer benign apps (Chapter B, §B.6.3). A behavioural deviation would appear within-app; this pattern appears only across apps. D_1 `ipc_intents` scores 0.793 malware-higher while the raw IPC share (scored directly, no centring) scores 0.575 benign-higher, and 10/18 PRODUCIBLE coordinates reverse direction under referencing (Table A.15).

The localisation gap. On the same encoder and benign-only training, scalar D_0 reaches 0.638, D_1 centroid 0.800, and supervised D_3 HGB 0.962 (Table A.17). The D-3 factorial (§A.6.7.1) shows the benign-only band is not a monotone reference ladder: per-node readout on the deviation representation ($D-C=+0.039$) is the only distinguishable step. The implied remedies of the readout-gap diagnosis fail under the three interventions in §A.8.3. Across method families the same pattern recurs: untrained or random-initialised encoders match or beat trained ones (§A.8.4). Extracting D_1 -level signal without malware labels at deployment remains open.

The observability mask. A reference graph on emulator data should either restrict the universe to observable categories or carry an explicit per-corpus observability mask $\mathcal{O}_c \subseteq \mathcal{U}$, so that a zero from “did not do it” is distinguishable from a zero from “could not happen here.” Nothing in the current ABRG schema expresses that difference (§A.10.1). That gap is future work.

A.8.6 Deployability at realistic prevalence

No benign-only method under the tested configuration is deployable at realistic prevalence. At a 1% false-positive target on the GAE split, OCPool mean-pool recalls 1.76% of malware (TPR=0.0176; Table A.20). The D_1 centroid recalls 0.24% (TPR=0.0024) with 0.33% precision at wild prevalence $\pi=0.01$ (Table A.20; Figure A.5). Yet D_1 has the higher floor AUC at 0.800 against OCPool’s 0.776. The readout that ranks higher on AUC recalls roughly seven times less malware at the operating point that would be deployed. Neither readout is usable at this prevalence.

No benign-only headline clears the trivial size floors by a comfortable margin. The D_1 centroid point estimate is 0.800 against a mapped-event floor of 0.7025 (Table A.4). Interval to interval, D_1 ’s nested percentile interval [0.757, 0.815] sits just above the floor’s bootstrap interval [0.6434, 0.7545]: the nested lower bound exceeds the floor upper bound by 0.0025, a real but slight margin over

volume. Against the floor point estimate alone the gap is 0.055. OCPool mean-pool, the pooling-family incumbent, has nested interval [0.699, 0.800] whose lower bound essentially meets the floor.

Windowed self-reference (E0/E2) never reached a reportable operating point either (§A.6.8, §A.6.9). What would change the picture lies outside this configuration. The supervision ladder in §A.6.6 prices what labels buy on identical tensors. Chapter C exercises the per-app cross-session reference AndroCT could not support (one trace per app, no timestamps) and reports strong within-cross separation (§C.3; k -curve §C.4.3). Neither extension reports a malware operating point and neither closes this gap.

The cost of the benign-only guarantee is measured on identical tensors rather than asserted; the supervision ladder prices it. Yang et al. [43] argue the same direction from molecular and protein graphs; this chapter reaches it from behavioural traces.

A.9 Conclusions

The AndroCT tensors carry strong malware signal under full supervision. On identical tensors the D_1 centroid readout (0.800) exceeds every trained deep one-class family in the sweep (OCGIN_{plus} 0.566, GLocalKD 0.559, OCGTL 0.550; §A.8.1). It also exceeds the best measured configuration of those families (0.680 under GLocalKD Eq. 13). Benign-only reconstruction does not convert that signal into a detector: the scalar autoencoder inverts; trained deep one-class objectives sit at or just below the range published for the matched OCGIN[†] configuration (≈ 0.572 – 0.635 excluding AIDS); and the best benign-only readout still fails a 1% false-positive operating point. Message passing does not help on the T22 GIN. A T1K one-class kernel reads structure without that stack (§A.6.5). Within-session self-reference does not detect at window granularity: windows are exchangeable and readouts sit below the mapped-event floor; cross-session evaluation is Chapter C (§C.1).

Table A.23 closes the eight research questions against the measurements in §A.6.

Table A.23. Research-question closure. Floor AUC in three decimals in the answer column; four-decimal sources in §A.6. Rung-2 cost uses pooled out-of-fold raw AUC, not the superseded per-fold floor mean.

RQ	Answer	Evidence
1 Signal present?	Yes.	Supervised HGB 0.976 on the 704-vector; adjacency alone 0.959 (§A.6.1).
2 Benign-only recon-struction?	No: objective and readout, not representation (0.638 $\rightarrow$ 0.962 on the same GAE).	Canonical dual-recon 0.638; windowing, whitening, and $h=2$ remain inverted (§A.6.2).
3 Objective, architecture, or representation?	Objective and readout; representation nonempty. Trained and untrained encoders indistinguishable across families.	Same GAE: benign-only scalar 0.638 $\rightarrow$ supervised D_3 0.962. Seven families: per-family Wilcoxon at $n=5$ underpowered ($p \geq 0.0625$); five of six favour untrained or no-training, sign test $p=0.031$ across five trainable families (§A.6.3, §A.6.7).
4 Topology?	T1K kernel reads structure; message passing does not help the T22 GIN—consistent across three poolings, Ward holdout, and per-seed evidence.	WL structure-only 0.627 vs edges-removed 0.500 (one-class, T1K); $M2 \geq M1$ at every GIN condition mean; add pooling unanimous; mean/max 2/5 seed reversals (§A.6.5).
5 Cost of the guarantee?	0.127 to unseen behavioural groups; a further 0.073 against the raw incumbent (0.072 against the matched T22 centroid).	Ladder 0.976 $\rightarrow$ 0.849 $\rightarrow$ 0.776 (raw); random-group control 0.971 pooled out-of-fold. Groups are a family proxy (§A.6.6, §A.10.1).
6 Where is the signal?	In the deviation vector under per-node readout; concentrated on <code>ipc_intents</code> as an architectural coordinate, not a behavioural deviation (§A.8.4); 10/18 coords reverse under referencing.	Benign-only: D-3 cell D D_1 0.800 (only distinguishable margin $D-C=+0.039$); cell A raw scalar 0.793 indistinguishable from D_1 on the full split but holdout 0.712 vs. 0.789; ipc ablation $\Delta=0.209$; uncentered raw IPC share 0.575 benign-higher vs. deviation 0.793 malware-higher. Supervised: D_3 0.962 (§A.6.7, §A.6.7.1).
7 Self-reference?	No at window granularity; same design detects reference identity and label signal.	Best raw 0.700 below the 0.703 floor (E0 0.624; E2 0.675). Same windows detect reference identity (d_{self} vs. d_{cross} +0.087, DeLong $p=2.7 \times 10^{-6}$) and label signal (shuffled 0.517 vs. ceiling 0.915–0.934; §A.6.8, §A.8.5).
8 Deployable?	No.	D_1 TPR 0.24% at 1% FPR; wild precision 0.33% at $\pi=0.01$ (§A.6.9, §A.8.6).

Beyond the closures in Table A.23, three measurements stand as the chapter’s transferable results.

On identical tensors, the supervision ladder prices the benign-only guarantee at 0.127 AUC to unseen behavioural groups (0.976 $\rightarrow$ 0.849 $\rightarrow$ 0.776; §A.6.6). Across seven method families, trained and untrained encoders are statistically indistinguishable: per-family Wilcoxon at $n=5$ cannot reach conventional significance ($p \geq 0.0625$), yet five of six favour untrained or no-training and a sign test across five trainable families gives $p=0.031$ (§A.6.3, §A.6.7). Corpus validation caught seven artefacts before they entered as results, two after briefly being headline numbers (0.814 residualised pooling; 0.734 V2 invocation floor); five surfaced while auditing a denominator or population, and the recurring pattern was label-aligned loss or silent subsetting moving a floor before any detector did (§A.7).

D_1 at 0.800 against a mapped-event floor of 0.7025, with nested lower bound 0.757, is not a comfortable separation from volume, and it is not a deployable detector (§A.8.6).

What remains untested on AndroCT is deferred, not promised. Generality beyond one public year-matched corpus is Chapter B.

Per-app cross-session self-reference is Chapter C (§C.1). Per-app references separate strongly from cross-

app ones (§C.3). Ordering and recency are nulls. The k -curve is `REFERENCE_DILUTION` with $k=1$ optimal (§C.4.3).

A description-built reference at $k=0$ fails: self-versus-cross AUC 0.501 versus ceiling 0.979; within-app 0.725 versus prevalence prior 0.952 (§C.5). Whether the `ipc_intents` concentration is corpus-specific is an open measurement on any later corpus that keeps a comparable node universe.

A.10 Threats to validity

Chapter A reports measurements under fixed pins (§A.5). The threats below bound how far those measurements generalise: what the AndroCT corpus does and does not represent, which ABRG design commitments were not exercised, where estimation or selection can understate uncertainty, and where a comparison-method reimplementing can change a reported figure. Verification scope and catalogue reload limits are process disclosure in §A.7.1, not threats.

A.10.1 Corpus and population

Single public corpus. All quantitative claims rest on one year-matched AndroCT 2017 release ($n=2403$ graph-eligible apps; §A.2). External validity to other years, markets, or instrumentation is untested. The positioning table in §A.8.1 compares against published graph-level detectors on molecular and protein benchmarks—a cross-domain scale reference, not evidence that AndroCT difficulty matches those corpora.

Family labels and the zero-day proxy. Rung-2 holdout uses Ward behavioural clusters ($k=30$) as a proxy for family holdout, because AndroCT ships no family labels and no VirusTotal reports were available to run AVClass: the chapter therefore demonstrates generalisation to unseen behavioural groups, not to unseen families (§A.6.6). Family labels are themselves noisy—AVClass assignments vary across tool versions and across the vendor labels they aggregate—and family membership does not guarantee behavioural distinctness, which is the axis a behavioural detector must generalise over. The proxy is therefore a corpus-imposed constraint rather than an obviously weaker operationalisation of the zero-day question; whether it biases results upward or downward relative to true family holdout is unknown. Volume diagnostics show the clusters are not length strata (between/within variance 1.90 on the non-volume tensor versus 0.026 on mapped-event count; fold AUC versus fold median volume $\rho=0.058$).

Emulator realism and observability. AndroCT traces were collected under a 600s Monkey protocol on a Nexus One AVD running Android 6.0 (DroidFux instrumentation; one trace per app). Monkey maximises event volume rather than behavioural relevance, so the stimulus is random rather than semantically informed (§B.1.1). Apps are year-matched to 2017 (§A.2.2). Those conditions bound realism relative to production devices and to guided stimulation; they are independent of the shared-hook design choice recorded in §A.2.1.

A zero coordinate conflates “the app did not perform this behaviour” with “the environment could not produce observable events for this category.” Nothing in the ABRG schema distinguishes the two; both encode as `act_v=0` with zero degree. Effective universe sizes from the observability audit: AndroCT 18/22 `PRODUCIBLE`; `v2_extended` 15/22 (342 reference-tier sessions, 59 apps). Occupancy tables: `results/observability_androct.csv` (graph-eligible $n=2403$) and `results/observability_v2_extended.csv`. Four coordinates are non-`PRODUCIBLE` on AndroCT: `sms` and `telephony` are `PERMISSION_GATED`; `clipboard` and `dynamic_code_loading` are `STIMULUS_LIMITED` under Monkey (§A.4.1).

These counts must not be read as the parser-gate “active universe” tallies in §A.2.2 and Table A.21 row 1 (21/18 benign/malware categories firing at least once anywhere in the parsed stream). That pair counts *ever-fired* categories per class on the full parse; the audit counts coordinates classified `PRODUCIBLE` under instrumentation evidence. They align on the rarest slots (`sms` never fires on either class in the parse) but diverge where a coordinate fires once or twice yet remains non-`PRODUCIBLE`: `telephony` (one benign app in the graph-eligible set, `PERMISSION_GATED`); `dynamic_code_loading` (ten malware apps fire, `STIMULUS_LIMITED` under Monkey); `clipboard` (zero on graph-eligible tensors, one app on the wider parse, `STIMULUS_LIMITED`). The 21/18 figures are correct for what they measure; 18/22 is the coordinate system that actually carries evidence on AndroCT.

On D-2’s per-coordinate nulls (§A.6.7.3), KS uniformity departure is partially explained by non-producibility: mean KS by bucket is 0.83 (PERMISSION_GATED), 0.77 (STIMULUS_LIMITED), 0.45 (PRODUCIBLE); Spearman(occupancy, KS)=−0.91 ($p \approx 6 \times 10^{-9}$). Two mechanisms are independently sufficient: environmental gating produces degenerate empirical CDFs by construction (sms KS=0.936, clipboard KS=0.851); sparse occupancy alone also breaks uniformity where the coordinate *is* PRODUCIBLE—notably audio (KS=0.801) and content_access (KS=0.714). Neither explanation replaces the other.

Attrition and label-aligned exclusions. Parser repair removed a label-aligned line-drop asymmetry (12.2% benign vs. 27.9% malware under the pre-fix regex); all reported tensors use the post-fix parse (§A.2.2). Zero-mapped exclusion remains asymmetric: 50 / 763 benign (6.6%) versus 39 / 1742 malware (2.2%). Benign package naming resolved only 172 of 1648 package-named archive files uniquely under AndroZoo year 2017 filters (§A.2.2). The withdrawn eligibility floor (≥ 320 mapped events) showed that population rules can invert trivial floors (§A.7, row 2); the live population keeps that lesson without applying the rule.

Abstraction and dual data sources. Category mapping is selective: among allowlisted callees in mapper diagnostics, empty-map rates are 0.965 (benign) and 0.959 (malware) (§A.2). Static features come from Androguard parsing of APKs; dynamic graphs come from released traces—two acquisition paths (§A.2.4). The API-level granularity sweep (§A.6.4) shows that a finer vocabulary raises supervised ceilings; the canonical 22-node representation trades coverage for cross-app comparability.

Training-set contamination. Yang et al. [43] identify the requirement for a verified-clean normal training set as a limitation of the dominant, partially unsupervised GLAD paradigm, and take it as the motivation for fully unsupervised alternatives that train on unfiltered mixtures. The 562 train-benign apps carry AndroCT’s benign labels; any mislabelled malware in that pool constitutes training-set contamination at an unknown rate. Wang et al. [38] stress-test this failure mode separately from their main protocol: anomaly/OOD graphs are mixed into the ID training set at 0/10/20/30%, drawn from a reserved contamination pool (§4.3, Appendix D.4). AUROC generally falls as contamination rises, with monotonic declines for GLocalKD, OCGTL, SIGNET and GOOD-D on PROTEINS, BZR and COX2; GLADC is comparatively stable and OCGTL declines sharply on PROTEINS (they report no aggregate rank-reordering statistic). That published sensitivity makes label noise in the benign training pool a live threat to benign-only results here, not a hypothetical one; the contamination rate in train-benign has not been measured on AndroCT.

A.10.2 Design commitments not exercised

AndroCT traces carry no timestamps. Consequently the recency node slot rec_v is identically zero on every app, the recency edge channel w_{rec} is never incremented, and the wall-clock proximity filter δ is not applied—only sequence proximity $k=5$ shapes edges (§A.4.3). The cost of enabling δ is quantified in Chapter B (§B.5); this chapter reports no number for it. On v2_extended, three Table A.1 rows are now measured at corpus level (no AUC): δ retention in §B.5; W_{REC} versus W_{CUM} in §C.4.2 (recency does not beat cumulative weighting); per-app within-cross separation in §C.3; session ordering in §C.4.1 (ORDERING_NEUTRAL); $k=1$ reference in §C.4.3. AndroCT itself still leaves those channels unexercised (no timestamps; $N=1$ per app). Multi-window trace partitioning ($N>1$) was tested as an intervention and is not part of the reference representation (§A.6.2). Per-app cross-session refinement is measured on v2 in Chapter C; state-based stimulus changes remain unavailable on AndroCT and are not constructible as a controlled comparison against v2 (different apps, AVDs, and sensors; §C.7). Negative results on message passing (it does not help), reconstruction, and support novelty therefore concern a *timestamp-free, single-window* slice of the full ABRG design; they do not adjudicate components that require temporal structure or repeated sessions.

A.10.3 Estimation, selection, and operating points

Vocabulary selection. When document-frequency vocabulary construction is part of the fitted pipeline (pooling-family OCPool configurations), vocabulary choice is part of the model, not a preprocessing detail.

Resampling scores without refitting vocabulary yielded a naive 95% interval width of 0.080 [0.771, 0.851] on one audited configuration; nested bootstrap with per-draw vocabulary refit widened that interval to 0.100 [0.699, 0.800]: the naive interval is 20% narrower than nested (nested is 25% wider; §A.7, row 4; §A.5.4). Headline nested intervals in §A.6.7 and source sheet T8 use the nested procedure where selection is fitted.

Class imbalance and deployment readouts. The canonical split holds 141 test-benign and 1700 test-malware apps (§A.5). Rank-based floor AUC is insensitive to base rates, but operating-point columns in §A.6.9 assume a wild prevalence $\pi=0.01$ for precision at a fixed false-positive target—an illustrative deployment scenario, not a measured field prevalence. Wild precision and recall at $\pi=0.01$ should not be read as calibrated deployment guarantees (§A.6.9).

Concentration on one coordinate. The deviation-profile headline concentrates on `ipc_intents` ($\Delta=0.209$ AUC when that node is zeroed in ablation; §A.6.7), but two Chapter A measurements now bound how that concentration should be read. The raw IPC quantity and the deviation coordinate move in opposite directions (uncentered raw IPC share 0.575 benign-higher versus deviation coordinate 0.793 malware-higher; per-coordinate referencing $\Delta_{\text{floor}}=+0.231$ with flip), so the deviation signal is not a restatement of raw IPC magnitude (§A.6.7). Under per-app self-reference, E2’s per-node ablation shows `ipc_intents` drop -0.0012 while `file_io` is the top contributor at $+0.014$, whereas the population-reference readout carries the 0.209 concentration (§A.6.8.3, §A.6.7). This is evidence that the concentration is primarily across-app rather than within-app at window granularity, consistent with a structural packaging/IPC mechanism; whether it persists under a different benign population, year, or collection regime remains untested, and no corpus in this thesis can settle that.

Paired seed variance and small differences. Most two-arm comparisons in this chapter are single-fit; DeLong and bootstrap intervals quantify sampling variability in the evaluation set. Where paired seed variance was measured, on the supervised GIN ablation, the Split-A mean-pool standard deviation across five seeds is 0.028, larger than several of the differences the chapter discusses. Differences smaller than roughly 0.03 AUC should be read as directional observations rather than as effect sizes, except where a seed distribution or paired interval is explicitly reported. The D-3 factorial cells other than $D-C$, the sparse-aggregator controls, and the E0/E2 self-reference readouts are the comparisons this bears on most directly.

A.10.4 Comparison method reimplementations

Reimplementation risk. The OCGIN arms, OCGTL, and GLocalKD are reimplementations: no upstream author code is vendored. OCGIN_Orig is documented as paper-faithful in `abrg/ocgin/README.md` but has not been validated against the authors’ reference implementation; any gap between published and measured OCGIN numbers (§A.8.1) is therefore confounded with implementation fidelity, not attributable to the corpus alone. OCGTL was written independently and cross-checked against the Bosch reference (Qiu et al. [30]; AGPL-3.0, commit 7b2295d, not vendored): centre handling and $\tau=1$ match the reference; batch size (64 versus 128), epoch count (300 versus 500), and the training loop (no StepLR, no early stopping) do not. $\mathcal{L}_{\text{GTL}}$ was not logged at run time and was recomputed post-hoc from saved checkpoints. These deviations do not plausibly produce mean pairwise encoder cosine ≈ 0.9999 on T22 and ≈ 0.97 on T1K under otherwise identical settings (§A.8.2); the encoder-agreement finding there is stated without hedging on batch or scheduler choice. GLocalKD was written independently and cross-checked against the Ma et al. [22] reference (`github.com/RongrongMa/GLocalKD`, commit 1c8c15f, not vendored; `abrg/glocalkd/_ref/` holds a partial inspection copy not imported at runtime): layer count ($3\times$ GCN) and equal-weight $\mathcal{L}_{\text{node}}+\mathcal{L}_{\text{graph}}$ match; learning rate (0.01 versus 10^{-4}), hidden/output (128/128 versus 512/256), epoch count (300 versus 150), batch size (64 versus 300), dropout (none versus 0.3), batch normalisation (off versus on), StepLR scheduler (absent versus step size 50, $\gamma=0.5$), and backend (PyG GCNConv versus custom adjacency GCN) do not. A paper-configuration validation arm on the T1K family-sweep winner (T1K, add pooling, full loss, s_{graph} , seed 46) shows how much those deviations can matter. Under the brief profile used in Table A.7, largest per-node class-mean gaps are $+0.000961$ and -0.000922 (floor AUC 0.690, score direction benign-higher). Under Ma et al.’s [22] reference hyperparameters (`glocalkd_papercfg_t1k_node/`; lr 10^{-4} ,

hidden/output 512/256, 150 epochs, batch 300, dropout 0.3, batch normalisation, StepLR) the same run gives +1.681 and -1.609 (floor AUC 0.586, malware-higher). Gap magnitude rises by roughly three orders of magnitude and the graph-level ranking flips; the brief profile was locating the same coordinates at a suppressed scale, not reporting a different phenomenon. One seed; the lesson is configuration sensitivity under reimplementation, not a matched comparison to the deviation readout. A separate T22 paper-configuration arm (`glocalkd_papercfg`) confirmed the untrained-beats-trained pairing survives reference training hypers (trained 0.5955 versus untrained 0.6633 on s_{graph} ; §A.6.3).

Appendix (Chapter A)

A.11 AndroCT GAE training pins

All rows share the Run 2 corpus cache (eligible $n=2403$; split 562 / 141 / 1700; $k=5$; shares-not-counts; w_{cum} edge channel; seed 42). The v2 campaign default $h=16$ (`abrg/config.py`) is out of scope for this table.

Table A.24. GAE encoder pins on AndroCT 2017. The reference method is the Run 5 row with $h=8$ (checkpoint `gae_androct_run5_h8.pt`).

Run	Objective	h	Other pins	Role in Chapter A
2	Stochastic PyG recon_loss	64	300 ep, lr 0.01, wd 0	§A.6.2 intervention
3	Full-adj. weighted BCE (structure only)	64	300 ep, lr 0.01, wd 0	§A.6.2 intervention
4	Dual: $\alpha\mathcal{L}_{\text{struct}} + (1 - \alpha)\mathcal{L}_{\text{feat}}$	64	$\alpha \in \{0.8, 0.5, 0.2\}$; best $\alpha=0.2$	Selects α for Run 5
5	Dual at $\alpha=0.2$	$\{2, 4, 8, 16, 64\}$	300 ep, lr 0.01, wd 0	Bottleneck sweep; $h=8$ carried forward
5 ($h=8$)	Dual at $\alpha=0.2$	8	$N=1$; benign-only train	Reference (D_0 – D_5)

A.12 Comparison-method pins

Pins for the objective-family sweep of §A.5.5. Each row is taken from the named artefact under `abrg/output/androct_2017/` (not hand-transcribed AUCs).

Table A.25. Comparison-method configuration pins (AndroCT 2017). [†]OCGTL: $K=6$ matches Qiu et al. [30]; $K=4$ is ours; batch 64 (ref. 128), 300 ep (ref. 500), no StepLR or early stopping (lr 0.001, $\tau=1$ match the reference).

Family	Encoder / objective	Key hyperparameters	Seeds
Reconstruction (GAE)	Dual recon. GCN	$\alpha=0.2$, $h=8$ (sweep $\{2, 4, 8, 16, 64\}$); 300 ep, lr 0.01, wd 0	42
One-class (OCGIN)	$4 \times$ GIN, emb 128; $\ f-\theta\ ^2$	300 ep, lr 0.01, wd 0, batch 32; θ frozen untrained mean; no edge weights	42–46
One-class (OCGTL)	$4 \times$ GIN + add; $L_{OCC}+L_{GTL}$	$K \in \{4, 6\}$, $\tau=1$; 300 ep, lr 0.001, batch 64, GraphNorm	42–46
Distillation (GLocalKD)	$3 \times$ GCN; random teacher	300 ep, lr 0.01, hid/out 128; $L_{node}+L_{graph}$; pool mean/add/max	42–46
Kernel / shallow	WL/Prop/SP/FGSD/ NetLSD/G2V/GL2V + OC heads	train-benign Gram/fit; gate size floor 0.7025; OCPool_mean raw 0.7765	42–46
Classical flat OC	Flat 704/220/484 vectors	Centroid E/C, IF, k NN, LW Mahalanobis, OCSVM RBF $\nu=0.1$	42–46
Pooling (OCPool)	Pool nodes; OCSVM (no edges)	add/mean/max pool; OCSVM RBF $\nu=0.1$	42–46
Supervised probes	GIN / HGB / LR	GIN: 4 layers, hid 64, modes M1/M2/M3; ≤ 200 ep, lr 0.001, wd $5 \cdot 10^{-4}$, drop 0.5	42–46

B.1 Framing

B.1.1 The stimulus problem

Dynamic analysis is bounded by what the exercised app does. Random input maximises event volume, not behavioural relevance. DL-Droid reports up to 97.8% detection from dynamic features alone using state-based input generation — a stimulus result, not a model result.

AndroCT traces used in Chapter A were collected under a 600s Monkey protocol. Those traces carry no wall clock, so the 600s figure is a stated duration, not a measured one (§B.4.5). Every structural result there is conditional on a stimulus policy this chapter did not choose. The question is what changes when the policy is replaced by nav-first BFS and LLM task execution on a modern corpus, and what does not.

B.1.2 What this chapter does

It specifies ContextDroid: a dynamic-analysis pipeline that combines deterministic BFS navigation with LLM-driven task execution (§B.2). It characterises the v2 corpus that pipeline produced: 342 usable sessions across 59 apps, of which 40 are graph-eligible (§B.3.1). Median usable sessions per graph-eligible app is 8; the 59-app median is 7. The median-8 figure is used only where the GAE-40 set is the stated population. It then compares the resulting 22-node graphs with AndroCT on identical metrics (§B.4), measures the wall-clock edge filter δ that Chapter A could not exercise (§B.5), and states what that comparison licenses about Chapter A (§B.6).

No detector is trained. No AUC is reported. v2 is benign-only.

B.1.3 Conventions

These conventions apply to every distribution table in this chapter.

Medians are the interpolated p_{50} written in the chapter’s `SUMMARY.md` files (linear interpolation on the order statistics; `abrg.androct.inventory._percentile`). Companion JSON Mann–Whitney records also store `median_x` / `median_y` as `sorted[n//2]`, which disagree at even n . Tests are taken from those records; reported medians are the interpolated figures.

The reused inventory helper names fields `benign_median` / `malware_median` and `n_benign` / `n_malware` even when the contrast is not malware. In the original-versus-new comparison those keys are the two batches; in the v2-versus-AndroCT comparison `n_benign` is 59 (v2 pooled) and `n_malware` is 2231 (AndroCT parsed benign). Those names are left as emitted; the records are read via `n_x` / `n_y` and `label_x` / `label_y`.

Tables use four decimals; prose uses three. Cosine distances that would print as 0.0000 at four decimals are given in scientific notation (§B.2.5).

A difference is material when $p < 0.05$ and Cliff $|\delta| \geq 0.147$ [33]. That rule is declared before every comparison table that uses it (§B.3.2, §B.4.2).

B.1.4 Contributions

1. ContextDroid, specified end to end: 58-hook Java-layer instrumentation, a two-phase stimulation policy (BFS navigation then LLM task execution), a session protocol with per-session `pmclear`, and an eight-conjunct reference-tier quality gate (§B.2).
2. A measured stimulus-quality result: pooled mapped-event rate 0.204 versus AndroCT’s 0.026 ($p=9.19 \times 10^{-14}$, Cliff $\delta=0.568$, large). Absolute mapped counts are not materially different ($p=0.150$). v2 total events are about a quarter of AndroCT’s (§B.4.2). BFS is 0.711 of pooled

usable steps (13,029 / 18,315). Mapped events cannot be joined to action-step windows, so that rate is not an LLM-planning result.

3. A measured coverage cost: v2 pooled median 2 active nodes and 2 edges, below all three AndroCT benign graph medians: train 5/14 ($n=562$), test 6/16 ($n=141$), parsed 6/16 ($n=2231$). Both structural contrasts are large (§B.4.2, §B.6.1). Signal density and behavioural breadth trade against each other under this policy.
4. The δ measurement Chapter A points to: retention rises from 0.272 (Q1) to 0.905 (Q4) by mapped-event count; pair-pooled retention 0.838 on 342 usable sessions is the operative answer to that deferred fraction, with an unweighted session-level mean of 0.489 on the 307 sessions with a defined retention. On the longest traces, Q4 pair-pooled retention is 0.905—roughly one k -candidate in ten still fails the clock; the fitted asymptote 0.941 summarises the length trend (§B.5). Evaluating the fit at AndroCT’s median mapped count is an out-of-sensor extrapolation, not a transferred percentage.
5. Retirement of a Chapter A threat, with an asymmetric mechanism. **sms** is dead in v2 (0/59) and in both AndroCT classes (0/2231 benign, 0/1736 malware). For v2 the planner action space contains no SMS-capable action, so the cause is identified in code. For AndroCT the cause remains an inference (§B.6.2).
6. A quality-gate design whose validation findings are documented: metric/no-op pathologies, explore-blind ratios, and a weakened failing rule. The plan says the weakened failing rule would have admitted 12 sessions. The 12 is plan-sourced, not re-counted on v2 (§B.2.6, §B.6.4).

B.1.5 Questions

1. B-Q1. What does two-phase guided stimulus change about trace composition? (§B.4.2)
2. B-Q2. Does it change the graph representation, and how? (§B.4.3, §B.4.4)
3. B-Q3. Are Chapter A’s structural findings artefacts of corpus age or stimulus policy? (§B.6.1)
4. B-Q4. What does the temporal edge filter δ cost? (§B.5)

B.2 ContextDroid

ContextDroid is the dynamic-analysis pipeline that produced the v2 corpus. This section specifies what was designed, what the code implements, and what v2 collection ran. Three labels are used throughout: *implemented and exercised* (code exists and ran on the v2 path), *implemented but not exercised* (code exists; v2 did not use it), and *designed but not implemented* (plan or documentation only, or a frozen constant that runtime does not use).

B.2.1 Design rationale and scope

v2 collection invoked `--armllm` at a configured duration of 420s: nav-first BFS navigation followed by LLM task execution (§B.2.3)—a different configuration from `llm_only`, not a reduced one. The protocol freeze titles the collector an “LLM-first methodology” and names `llm_only` as the primary mode. That is the design intent v2 departs from. The working-tree documents describe that intended design, not the frozen v2 configuration developed below.

The current README opening sentence states that the repository “runs deterministic + monkey-driven interaction.” The methodology document lists as high-level step 4 “Run deterministic stimulation followed by monkey-driven UI events.” Monkey remains a pre-simulation warmup of 10 events with seed 42, outside the timed loop (§B.2.3); it is not the timed stimulation arm. The optional `llm_plus_monkey` comparison path is implemented but was not exercised for v2.

A second freeze is unused. `protocol_constants.md` and `protocol_config.py` still set session duration to 120 s. The analyzer CLI default is 180 s; the v2 wrapper sets 420 s; every usable original and extend session records `duration_sec=420`. The 120 s figure is designed (Phase 0) but not v2.

The governing remediation principle, quoted from the collector’s own plan, is: “Instrument before you fix. Fix before you multiply.” The same record attributes three incidents to “optimizing something no metric could see.” Those incidents reappear as quality-gate validation findings in §B.2.6.

B.2.2 Instrumentation

The hook script is `frida_scripts/hook_apis.js`, 856 lines, version tag “3” in the `hook_loaded` event. Its SHA256 is `3192c7d6f618bb455588a7db7b26b20ccb9028af22cf1d8d8f0db99c28009ffc`, identical on all 168 original usable sessions and all 172 extend usable sessions.

Direct `logEvent(...)` string literals in the script number 58 API names. Four further names are emitted through `logContentUriEvent` and then `logEvent` as `ContentResolver.query`, `.insert`, `.update`, and `.delete`, giving 62 distinct API names in traces. `Context.startActivity` is logged twice: once under `ipc_intents` (kept as a graph node) and once under `navigation` (dropped from graphs). The full 63-row inventory, including that duplicate category row, is Appendix B.9 (Tables B.14–B.15). Table B.1 summarises the 22 graph categories: hook-row count and how many of those rows fired anywhere in the v2 corpus.

Table B.1. Hook coverage of the 22-node graph universe. Counts are CSV rows in `docs/contextdroid_hooks.csv` per graph category. Four `ContentResolver` names are rows under `content_access` but not among the 58 `logEvent` literals; `Context.startActivity` is one API counted here under `ipc_intents` only (the `navigation` duplicate is dropped with that category). Corpus-wide, 40 distinct APIs fired and 22 never fired, including `hook_loaded` and `Method.invoke` outside this table.

category	hooks	fired in v2
accounts	2	0
audio	2	2
camera	2	1
clipboard	1	0
content_access	4	4
crypto	5	4
database	3	3
device_info	3	0
dynamic_code_loading	2	0
file_io	4	3
ipc_intents	4	4
location	1	1
media	3	0
native_code	3	1
network	7	5
notifications	2	2
package_manager	2	1
process	2	2
sms	2	0
storage	4	4
telephony	1	0
webview	1	1
total	60	38

The hook layer uses 25 categories. The graph universe is the 22 in Table B.1; `lifecycle`, `reflection`, and `navigation` are collected and then dropped. Seven graph categories have zero fired hooks in v2 (`accounts`, `clipboard`, `device_info`, `dynamic_code_loading`, `media`, `sms`, `telephony`). That is hook-level non-fire, not yet the per-app fire rates of §B.4.3.

Each event record has fields `type`, `timestamp`, `api`, `category`, and `args`. Frida timestamps are `Date.now()` in the JavaScript runtime (milliseconds). Action-log timestamps are host `int(time.time()*1000)`. The two clock domains are not synchronised in code. Mapped-event attribution to action-step windows was therefore not attempted; the phase split in §B.2.3 is a step census.

Attach is to a running PID after `ensure_app_running`. `Spawn(-f)` is implemented but was not exercised on the default `analyze_apk` path. Behaviour before `onCreate` is not observed. Attach is retried up to three times with a 12s host timeout. Mid-session health-check can reattach; two extend sessions failed with `failed_frida_reattach` (`anonvpn.anon_next.android` slots s4 and s5). `safeHook` swallows install errors so a failed hook cannot crash the app—a deliberate coverage/robustness trade.

B.2.3 Stimulation: two timed phases

v2 stimulation is nav-first: a deterministic breadth-first (BFS) navigation phase, then LLM-driven task execution—the two phases of the timed loop after Frida attach. Naming the pipeline “LLM-guided exploration” would describe the execute arm and erase the phase that produces most of

the steps. Stimulus regime has been compared on Android malware before: Onwuzurike et al. [25] evaluate Monkey against human interaction on a shared app set.

Warmup (untimed). Before the timed loop, and before Frida attach, the analyzer grants declared permissions via `pmgrant`, resolves setup dialogs, runs 10 Monkey events with seed 42, resolves dialogs again, and takes a verified-start uiautomator dump. That burst runs before Frida attach, so it produces no captured trace events. That Monkey burst is implemented and exercised as warmup. It is not the timed arm.

Phase 1 — BFS navigation. While elapsed time is below the explore budget, each step calls `choose_explore_action` and does not call the LLM. The logged `prompt_hash` is the label `bfs_navigation_phase`. Candidate buckets are `nav` / `tab` / `expand` / `other`, with recovery actions `bfs_return_to_hub` and `bfs_avoid_back_loop`. The explore budget is $\min(\text{duration} \times \text{ratio}, \text{duration} - \text{reserve})$, then floored at `CONTEXTDROID_LLM_EXPLORE_UNTIL_SEC_FLOOR` (v2: 120s) and clamped to $[25, \text{duration} - 30]$. v2 sets ratio 0.35; `reserve` is `CONTEXTDROID_LLM_EXECUTE_RESERVE_SEC`, defaulting to $\max(90, 0.52 \times \text{duration})$ and clamped to $[45, \text{duration} - 35]$. At `duration_sec=420` that yields `reserve=218.4s` and an explore cutoff of 147s—matching 0.35×420 because the ratio slice binds before the floor. Session metadata record `duration_sec`, the floor, and the ratio on every usable session; `explore_until_sec` itself is not persisted per session.

Phase 2 — LLM task execution. After the explore cutoff, steps use Ollama `llama3.2` at temperature 0.0, with `options.seed` set when `agent_seed` is an integer. Prompt families are `execute` / `UX-goal` and `primary-UX`. The export records planner actions and the generated `llm_ux_goals` list but no per-goal completion outcome, so what Phase 2 achieved session by session is not measurable from the export.

Phase split. Action logs are absent from the `v2_extended` export (Frida JSONL and metadata only). They remain in the collection trees and in the packaged original 168 under `datasets/v2/sessions/`. The census below joins every export session to its collection `*_llm_actions.jsonl`; all 342 usable sessions have an action log.

Table B.2 reports the step census. On the pooled analysis set, 13 029 of 18 315 steps carry `prompt_hash=bfs_navigation_phase` (fraction 0.711). That count equals the `pipeline_phase=explore` count exactly; `execute` and `primary-UX` make up the remainder. Per-session BFS fractions have median 0.732 and IQR $[0.688, 0.759]$ ($n=342$). The step fraction is near-identical across collection windows a month apart (0.711 original, 0.712 extend): the phase split is a property of the policy and budget, not of the apps or the collection conditions. The median explore/execute step ratio (primary-UX excluded from the denominator) is 6.37, IQR $[5.26, 8.25]$ —counting all execute-phase steps, including `advance_goal`; on device-interacting execute steps alone the median is 18.5. This is a *step* fraction, not a time fraction: BFS steps have no LLM round-trip, so the configured explore-time ratio (v2: 0.35) is not the quantity in Table B.2. The mapped-rate comparison in §B.4.2 cannot be attributed to LLM planning.

Table B.2. Stimulation phase split from collection `*_llm_actions.jsonl`, joined through the `v2_extended` export index. BFS steps are those with `prompt_hash=bfs_navigation_phase`, which coincides with `pipeline_phase=explore`. Per-session fraction is BFS steps / session steps. Canary $n=2$: median only, IQR omitted.

population	n sess.	steps	BFS	exec.	prim. UX	sess. BFS med. [IQR]
pooled usable	342	18 315	13 029	2 662	2 624	0.7321 [0.6875, 0.7591]
original	168	9 063	6 446	1 368	1 249	0.7344 [0.6858, 0.7602]
extend	172	9 146	6 511	1 274	1 361	0.7302 [0.6875, 0.7588]
canary	2	106	72	20	14	0.6764

Frida events and action logs use unsynchronised clocks (§B.2.2). That is why mapped events are not allocated to phase windows and why the mapped-rate comparison in §B.4.2 cannot be attributed to LLM planning. On the original 168, the first Frida event precedes the first action by median 5.98s (IQR 0.551s), which is consistent with attach-before-loop, but the start-offset minimum is -72 s and the end-offset IQR is 21s.

Observation. The agent sees uiautomator XML, not screenshots (no `screenshot` in `llm_agent/`). `dump_clean_screen` keeps enabled elements that are clickable, long-clickable, or focusable text-entry, with fields `package`, `resource_id`, `content_desc`, `text`, `class_name`, `bounds`, `clickable`; the list is token-trimmed and hashed. LLM prompts receive the result as `CLEAN_SCREEN_ELEMENTS`. Rendered prompts are hashed, not stored, so the observation shown on a given execute step is not recoverable from the action log.

Action space. The planner may emit six types: `tap`, `input`, `back`, `wait`, `advance_goal`, `swipe`. Invalid types degrade to `wait` with `planner_contract_invalid_action_type`. `advance_goal` performs no device I/O and returns success. The remediation record identifies this as a metric pathology; the fix was to the metrics that consumed the success bit, not to the action. Table B.2’s `execute` column is a step count, not a device-interaction count: `advance_goal` is a goal-advance signal to the planner’s own state, and on the pooled usable set it accounts for 1,323 of 2,662 `execute`-phase steps (per-session median 0.667, IQR [0.571, 0.800]). That is the same distinction already drawn between step fraction and time fraction—BFS is 0.711 of all steps and 0.767 of interacting steps (13,029 / 16,992), so nav-first dominance is larger on steps that actually touch the device than the raw step census alone suggests. Absent from the planner set—and therefore unreachable by policy—are permission grants in the timed loop (pre-granted via `pmgrant`), credential flows, purchases, `long_press`, a named `scroll` action, and intent injection. That is the code-level explanation for the dead categories in §B.4.3.

One internal inconsistency: the explore prompt template lists `action_type(tap|input|back|wait)`, omitting `swipe` and `advance_goal`, while the planner `frozenset` includes them. Under nav-first v2 the explore template is not the explore policy—BFS is—so the omission did not bind explore steps. Under the implemented `llm_only` arm with default nav-first, explore steps are BFS-driven and the explore LLM prompt is not reached; with nav-first disabled the timed loop has no explore phase and the legacy `execute` prompt lists all six action types.

B.2.4 Session protocol

Per session, in order: snapshot restore or skip-load; ensure `frida-server`; `isolate_emulator_state` (HOME, then `amforce-stop` other third-party packages); device-identity guard; `adbinstall-r`; `pmclear`; launch; warmup (§B.2.3); force-launch; Frida attach; timed session; uninstall; write

metadata. Optional strace is implemented; skip is recorded in metadata. Timed Monkey as an arm is implemented but not exercised.

Sessions of the same app are independent of *that package’s app data* when `pmclear` succeeds. They are not AVD resets. `pm_clear_rc=0` on all 168 original usable sessions and all 172 extend usable sessions. Fairness/strict-clean-start fails the session if the clear returns non-zero. `snapshot_restored=true` in metadata does not prove a snapshot load. If `CONTEXTDROID_SKIP_SNAPSHOT_LOAD` is set and `sys.boot_completed==1`, `_restore_snapshot` returns `True` without loading. The extend launch exports that skip flag; all 172 extend usable sessions still record `snapshot_restored=true`. That bounds AVD-level reset, not app-data isolation: every usable session cleared its target package successfully. Persisting across sessions on the same AVD, with no `-wipe-data`: emulator userdata, other installed packages (isolation force-stops them but does not uninstall them), device-level accounts, and external-storage files not owned by the cleared package. This is the cross-session isolation that Chapter C’s per-app reference analysis conditions on; it is a protocol fact, not a finding of that chapter (§C.1). `-wipe-data` is implemented on the malware AVD path and was not exercised for v2.

B.2.5 Session mode schedule

Collection is configured as `SESSIONS_PER_APP=3` with `SESSION_MODE_SCHEDULE=identical, identical, varied`. Identical-mode sessions of an app are intended to share `agent_seed`. Table B.3 is the usable-set census. `session_mode` and `agent_seed` are collection-metadata fields; they are absent from the export `metadata.json` schema and were read from `source_meta_path`.

Table B.3. Usable sessions by `session_mode`. Canary usable sessions have no `session_mode` field and are counted as `MISSING`, not dropped. Identical-mode sessions occur only in the original batch.

batch	identical	varied	MISSING
original	112	56	0
extend	0	172	0
canary	0	0	2
pooled usable	112	228	2

112 of 342 usable sessions (0.327) are identical-mode replicates, all in the original batch. Original slots follow the schedule (s1 identical, s2 identical, s3 varied: 112 / 56). The extension added no identical-mode sessions: every usable extend slot is labelled `varied`. The pooled set is 0.327 replicated, not two-thirds.

Among apps with at least two usable identical-mode sessions ($n=54$ apps, 54 pairs), every pair shares `agent_seed`; none differ. Distinct `screen_hash` overlap on those pairs has median Jaccard 1.000, IQR [0.889, 1.000].

Table B.4 reports within-app pairwise graph distance on the Chapter B sequence-proximity graphs (k-burst 5, zero static stub), flattened as the Chapter C w_{cum} session vector (node block plus outgoing-share adjacency). Cosine is a distance ($1 - \text{similarity}$). Identical-identical pairs sit at median cosine ≈ 0 and median Frobenius 0. Two contrasts on that combined cosine, both small and material under the declared $p < 0.05$ and $|\delta| \geq 0.147$ rule: identical-identical versus identical-varied, Mann-Whitney $U=8952$, $p=0.00343$, Cliff $\delta = -0.243$; identical-identical versus varied-varied, $U=9820.5$, $p=0.000235$, $\delta = -0.303$ (varied-varied median 7.18×10^{-5}). The matching Frobenius test of identical-identical versus identical-varied is $U=8680$, $p=0.00115$, $\delta = -0.266$.

Table B.4. Within-app pairwise graph distance by session-mode pair type, usable sessions ($n=342$ tensorised). Cosine is $1 - \text{similarity}$ on the combined flattened vector and is reported in scientific notation: four-decimal convention would print 0.0000 for every cosine median. Frobenius uses four decimals. Eleven of 54 identical-identical pairs are both zero-edge, which forces distance 0.

pair type	n pairs	cosine med.	Frobenius med.	both 0-edge
identical-identical	54	2.2×10^{-16}	0.0000	11
identical-varied	438	3.21×10^{-5}	0.0207	22
varied-varied	522	7.18×10^{-5}	0.0305	0

Identical-mode pairs of the same app are near-replicates: they share a seed, they see essentially the same screen set, and their graphs sit at distance 0. They are not independent observations, and Chapter C must treat them as such. The contrast with varied pairs is real but small. Absolute distances are tiny for every pair type because the graphs are sparse, and 11 of 54 identical pairs are both zero-edge; there is little structure to differ over.

B.2.6 Quality gates

Gate validation surfaced findings before and during collection: metrics that could not see engine skips or explore-blind ratios, and a weakened flailing rule (§B.2.1), were acted on before they entered as results. A session enters the exported analysis set only if all eight conjuncts hold:

```
analyze_ok ∧ sim = success ∧ faith ∈ {FAITHFUL, PARTIAL}
  ∧ c0_pass ∧ meaningful_22 > 0 ∧ ¬flail ∧ ¬auth_gated ∧ ¬network_degraded.
```

`c0_pass` requires at least three named functional explore taps *or* at least two functional explore screen hashes. `meaningful_22` counts Frida events whose category lies in the 22-node universe and whose API is not `hook_loaded` or `Method.invoke`. The notes file restates C0 with the word “effective”; the predicate uses the named-tap and screen-hash counters, not that adjective.

Flailing is `quality_rules.detect_suspect_flailing`, six sub-rules plus empty-action cases: `mechanical_majority` ($\geq 50\%$ of successful actions in `{back, wait, swipe, scroll}` and named functional taps < 3); `explore_back_wait_dominant` (explore back/wait ratio > 0.50 and explore functional taps < 3); `dominant_screen` (execute+primary hash dominance ≥ 0.80 and named < 3); `same_element_cycle` (session hash dominance ≥ 0.80 , named ≥ 3 , and top named target $\geq \max(3, 75\% \text{ of named})$); `low_all_phase_direct_ratio` (`sim=success` and all-phase direct ratio < 0.40).

An earlier `_flailing_new` dropped `same_element_cycle` and guarded `dominant_screen` when named taps were ≥ 3 . The remediation record states that the weakened rule would have admitted 12 flailing sessions. Those rules were merged back into a single `detect_suspect_flailing`. The count of 12 is plan-sourced (`remediation_plan.md`), not re-counted on v2. This is the same class of finding as Chapter A’s corpus artefacts (§A.7): validation caught what the gate was meant to reject before it entered as a result.

B.2.7 Safety posture

v2 is a benign corpus. Malware-tier controls exist in code and did not run because no malware was collected. Table B.5 records each control’s three-state status; the exercised column distinguishes controls on the benign launch path from malware-tier controls that were not required for this corpus.

Table B.5. Safety controls on the v2 benign collection. “Exercised” means the v2 launch path invoked the control, not that a malware sample was presented to it. Malware-tier rows read “not required” because v2 is benign-only, not because the control is unimplemented. Device-guard *pass* is not a per-session metadata field; only failure sets `analysis_status`.

control	implemented	exercised for v2
<i>Benign collection path</i>		
Device guard (hard assert, watchdog)	yes	yes — 0 <code>failed_device_guard</code> in 388
AVD isolation, force-stop others	yes	yes
<code>pmclear</code> per session	yes	yes
<i>Malware tier (not required for v2)</i>		
Network sink + guest iptables DNAT	yes	not required
<code>gate_p4</code> preflight	yes	not required
<code>-wipe-data</code>	yes	not required
<code>adb_pinned.sh</code>	yes	not required
Encrypted vault	yes	not required
<i>Design only</i>		
Host <code>pfctl</code> backstop	design only	design only

Device-guard *pass* is not recorded per session. The check is the absence of `failed_device_guard` from the 388-session status census.

B.2.8 What ContextDroid is not

ContextDroid did not collect malware for v2. The malware-handling infrastructure—network sink, `gate_p4`, `-wipe-data`, pinned adb, encrypted vault—is implemented and was not exercised (Table B.5). Chapter A’s malware evaluation uses a public corpus for the sensor-symmetry reason in §A.2.1, not because ContextDroid lacked a malware path on disk.

No detector is trained in this chapter, and no discrimination metric is reported. v2 is a benign stimulus-and-representation study.

B.3 The v2 corpus

The v2 export is a benign ContextDroid collection: 388 indexed sessions, 342 of which pass the eight-conjunct reference tier of §B.2.6. This section records the analysis population, the July/August extension and whether those sessions may be pooled, and the recoverable versus UNRECOVERABLE collection identity.

B.3.1 Population

Table B.6 is the session census. Original collection contributes 168 sessions, all of which pass. The August canary and extension contribute 220 further indexed sessions, of which 174 pass and 46 fail. Usable $n=342$ is original pass plus new pass ($168+2+172$).

Table B.6. v2 session inventory by collection batch. Usable means all eight reference-tier conjuncts hold (§B.2.6).

batch	indexed	usable	fail
original	168	168	0
canary	3	2	1
extend	217	172	45
all	388	342	46

Fifty-nine apps have at least one usable session. Graph eligibility is stricter: a session is graph-eligible when its timed export graph has at least two active nodes and at least one edge; an app is graph-eligible when at least one of its usable sessions meets that floor. The floor is evaluated on export `build_session_graph` ($k=5$, $\delta=5$ s): an app counts as analysable only on edges that survive the temporal filter, so the population is not inflated by transitions δ would reject. Under Run 2 `update_graph_sequence` (no δ) the same floors would admit 46 graph-eligible apps and 307 eligible sessions rather than 40 and 288; six additional apps would qualify (`ademar.bitac`, `app.alextran.immich`, `at.manuelbichler.octalsuntime`, `at.tomtasche.reader`, `biz.bina`, `rysolutions.stress`, `br.odk.knights`). This chapter retains the δ -filtered counts; Chapter C’s per-app design runs inherit the 40-app graph-eligible gate (§C.2) while building E4 session graphs with Run 2 sequence (§B.4.2). Forty apps are graph-eligible on the original usable set, and the same forty remain eligible after the extension (none enter, none leave). Those 40 apps hold 290 of the 342 usable sessions and are the graph-eligible population. Analysis n is stated per section. Median usable sessions per graph-eligible app is 8.

The pre-extension GAE snapshot campaign on dataset v2 (`norm_ab_v2`) materialised 731 total snapshots, 565 GAE-eligible, edge sum 2888.0, mean 5.112 edges per eligible snapshot (`abrg/output/norm_ab_v2/comparison.json`); that 60s-window builder is distinct from the whole-session `update_graph_sequence` counts in §B.4.2.

The two eligibility levels are not interchangeable. Of 342 usable sessions, 288 meet the session-level floor; 54 do not. Fifty-two of those 54 belong to the nineteen apps that never meet the floor; the remaining two are `at.linuxtage.Eventfahrplan` extend sessions `84bca8b5903c_11m_s4` and `..._s5`, which have $n_{\text{active}}=2$ and $n_{\text{edges}}=0$ but sit on an otherwise-eligible app. That app remains eligible through its other seven usable sessions ($n=9$ usable in total).

Table B.7 is the full per-app session-count distribution, not a top- k slice. The 59-app view (median 7) is the inventory from which the analysis population was selected. Counts {7:5, 8:16, 9:10} are 31 apps, all inside the graph-eligible 40; they are not the eligible set.

Table B.7. Usable sessions per app after the extension. The 59-app column is every package with ≥ 1 usable session; the GAE-40 column is the graph-eligible analysis population.

usable sessions	apps (59)	apps (GAE-40)
1	2	0
2	1	0
3	21	5
5	2	2
6	2	2
7	5	5
8	16	16
9	10	10
apps	59	40
sessions	342	290
median sessions / app	7	8

The distribution has no app with exactly four usable sessions. The original schedule is three sessions per app, and no package received exactly one additional usable session from the extension.

Nineteen apps, 52 usable sessions, never meet the floor—the apps at one, two, or three usable sessions in Table B.7 that sit entirely outside the GAE-40 column (2+1+16 apps; 52 sessions). All 19 are original-only: they received no canary or extend slots. On every usable session, $\max n_{\text{edges}}=0$. Some reach $n_{\text{active}}=2$ and still form no edge. That is a coverage finding about the stimulation policy, not only an exclusion rule: 19 of 59 apps with usable sessions (0.322) produce no transitions at all. Graph eligibility is a property of the apps, not a filter applied downstream: §B.4’s per-app comparison pools all 59 apps with usable sessions, and the 19 never-eligible apps contribute zero-edge graphs to those distributions. They are part of the sparsity result, not omitted from it.

Before the extension the same 59 apps already had 168 original usable sessions (three apps with one, three with two, 53 with three). The August slots were added only on the graph-eligible 40 ($20 \text{ apps} \times 6 \text{ new slots}$ and 20×5 , totalling 220, equal to 3 canary +217 extend).

B.3.2 The extension and poolability

Original sessions were collected in July 2026; the canary and extension in August 2026. Whether the 342 usable sessions may be treated as one sample has two parts: batch heterogeneity, and within-app replication. Chapter C inherits both.

Batch heterogeneity. Table B.8 compares original usable sessions ($n=168$) with new usable sessions (canary + extend, $n=174$). Graph-structure rows (active nodes, edges, density) use the timed export builder `build_session_graph` on per-session metrics (`run1_corpus/old_vs_new.json`); they are not Run 2 `update_graph_sequence` medians (5.0 edges on whole sessions; §B.4.2). The materiality rule is declared before the numbers: $p < 0.05$ and Cliff $|\delta| \geq 0.147$ (Romano et al. [33] small threshold). Wall duration does not differ ($U=13867$, $p=0.413$, $\delta = -0.051$). Mapped events, total events, active nodes, edges, and density do, all at small effect size ($|\delta|$ from 0.160 to 0.287). Mapped-event spread is wide in both batches (original IQR [10, 199]; new IQR [48, 268]). New sessions run as long as original ones and yield more mapped events and slightly denser graphs; the batches are not replicates.

Table B.8. Original usable versus new usable (canary + extend). Medians are the interpolated p_{50} of `_summarize_dist` (linear interpolation on the order statistics; even n is not the upper central order statistic). U , p , and Cliff δ are from the companion Mann–Whitney records. Material: $p < 0.05$ and $|\delta| \geq 0.147$. Wall duration is the only row that fails the rule. Active nodes, edges, and density: export

build_session_graph per session (not Run 2 sequence).					
metric	orig.	new	U	p	δ
mapped events	57.5	119	10876.5	4.29×10^{-5}	-0.2558
total events	385	701.5	12277.5	0.0105	-0.1600
active nodes	2	3	11323	1.95×10^{-4}	-0.2253
edges	2	2.5	10421.5	3.38×10^{-6}	-0.2870
density	0.0043	0.0054	10421.5	3.38×10^{-6}	-0.2870
wall duration (s)	460.721	462.067	13867	0.413	-0.0512

Canary. Three apps received one new session each in August 2026: `app.comaps.fdroid`, `app.organicmaps`, and `ai.susi`. Two recorded verdicts exist. Under the pre-declared band—mapped events in $[0.25 \times, 4 \times]$ of that app’s existing median, no novel category, no silent-reliable category—all three pass (mapped 153, 125, 348 against medians 157, 142, 408). Under existing min–max

ranges the same three fail as a set. `app.comaps.fdroid` is outside on edges (6 vs 7–8) and elapsed time (226.6 s vs 455.9–472.2 s). `app.organicmaps` is outside on total events (867 vs 939–1004) and mapped events (125 vs 142–145). `ai.susi` is outside on total events (5048 vs 5108–7082), mapped events (348 vs 401–421), and elapsed time (544.6 s vs 466.3–474.7 s).

The operative criterion is the $0.25 \times 4 \times$ mapped band. Min-max over two or three prior sessions is an extremely tight bound on a corpus whose session-to-session mapped-event IQR on the original usable set is [10, 199]. The identity-check canary and the reference-tier gate are distinct: the band canary is a PASS, while `app.comaps.fdroid s1` still fails the eight-conjunct tier (`partial:bad_handoff`) and is one of the 46.

Replication structure. The session-mode census of §B.2.5 is part of the poolability argument, not only a protocol note. Of 342 usable sessions, 112 are identical-mode (0.327), 228 varied, and 2 MISSING (the two usable canaries, which lack a `session_mode` field and are not dropped). All 112 identical-mode sessions are original; every usable extend session is varied (172/172). Original slots follow the configured schedule (112 identical / 56 varied). Among 54 apps with at least two usable identical-mode sessions, all 54 pairs share `agent_seed`.

Those 112 are near-replicates: same seed, median screen-hash Jaccard 1.000, median pairwise graph distance at 0 on the combined cosine and Frobenius of §B.2.5. They are not independent draws of the app. The 342 usable sessions—and the 290 on graph-eligible apps—are therefore a mixture of two collection windows whose five structural metrics differ, nested inside apps that contribute seed-sharing identical-mode pairs in the original window only. Chapter C’s unit is the session; it inherits both the batch mixture and the replication.

Failure concentration. The 46 reference-tier failures fall on 13 of the 40 extension apps (median fail rate 0 among the 40, IQR [0, 0.333]). Maximum failures on one app is 6; among apps with at least one failure the median count is 3. Four apps failed every new slot and remain at their three original usable sessions: `app.fedilab.mobilizon`, `app.ladefuchs.android`, `app.udderance`, and `cc.kafuu.bilidownload`. Their Chapter C curves are those three sessions.

Table B.9 lists failure-family counts. The families are the eight-conjunct gate of §B.2.6, not `analysis_status` alone: 344 sessions record `analysis_status=success` against 342 usable, and `flag:webview_dominant` appears on 9 sessions while the fail-family `webview_dominant` count is 6 (the extra three are original `app.udderance` sessions that passed the reference tier). Fail-family counts are the ones used here.

Table B.9. Reference-tier failure families ($n=46$). One canary session is included: `app.comaps.fdroid s1`, `partial:bad_handoff`.

family	n
<code>partial:bad_handoff</code>	14
<code>partial:ux_quality_gate</code>	12
<code>partial:no_goal_progress</code>	7
<code>webview_dominant</code>	6
<code>flailing:dominant_screen</code>	4
<code>failed_frida_reattach</code>	2
<code>flailing:same_element_cycle</code>	1

Two extend sessions of `anonvpn.anon_next.android (s4, s5)` are the only non-zero analysis exit codes in the export (exit 12, `failed_frida_reattach`), matching §B.2.2.

B.3.3 Provenance

Collection identity is recovered from session metadata, launch scripts, and the operator override. The hook script SHA256 (`3192c7d6...`) is identical across both collection windows—168 original and 172 extend usable sessions—so the instrumentation did not change between them. Hook version tag 3, session duration 420s, planner `llama3.2`, emulator API 29, extend/canary AVD `abrg_benign`, and the reset protocol (`snapshot_restored=true` plus `pmclear`) are all recorded.

For the July window—168 of 342 usable sessions—several environment fields were not captured and are UNRECOVERABLE: Frida client version, emulator system image, July AVD name in session metadata, LLM planner digest, frozen `prompts.py` SHA, adb/platform-tools pin, Python and pip pins, host OS, and hook git commit. The August window has full environment provenance.

Frida server version for July is established by inference rather than record. Client 17.9.3 was installed 30 April 2026 with no upgrade trail through the July window; Frida enforces client/server compatibility at attach, and July sessions attached, so the server was 17.9.3-compatible; no frida commands appear in July shell history; current `tools/frida-server-android-arm64` strings report 17.9.3. An operator override note of 14 August 2026 applies only to frida-server version; the 5 August 2026 mtime on the checked binary is a file write, not a verified content change.

Reproducibility notes. Distribution medians in this chapter are the interpolated p_{50} written in `SUMMARY.md` (linear interpolation; `abrg.androct.inventory._percentile`). Companion JSON Mann–Whitney records also store `median_x` / `median_y` as the upper central order statistic (`sorted[n//2]`), which disagree at even n (mapped original 57.5 vs 59; edges-new 2.5 vs 3.0). Tests are taken from those records; reported medians are the interpolated figures. The same helper names those JSON fields `benign_median` and `malware_median`; the comparison is original versus new (canary + extend), not a malware contrast. The same helper in §B.4.5’s comparison JSON names `n_benign` = 59 (v2 pooled) and `n_malware` = 2231 (AndroCT parsed benign); that is not a malware contrast either. They apply to every distribution table in this chapter and are stated as conventions in §B.1.3.

B.4 The representation comparison

This section compares v2 graphs with AndroCT benign traces on the same 22-node universe and the same sequence builder. Two AndroCT populations appear and are never interchangeable: graph, fire-rate, and yield columns use the parsed-non-empty archive ($n=2231$ benign), while static features use Chapter A’s graph-eligible tensor cache ($n=703$ benign). The 2231 includes apps that never entered Chapter A’s split; the 703 has been through fetch, VT-resolution, and mapped ≥ 1 filters that dropped 1,331 unresolvable package-named traces. No detector is trained.

B.4.1 Units

AndroCT’s unit is one whole-trace graph per app. v2 reports two units and does not conflate them: per-session ($n=342$, Chapter C’s unit) and per-app pooled ($n=59$). The 59 are every app with at least one usable session, including the 19 never-eligible apps of §B.3.1, which contribute zero-edge graphs.

Pooling is concat-then-build. Each app’s usable sessions contribute their mapped category streams in `session_index_within_app` order (start-time order); one `update_graph_sequence` call ($k_{\text{burst}}=5$) is made on the concatenation. That matches AndroCT’s unit: one ordered whole-

trace category stream, one graph. Session graphs are not built and then merged; no edge-weight combination rule is applied.

B.4.2 Trace composition

The materiality rule is the same as §B.3.2: $p < 0.05$ and Cliff $|\delta| \geq 0.147$ [33]. Tests in Table B.10 compare per-app pooled v2 ($n=59$) with parsed AndroCT benign ($n=2231$). Session medians are descriptive.

Both sides of Table B.10 use the same builder: `graph_from_categories` $\rightarrow$ `update_graph_sequence` at $k_{\text{burst}}=5$ with zero static and no δ or recency; v2 pooled graphs concat per-app session streams first, AndroCT parsed-benign graphs apply it to one whole trace per app. Unit, sensor, and population differ: multi-session concat versus a single 600 s Monkey trace, ContextDroid Frida hooks versus DroidFax logcat, and 59 apps versus 2231 parsed traces.

Pooled mapped-event rate is 0.204 against AndroCT’s 0.026 (roughly eight times higher, large effect) on graphs that are sparser on every structural measure (v2 pooled median 2 active nodes and 2 edges from Run 2 concat-then-build against AndroCT parsed 6 and 16, both large). Absolute mapped counts do not differ materially ($p=0.150$, $\delta=0.110$) while v2 total events are about a quarter of AndroCT’s (1703 vs 6878, small effect). Thirty-two of 59 pooled graphs have at most two edges (0.542); 19 of those 59 are the never-eligible apps with no transitions at all (§B.3.1). That is one trade under this policy: relevance up, breadth down — not a mapped-rate win with a sparsity caveat attached afterwards.

Table B.10. Trace composition. v2 session $n=342$; v2 pooled $n=59$ (all apps with ≥ 1 usable session, including 19 zero-edge graphs). Session edge/active-node medians: Run 2 `update_graph_sequence` ($k=5$, whole session). Pooled edge/active-node medians: Run 2 concat-then-build on the same builder. Export-index `build_session_graph` medians are 2.0 edges on all 342 sessions (not in columns). AndroCT benign parsed-non-empty ($n=2231$): Run 2 `update_graph_sequence` on one whole trace per app (same builder and pins as v2 pooled; not Chapter A’s graph-eligible $n=703$). Tests are pooled versus AndroCT.

Medians are interpolated p_{50} . Material: $p < 0.05$ and $|\delta| \geq 0.147$.

metric	v2 sess.	v2 pooled	AndroCT	test (pooled)
mapped events	84	372	185	$p=0.150$, $\delta=0.110$
total events	524	1703	6878	$p=2.09 \times 10^{-4}$, $\delta=-0.282$
mapped rate	0.2436	0.2043	0.0259	$p=9.19 \times 10^{-14}$, $\delta=0.568$
active nodes	3	2	6	$p=3.61 \times 10^{-12}$, $\delta=-0.527$
edges	5	2	16	$p=9.08 \times 10^{-11}$, $\delta=-0.493$
≤ 2 edges	158/342	32/59	455/2231	—

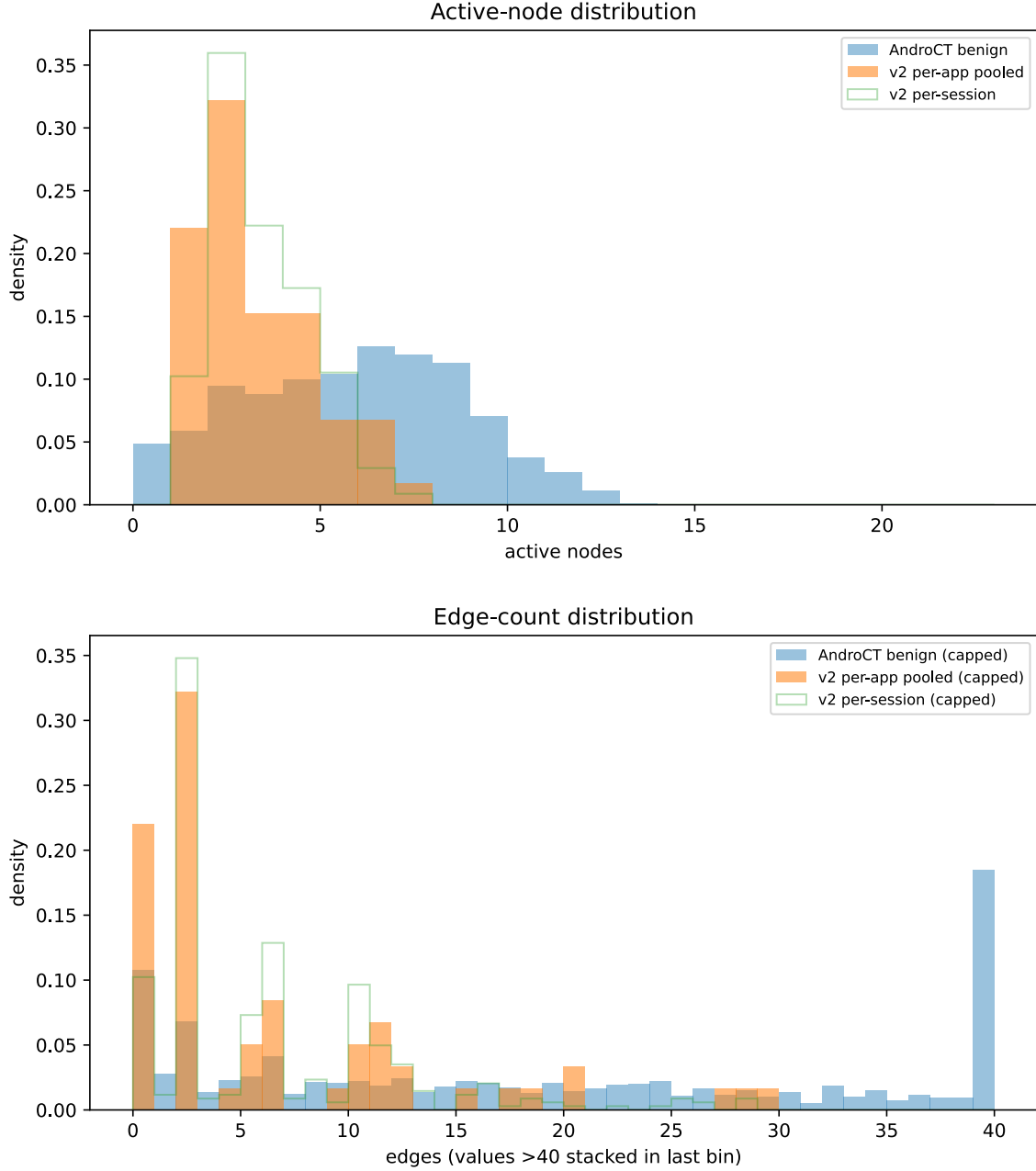


Figure B.1. Active-node and edge-count densities. AndroCT benign parsed-non-empty ($n=2231$; Run 2 `update_graph_sequence` on one whole trace per app), not Chapter A’s graph-eligible $n=703$. v2 pooled $n=59$ (Run 2 `concat-then-build`); v2 session $n=342$ (Run 2 `update_graph_sequence`). Edge counts >40 are stacked in the last bin. v2 is compressed against AndroCT’s long tail, not a location shift of the same distribution.

The medians in Table B.10 do not show that shape. Figure B.1 does: AndroCT has a long tail; v2 sits in a narrow low band on both axes.

Three v2 edge builders coexist and are not interchangeable. `build_session_graph` on the timed export index yields median 2.0 edges ($n=342$ sessions). Run 2 `update_graph_sequence` on whole sessions yields median 5.0 edges ($n=342$). The original `norm_ab_v2` 60s-window GAE campaign yields median 3.0 / mean 5.112 edges per GAE-eligible snapshot ($n=565$ snaps; §B.3.1).

E4 `provenance.json` records 194/342 sessions whose export graph differs in edge count from the Run 2 sequence graph (148/342 agree); the mechanism is the $\delta=5$ s filter §B.5 measures at pair level — export applies it on ordered pairs inside the k -window and `update_graph_sequence` does not, so δ can only remove candidates. On every mismatched session Run 2 sequence edges are never lower than export (median surplus +2, range +1 to +15); active-node sets match on all 342 sessions. Edge statistics must name the builder before cross-section comparison.

Builder audit (Chapter B v2 density figures). Table B.10 session columns: Run 2 sequence (median edges 5). Pooled columns: Run 2 concat (median edges 2). Figure B.1: same. Table B.8 edges (2 / 2.5): export `build_session_graph` (§B.3.2). §B.3.1 graph-eligibility rule uses export timed graphs (40 apps, 288 sessions; counterfactual 46/307 under Run 2 sequence). §B.5 edge retention: timed export builder. §B.7: E4 Phase 0/−1 corpus pin; design runs in §C.1.

The AndroCT 6-node / 16-edge medians are the parsed population. They are not the graphs Chapter A’s GAE fitted. Train-benign graphs there have median 5 nodes and 14 edges ($n=562$); test-benign has 6 and 16 ($n=141$). A pooled median on the 703 eligible benign apps was not computed. v2 pooled (2 and 2) is sparser than all three; that comparison is taken up in §B.6.1.

The mapped-rate side of that trade cannot be attributed to LLM planning. BFS navigation is 0.711 of pooled usable *steps* (13,029 / 18,315); per-session median 0.732, IQR [0.688, 0.759], stable across batches (original 0.711, extend 0.712). That is not 0.711 of mapped events. Frida `Date.now()` and `host.time.time()` are unsynchronised (§B.2.2), so mapped events cannot be joined to action-step windows; half of execute-phase steps, moreover, perform no device interaction (§B.2.3). The step fraction does not license the claim that BFS produced 71% of the mapped signal.

B.4.3 Category coverage

Table B.11 is the per-app fire rate (fraction of apps with at least one mapped event in the category) for all 22 categories, ranked by absolute difference from parsed AndroCT benign. v2 denominator is 59; AndroCT is 2231. Figure B.2 is the same rates as grouped bars.

Table B.11. Category fire rate (apps with ≥ 1 mapped event), all 22 categories. v2 $n=59$; AndroCT benign parsed-non-empty $n=2231$. Left column ranks 1–11 by $|v2 - \text{AndroCT}|$; right column 12–22.

$\Delta = v2 - \text{AndroCT}$.							
category	v2	And.	Δ	category	v2	And.	Δ
package_manager	0.0169	0.7517	−0.7347	media	0.0000	0.0332	−0.0332
native_code	0.1525	0.7808	−0.6283	audio	0.0339	0.0031	+0.0308
ipc_intents	0.3051	0.6800	−0.3749	content_access	0.0508	0.0717	−0.0209
process	0.0339	0.3537	−0.3198	location	0.0169	0.0323	−0.0153
crypto	0.2203	0.5137	−0.2933	notifications	0.0339	0.0188	+0.0151
storage	0.7458	0.5365	+0.2092	accounts	0.0000	0.0126	−0.0126
device_info	0.0000	0.1744	−0.1744	camera	0.0169	0.0076	+0.0093
network	0.2034	0.3501	−0.1467	dynamic_code_loading	0.0000	0.0018	−0.0018
webview	0.1017	0.2030	−0.1014	clipboard	0.0000	0.0004	−0.0004
file_io	0.7119	0.6531	+0.0588	telephony	0.0000	0.0004	−0.0004
database	0.1525	0.1972	−0.0447	sms	0.0000	0.0000	0.0000

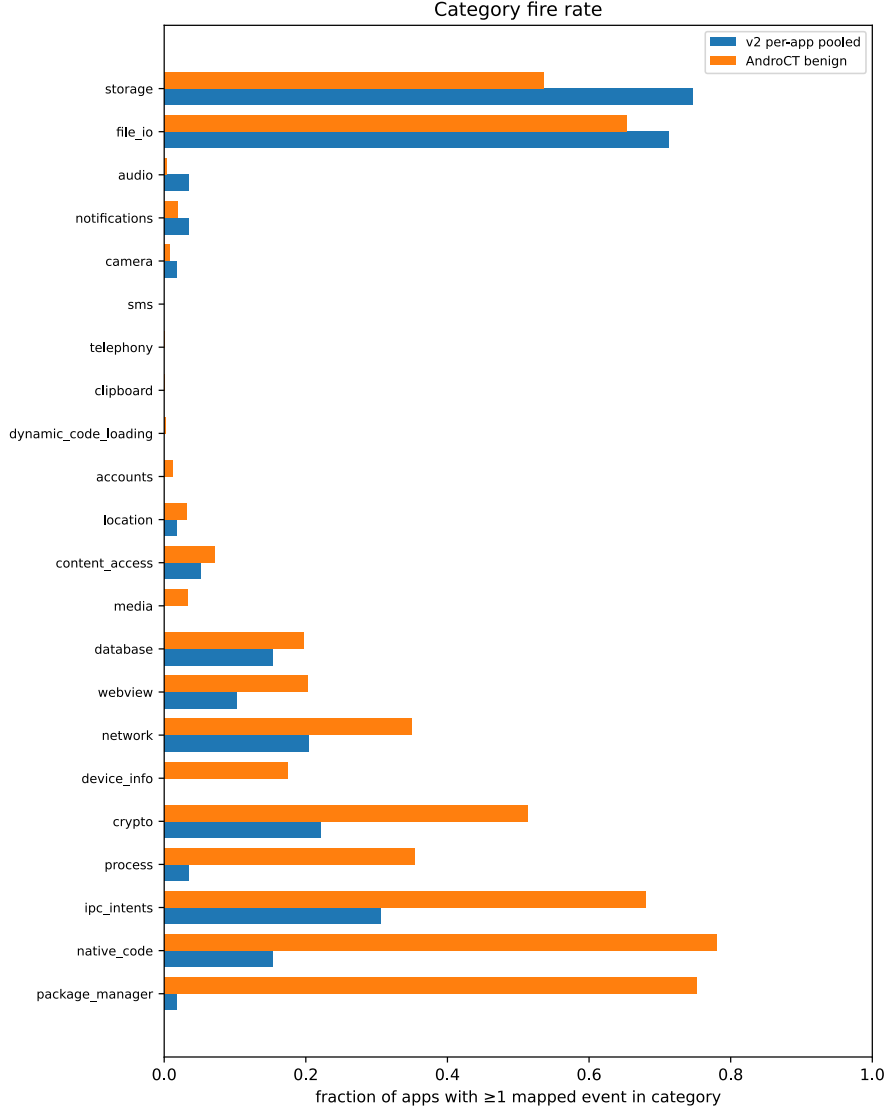


Figure B.2. Category fire rate, all 22 categories. v2 per-app pooled ($n=59$) against AndroCT benign parsed-non-empty ($n=2231$).

The largest gaps are categories that fire under high-volume Monkey and almost never under guided traversal: `package_manager` and `native_code`. Five categories have a positive difference. `storage` (+0.209) and `file_io` (+0.059) are the two that matter in magnitude; `audio`, `notifications`, and `camera` are +0.031, +0.015, and +0.009. Those two file-side excesses are a small positive inside a section of negative differences: guided exploration reaches file operations that random tapping does not. The right-hand column is the similarities: several categories fire at nearly the same rate in both corpora, and the dead seven sit at or near zero on both sides. `ipc_intents` sits in between and needs its own note: it carries Chapter A’s entire benign-only D_1 signal (§A.6.7 — zeroing that block drops floor AUC from 0.800 to 0.591) and fires on 18 of 59 v2 apps against 1517 of 2231 AndroCT benign apps. What that implies for transferring the headline is in §B.6.3.

Seven categories are dead on v2 pooled apps (0/59). Four are unreachable from the planner action space `_ACTION_TYPES_PLANNER = \{tap, input, back, wait, advance_goal, swipe\}`: `sms`, `telephony`, `clipboard`, and `dynamic_code_loading`. No SMS, call, clipboard, or dex-loading action exists in that set. Those four are unreachable by construction, not unreached by chance.

`sms` is also dead in both AndroCT classes under Monkey (0/2231 benign, 0/1736 malware). Two stimulus policies, same outcome; the v2 mechanism is identified in the action set (§B.6.2).

The other three dead categories — `accounts`, `device_info`, `media` — are not explained by that frozen set. The spec marks their non-firing UNDETERMINED (whether the AVD has accounts; identifier APIs on API 29 vs emulator null; media hardware/codec). The action-space story does not absorb them. `camera` is not dead: it fires on 1 of 59 v2 apps (Camera2 `CameraManager.openCamera`; Camera1 `Camera.open` never fired).

B.4.4 Static features

Static slices are compared on Chapter A’s graph-eligible tensors, not on the parsed 2231. All 59 v2 apps resolve; 0 are all-zero. AndroCT benign: $n=703$, 0 all-zero. Static-vector L_2 median is 7.664 on v2 (IQR [5.352, 15.335]) against 9.731 on AndroCT (IQR [5.970, 18.034]).

Table B.12 records per-app mean coordinates. Dangerous and signature permission gates and the epoch slot sit at 0 in both corpora at the median. Declared-permission occupancy is lower on v2 (0.636 vs 0.773). These are APK-derived features, not stimulus features; they confirm that the v2 packages are not an empty static set, not that ContextDroid exercised more of the graph.

Table B.12. Static-slice per-app means. v2 $n=59$ (all resolved, 0 all-zero). AndroCT benign is Chapter A graph-eligible tensors ($n=703$, 0 all-zero), not the parsed $n=2231$ of Tables B.10–B.11. Medians are interpolated p_{50} .

coordinate	v2	AndroCT $n=703$
s_v	0.4091	0.4273
<code>declared_v</code>	0.6364	0.7727
<code>gate_v[0]</code> normal	0.0909	0.1364
<code>gate_v[1]</code> dangerous	0.0000	0.0000
<code>gate_v[2]</code> signature	0.0000	0.0000
<code>reach_v</code>	0.4091	0.5909
<code>epoch_v</code>	0.0000	0.0000

B.4.5 Event yield

Three candidate explanations for lower absolute volume, and what separates them.

Narrower hook coverage. Twenty-two of 58 hooked API names never fire corpus-wide (§B.2.2, Table B.1). The spec’s hook-level inventory (its §9.1) records reasons only where code or docs state one. Several are overload or generation misses rather than missing coverage: `Camera.open` (Camera1) is silent while `CameraManager.openCamera` (Camera2) fires; `Cipher.getInstance` is silent while `doFinal` fires; `FileOutputStream.<init>` is silent while `FileInputStream.<init>` fires. Those are not coverage failures of the 22-node universe.

Fewer surfaces. Distinct `screen_hash` values are absent from the export and were counted from collection `*_llm_actions.jsonl`. Usable sessions have median 5 distinct screens, IQR [4, 12] ($n=342$). Spearman correlation with mapped events is $\rho=0.250$, $p=2.90\times 10^{-6}$ on the pooled set. It does not replicate by batch: original $\rho=0.407$, $p=4.21\times 10^{-8}$ ($n=168$); extend $\rho=0.058$, $p=0.447$ ($n=172$). Screen variety tracks mapped volume on the July sessions and does not on the August ones — another batch-heterogeneity fact, not a pooled yield law.

Lower event density. v2 mapped events per second have median 0.184 (IQR [0.058, 0.545], $n=342$) against AndroCT 0.308 (IQR [0.055, 1.422], $n=2231$). The AndroCT denominator is the stated 600s protocol; those traces carry no wall-clock. v2 mapped events do not correlate with observed session wall duration ($\rho=0.074$, $p=0.173$, $n=342$): more time does not produce more mapped events, which points at the policy rather than the budget.

Reproducibility note. Mann–Whitney records in `run2_comparison/comparison.json` reuse the inventory helper: `n_benign` is 59 (v2 pooled) and `n_malware` is 2231 (AndroCT parsed benign). Those keys are not a malware contrast. The same trap appears in §B.3.3 for original versus new. Those names are left as emitted; the records are read via `n_x` / `n_y` and `label_x` / `label_y`.

B.5 The temporal filter δ , measured

Chapter A records that AndroCT traces carry no timestamps, so δ is unexercised there, and points here for the cost on a timestamped corpus (§A.4.3). v2 traces have timestamps. This section measures that cost. The graphs compared in §B.4 were built with `update_graph_sequence` and did not apply δ ; the measurement here is on the timed builder `abrg.graph.update_graph` ($k=5$, $\delta=5$ s) that Chapter C uses. The same filter appears at graph level in §B.4.2: 194/342 sessions differ in edge count between export and Run 2 sequence builders (148 agree), always with sequence $\geq$ export, median surplus two edges, and identical active-node sets.

In the full design, an ordered pair of mapped events is an edge candidate only if both conjuncts hold: sequence distance at most k and wall-clock gap at most δ . Self-transitions are not candidates. The filter can only discard pairs; the gap $t_v - t_u$ is a predicate, not a tensor coordinate. Retention is the fraction of k -candidates that also satisfy δ . That is a pair count, not a unique-edge count: the same category pair may occur many times, and each occurrence is counted. Chapter A’s deferral asked for that fraction, so that is what is reported.

The population is every usable session: $n=342$ sessions across 59 apps. It is not the Stage 2 gate of 35 apps with ≥ 5 usable sessions, and it is not restricted to GAE-eligible graphs. Overall retention is pair-pooled ($\sum n_\delta / \sum n_k$ over those 342 sessions). Thirty-five of the 342 usable sessions ($\approx 10\%$) have $n_k=0$ (no hetero-category pair inside the k -window); they contribute nothing to the pooled ratio—a different “35” from the Stage 2 app count above.

B.5.1 Retention on v2

Retention is strongly stratified by session mapped-event count (Table B.13): rank-based quartiles, 86/85/86/85 sessions, run from 0.272 in Q1 (1–26 mapped events) to 0.905 in Q4 (254–1941). Q3 and Q4 both list 254 because nine sessions sit at that count and the split is by rank, not by a unique value cut. Q4 holds 0.793 of all k -candidates (72,831 / 91,816), so longer traces dominate the pair pool. That stratification is the headline finding: δ survival is a function of trace length, not a single corpus constant.

Pair-pooled retention—the answer to Chapter A’s deferred fraction—is 0.838 (76,970 of 91,816 k -candidates), a long-trace number for the reason above. An unweighted mean of the 307 session retentions with $n_k \geq 1$ is 0.489, showing what a typical session retains and why it differs from the pooled figure. The fitted curve below drops the 35 $n_k=0$ sessions and is fit on those 307.

Table B.13. Pair retention: of $k \leq 5$ hetero-category candidates, the fraction that also satisfy $\delta \leq 5$ s. Population: all usable sessions ($n=342$, 59 apps), not the 35-app ≥ 5 gate. Quartiles are rank-based on mapped-event count; n_{events} bands may touch. Retention in each row is pair-pooled inside the bin.

quartile	mapped n	n sess.	n_k	retention
1	1–26	86	1402	0.2725
2	29–80	85	5102	0.4463
3	88–254	86	12481	0.6719
4	254–1941	85	72831	0.9052
all	—	342	91816	0.8383

The measurement is on the full-session mapped stream. Chapter C then splits that stream into 60 s processing windows before `update_graph`. Pairs that sit inside k on the full stream but across a window boundary are counted here and would not form edges in those session graphs. This section answers the Chapter A question (the δ conjunct), not the windowing question.

B.5.2 The fitted asymptote

At the longest trace lengths in the corpus, pair-pooled retention is 0.905 (Table B.13, Q4: 85 sessions spanning 254–1,941 mapped events, 72,831 of 91,816 k -candidates—0.793 of the pooled candidate pool). Roughly one k -candidate in ten still fails the clock on those traces. Enabling δ does not become a no-op as traces lengthen.

The curve $a - b \exp(-cn)$ summarises that trend across the 307 sessions with $n_k \geq 1$ ($a=0.941$, $b=0.667$, $c=0.00228$; bounded nonlinear least squares, `scipy.optimize.curve_fit`). $R^2=0.432$ and RMSE 0.221 on retention were computed from the persisted per-session values rather than recorded at fit time.

The fitted asymptote is $a=0.941$ (95% CI [0.82, 1.06], recomputed from the same data). The interval does not exclude full retention, so the asymptote is a summary of where the trend points rather than an independent bound.

At the data maximum of 1,941 mapped events—the longest v2 session—the curve is 0.933.

B.5.3 What this does — and does not — say about Chapter A

AndroCT still has no timestamps, so δ remains unmeasured there. AndroCT benign parsed-non-empty traces have median 185 mapped events (§B.4.2). That count sits in v2’s Q3 mapped band (pair-pooled retention 0.672), not at the asymptote. At that count the fit indicates that roughly half of k -candidates would fail the clock (point estimate 0.50). That figure is a labelled counterfactual, not a measurement; the evaluation is a cross-sensor extrapolation from v2 sessions under ContextDroid’s stimulation and hook set; and it inherits the fit’s session-level scatter ($R^2=0.432$, RMSE 0.221—computed from the persisted per-session retentions, not recorded at fit time). (The Q3 pooled 0.672 is a different aggregation — a band 88–254 in which longer sessions dominate — not the session-level curve at 185.)

The consequence for Chapter A is stronger than a uniform trim. δ ’s absence is not a uniform one-in-ten loss on long traces (Q4 measured retention 0.905). At AndroCT’s median mapped-event count the filter would plausibly remove around half of candidate pairs, but that figure is an out-of-sensor extrapolation and this chapter does not claim it as measured.

B.6 What this says about Chapter A

Chapter A’s structural findings were measured on AndroCT. This section states what the v2 comparison licenses about those findings, and what it does not.

B.6.1 Sparsity is not a corpus-age artefact

v2 pooled graphs have median 2 active nodes and 2 edges ($n=59$, including 19 never-eligible zero-edge apps). That is below all three AndroCT benign populations already distinguished in §B.4.2: train-benign median 5 nodes and 14 edges ($n=562$), the graphs Chapter A’s GAE fitted; test-benign 6 and 16 ($n=141$); parsed-non-empty 6 and 16 ($n=2231$). No pooled median on the 703 graph-eligible benign tensors exists. The claim does not depend on which of those three one compares against, which is what makes it robust: a modern F-Droid corpus, API 29, purpose-built Frida hooks, and nav-first guided stimulation still produces the sparser graphs.

Chapter A’s reconstruction results condition on that sparsity. v2 does not explain it away as an AndroCT-age or Monkey-only artefact. What it establishes is that the sparse regime those results were measured under survives a modern instrument and a different stimulus policy—not that the structural null is strengthened, which v2 cannot test without a malware arm. Mapped-event *rate* is higher on v2 (0.204 vs 0.026 on the parsed comparison); absolute mapped counts are not materially different; the graphs are still thinner (§B.4.2). Relevance up, breadth down, on every AndroCT population that has a graph median.

B.6.2 A retired threat, with an asymmetric mechanism

sms fires on 0 of 59 v2 pooled apps, 0 of 2231 parsed AndroCT benign apps, and 0 of 1736 parsed AndroCT malware apps. Chapter A’s inventory records the same zero on the graph-eligible parse (§A.2). Two stimulus policies, eight years apart, same category dead.

For v2 the cause is identified in code. The planner frozen set contains no SMS-capable action (§B.4.3, §B.2.3). Hooks for `SmsManager.sendMessage` and `sendMultitpartTextMessage` exist and never fire because nothing in the action space can reach them. That is unreachable by construction, not unreached by chance.

For AndroCT the cause is not identified in code. Runtime-permission gating of `SEND_SMS` on API 23+ is a plausible mechanism, and the evidence is consistent with it; neither corpus measures whether those APKs requested the permission, whether the emulator granted it, or whether a send UI was ever shown. The alternative is that neither Monkey nor BFS-plus-LLM traversal happens to reach SMS-sending flows. Those two explanations are not separated here. The asymmetry is the finding: v2’s non-firing is a policy fact; AndroCT’s is an inference.

B.6.3 The node that carries Chapter A’s headline

Chapter A’s benign-only headline is a D_1 centroid whose lift is concentrated on `ipc_intents`: zeroing that deviation coordinate drops floor AUC from 0.800 to 0.591 ($\Delta=0.209$; §A.6.7). That category fires on 18 of 59 v2 apps against 1517 of 2231 AndroCT benign apps (§B.4.3). Chapter A reads the concentration as architectural rather than behavioural (§A.8.4): a behavioural coordinate would be expected under any stimulus that exercises the app. v2 corroborates that reading—it is not a test, because there is no malware arm—but the same node is substantially less active under nav-first guided stimulation. The comparison therefore does not license transferring that detector: a signal that lives on IPC intents is a property of the AndroCT graphs, not a property shown to survive this policy.

B.6.4 A shared methodological pattern

Chapter A’s §A.7 records seven corpus artefacts caught by auditing denominators and populations; five of the seven surfaced that way rather than by inspecting a score. ContextDroid’s remediation record is a different class—pipeline gates and metrics, not measurement readings that would have entered as results—but both were surfaced by auditing denominators, populations, and gate logic rather than by inspecting an output score: the same validation findings as §B.2.6—metrics that could not see engine skips, explore-blind ratios, and a weakened flailing rule that the plan says would have admitted 12 sessions. The count of 12 is plan-sourced, not re-counted on v2. The same governing principle is recorded in §B.2.1.

B.6.5 What this chapter cannot say

No malware, so no detection claim transfers. The comparison shows that the 22-node representation under modern instrumentation is at least as sparse as all three AndroCT graph medians already distinguished in §B.6.1, and more extreme on pooled v2. It does not show that Chapter A’s detection numbers would replicate on v2: there is no malware arm, and the node that carries the headline is substantially less active under this policy (§B.6.3). That needs a sensor-symmetric malware arm: future work.

B.7 v2 extended corpus characterisation

The v2_extended corpus has *no malware*; no AUC is reported here. Corpus-level inventory and observability below use Frida JSONL timing on ContextDroid sessions, contrasted with AndroCT’s DroidFax traces only where explicitly noted. Design-commitment measurements on the $n=26$ apps with at least eight usable sessions—ordering, recency, and cold-start self/cross ratios—are in Chapter C (§C.1).

B.7.1 Corpus viability

Phase 0 inventory: 342 usable sessions, 59 apps, 40 graph-eligible, 26 apps with ≥ 8 sessions. Median inter-session gap is 17509.528 s (≈ 4.86 h); $p_{10}=455.89$ s (≈ 7.6 min), $p_{90}=2.51 \times 10^6$ s (≈ 29 d); all 26/26 eligible apps span multiple calendar days.

Intra-session timing (first measurement in the project): median session wall duration 460.882 s, median inter-event gap 7.0 ms, idle fraction at 1 s bins 0.959. At $W=120$ s the median empty-mapped fraction is 0.0 but zero-edge under $k=5$ is 0.50; shorter windows are worse. Session-level windowing is not viable on v2; the session is the unit of analysis.

Phase -1 builder provenance: three builders reconciled; E4 design runs use `update_graph_sequence` (Run 2 stage). Edge-count summaries and the chosen builder pin are in `abrg/output/v2_extended/e4_ordering/provenance.json`.

B.7.2 Observability on the v2 AVD

Sessions were collected on a local Android emulator (AVD): no cellular radio, no SIM, and no incoming calls or messages. Seven categories never fire on reference-tier sessions (0/342): `accounts`, `clipboard`, `device_info`, `dynamic_code_loading`, `media`, `sms`, `telephony`. Effective PRODUCIBLE universe: 15/22.

Classification (§A.10.1): `sms` and `telephony` are `HARDWARE_ABSENT` (no telephony stack on the AVD; planner action frozenset excludes SMS/call actions). `clipboard` and `dynamic_code_1`

loading are `STIMULUS_LIMITED` (hooks exist; planner excludes dex-loading; guided traversal does not reach clipboard APIs). `accounts`, `device_info`, and `media` remain `UNDETERMINED`—zero occupancy, cause not established (AVD account state, identifier APIs, media codec path).

AndroCT and v2 nominally share Table A.3 but not the same effective coordinate system (18/22 versus 15/22 `PRODUCIBLE`). Occupancy: `results/observability_v2_extended.csv`.

B.8 Threats to validity

The claims in this chapter are a case study of one instrument, one benign corpus, and one comparison against AndroCT. They are not a population estimate, and they do not transfer a detector. Analysis n is stated per section: 342 usable sessions across 59 apps, of which 40 are graph-eligible. The comparison population of §B.4 is those 59 apps; the graph-eligible subset is 40 of them.

B.8.1 The instrument

Attach, not spawn. Frida attaches to a running PID after `ensure_app_running` (§B.2.2). `Spawn` is implemented and was not exercised on the v2 path. Behaviour before `onCreate` is unobserved.

Hooks. Twenty-two of 58 hooked API names never fire corpus-wide (§B.4.5). Several are overload or API-generation misses rather than missing nodes of the 22-category universe (`Camera.open` silent while `CameraManager.openCamera` fires). Seven graph categories have zero fired hooks (§B.2.2). That bounds coverage; it is not a count of apps that never used those APIs under a different sensor.

Clocks. Frida events use `Date.now()` in the JavaScript runtime; action logs use host `time.time()` (§B.2.2). The domains are unsynchronised, so mapped events cannot be joined to action-step windows. BFS is 0.711 of pooled steps, not of mapped events (§B.4.2). The mapped-rate result is a property of the two-phase policy as configured; the step census does not license attributing it to either phase individually.

Action space. The planner frozenset is six types (§B.2.3). `advance_goal` returns success without device I/O; consuming metrics were fixed, the action was not. SMS, telephony, clipboard, and dex-loading actions do not exist in that set, so those categories are unreachable by construction (§B.4.3). Accounts, device-info, and media remain `UNDETERMINED`. One configuration produced the traces: nav-first BFS at explore ratio 0.35 with a 120 s floor, then llama3.2 at temperature 0, configured duration 420 s, Monkey warmup 10 events at seed 42. No ablation over planner, prompt, ratio, or phase split was run.

Documentation. The current README and methodology still describe a monkey-driven pipeline (§B.2.1). What v2 ran is `--arm11m`. A reader of those files alone will mis-identify the stimulus.

B.8.2 The corpus

Benign, F-Droid, small n . Fifty-nine apps, all benign. No malware arm, so no detection claim transfers. That $n=59$ is a case-study size, not a market.

Two collection windows. Original (July) and extend (August) differ on five of six session metrics at small effect size; wall duration does not (§B.3.2). Chapter C pools them regardless. Failure is concentrated: 46 failures on 13 of 40 extension apps; four apps failed every new slot and keep three original sessions (§B.3.2).

Replication. Identical-mode sessions are near-replicates, not independent draws. 112 of 342 usable sessions (0.327) are identical-mode, all original; 54 of 54 identical pairs share `agent_seed` and sit at median graph distance 0 (§B.2.5). The pooled fraction is not two of every three. Chapter C’s session unit inherits both the batch mixture and this nesting.

Canary. The operative 0.25×–4× mapped band is a PASS; existing min–max on two or three prior sessions is a FAIL (§B.3.2). Behavioural equivalence was substituted for an unrecovered July frida-server binary. The canary does not close that gap.

Independence of sessions. `pmclear` clears that package’s app data when it succeeds. Device state, other packages, and accounts persist (§B.2.4). Sessions of one app are not independent draws of a clean device.

B.8.3 The comparison

Three AndroCT populations. Graph, fire-rate, and yield columns use parsed-non-empty benign traces ($n=2231$). Static features use Chapter A’s graph-eligible tensors ($n=703$). Chapter A’s GAE fitted train-benign graphs of median 5 active nodes and 14 edges ($n=562$); test-benign is 6 and 16 ($n=141$). No pooled median on the 703 eligible benign apps was computed (§B.4.2). Those three populations are not interchangeable. v2 pooled (2 and 2) is sparser than all three (§B.6.1). The 6/16 figures are the parsed-non-empty population, which coincides with test-benign, not the training graphs.

Units. AndroCT is one whole-trace graph per app. v2 reports per-session ($n=342$) and per-app pooled ($n=59$), including 19 never-eligible zero-edge apps in the 59 (§B.4.1). Session medians are descriptive; tests attach to the pooled unit. Mixing those units is a false comparison.

Denominators. AndroCT mapped-per-second uses the stated 600s protocol; traces carry no wall-clock (§B.4.5). v2 rates use observed session wall. If AndroCT runs were shorter than 600s, AndroCT rates are understated and the rate gap is overstated.

Builders. §B.4 graphs were built with `update_graph_sequence` (no δ). §B.5 measures the timed builder. Edge counts in the comparison are not δ -filtered, and they are not Chapter C session graphs.

B.8.4 The measurement

δ does not transfer as a percentage. Pair retention on v2 is 0.838 pooled and 0.489 unweighted (§B.5.1). The fitted asymptote is 0.941, not 1. Evaluating that curve at AndroCT’s median mapped-event count (185) gives 0.504: a labelled out-of-sensor counterfactual, not a measurement (§B.5.3). The chapter does not license “~6% at AndroCT densities, roughly uniformly.” Chapter A’s §A.4.3 points at that measurement and carries no number for the cost.

Pair count is not unique-edge count. Retention counts hetero-category k -pairs, then those that also satisfy δ . Recurring category pairs are counted each time. The measurement is on the full-session mapped stream; Chapter C then windows at 60s (§B.5). Neither identification is the fraction of unique edges in a Chapter C graph.

UNRECOVERABLE July identity. Frida client and server versions, server binary SHA, emulator system image, July AVD name, planner digest, July `prompts.py` SHA, and the hook git blob are UNRECOVERABLE as frozen artefacts (§B.3.3). The frida-server override is an argument from client/server compatibility at attach, not a hash. Device-guard *pass* is not a per-session metadata field; 0 of 388 indexed sessions recorded `failed_device_guard` (§B.2.7). Absence of a fail bit is not a recorded pass.

Google accounts on the AVD. Whether Google accounts on the AVD persist is UNDETERMINED from the analyzed code: the session sequence does not call `pmclear` on `com.google.android.gms`, and July system-image identity is UNRECOVERABLE (§B.3.3).

Snapshot flag. On the extension, `snapshot_restored=true` is compatible with skip-load and does not prove a snapshot was loaded (§B.2.4). Cold-start fairness is therefore unverified for those 172 sessions.

Appendix (Chapter B)

B.9 ContextDroid hook inventory

Tables [B.14–B.15](#) list every row of `docs/contextdroid_hooks.csv` (63 rows; 62 distinct API names). The 58 `logEvent` literals plus four `ContentResolver` names emitted through `logContentUriEvent` are the 62 names; `Context.startActivity` appears twice, under `ipc_intents` and `navigation`. “Exercised” / “not exercised” here abbreviate *implemented and exercised* / *implemented but not exercised*. No hook in this inventory is design-only.

The in-text summary by graph category is Table [B.1](#) in §[B.2.2](#).

Table B.14. ContextDroid hook inventory, rows 1–32 of docs/contextdroid_hooks.csv. Fire/non-fire follows the v2 corpus API set in SUMMARY.md.

Continued in Table B.15.

hook	category	fired in v2	v2 state
hook_loaded	lifecycle	yes	exercised
SmsManager.sendMessage	sms	no	not exercised
SmsManager.sendMultipartTextMessage	sms	no	not exercised
URL.openConnection	network	yes	exercised
HttpURLConnection.connect	network	no	not exercised
okhttp3.RealCall.execute	network	yes	exercised
FileOutputStream.<init>	file_io	no	not exercised
FileInputStream.<init>	file_io	yes	exercised
File.delete	file_io	yes	exercised
Cipher.getInstance	crypto	no	not exercised
SecretKeySpec.<init>	crypto	yes	exercised
Method.invoke	reflection	yes	exercised
Runtime.exec	process	yes	exercised
ProcessBuilder.start	process	yes	exercised
ContentResolver.query	content_access	yes	exercised
ContentResolver.insert	content_access	yes	exercised
ContentResolver.update	content_access	yes	exercised
ContentResolver.delete	content_access	yes	exercised
ClipboardManager.setPrimaryClip	clipboard	no	not exercised
Camera.open	camera	no	not exercised
CameraManager.openCamera	camera	yes	exercised
PackageManager.getInstalledPackages	package_manager	yes	exercised
PackageManager.getInstalledApplications	package_manager	no	not exercised
LocationManager.requestLocationUpdates	location	yes	exercised
TelephonyManager.getDeviceId	device_info	no	not exercised
TelephonyManager.getSubscriberId	device_info	no	not exercised
TelephonyManager.getSimSerialNumber	device_info	no	not exercised
TelephonyManager.getCallState	telephony	no	not exercised
DexClassLoader.<init>	dynamic_code_loading	no	not exercised
PathClassLoader.<init>	dynamic_code_loading	no	not exercised
System.loadLibrary	native_code	yes	exercised
Runtime.loadLibrary	native_code	no	not exercised

Table B.15. ContextDroid hook inventory, rows 33–63 of `docs/contextdroid_hooks.csv` (continued from Table B.14).

hook	category	fired in v2	v2 state
<code>Runtime.load</code>	<code>native_code</code>	no	not exercised
<code>Cipher.doFinal</code>	<code>crypto</code>	yes	exercised
<code>MessageDigest.getInstance</code>	<code>crypto</code>	yes	exercised
<code>MessageDigest.digest</code>	<code>crypto</code>	yes	exercised
<code>okhttp3.RealCall.enqueue</code>	<code>network</code>	yes	exercised
<code>AccountManager.getAccounts</code>	<code>accounts</code>	no	not exercised
<code>AccountManager.getAccountsByType</code>	<code>accounts</code>	no	not exercised
<code>AudioRecord.<init></code>	<code>audio</code>	yes	exercised
<code>MediaRecorder.setAudioSource</code>	<code>audio</code>	yes	exercised
<code>SharedPreferences.getString</code>	<code>storage</code>	yes	exercised
<code>SharedPreferences.getInt</code>	<code>storage</code>	yes	exercised
<code>SharedPreferences.putString</code>	<code>storage</code>	yes	exercised
<code>SharedPreferences.apply</code>	<code>storage</code>	yes	exercised
<code>SQLiteDatabase.rawQuery</code>	<code>database</code>	yes	exercised
<code>SQLiteDatabase.execSQL</code>	<code>database</code>	yes	exercised
<code>SQLiteDatabase.insert</code>	<code>database</code>	yes	exercised
<code>MediaPlayer.setDataSource</code>	<code>media</code>	no	not exercised
<code>MediaPlayer.start</code>	<code>media</code>	no	not exercised
<code>MediaPlayer.prepare</code>	<code>media</code>	no	not exercised
<code>Context.startActivity</code>	<code>ipc_intents</code>	yes	exercised
<code>Context.startActivity</code>	<code>navigation</code>	yes	exercised
<code>Context.sendBroadcast</code>	<code>ipc_intents</code>	yes	exercised
<code>Context.startService</code>	<code>ipc_intents</code>	yes	exercised
<code>Context.bindService</code>	<code>ipc_intents</code>	yes	exercised
<code>WebView.loadUrl</code>	<code>webview</code>	yes	exercised
<code>NotificationManager.notify</code>	<code>notifications</code>	yes	exercised
<code>NotificationManager.notifyAsUser</code>	<code>notifications</code>	yes	exercised
<code>Socket.connect</code>	<code>network</code>	yes	exercised
<code>AssetManager.open</code>	<code>file_io</code>	yes	exercised
<code>retrofit2.OkHttpClient.execute</code>	<code>network</code>	yes	exercised
<code>volley.RequestQueue.add</code>	<code>network</code>	no	not exercised

C.1 Framing

The thesis question admits two readings. “Like what it is” asks whether an app’s behaviour matches a *population* reference built from other apps’ benign traces; that reading is Chapter A’s, and it requires malware labels for evaluation. “Like its own past” asks whether an app matches a *self*-reference built from its earlier sessions; that reading is this chapter’s, and it requires repeated sessions.

ABRG’s design has two components: a per-app reference, and temporal adaptation of that reference. This chapter tests both. It also tests two constructions of the reference itself: from the app’s own sessions, and from its stated purpose. That yields four questions in order: what is the reference, does it carry signal, does it need to be temporal, and can it be built before any session exists.

Reference from past sessions carries strong signal. Within-app Frobenius median 1.28 against cross-app 38.3 (Cliff’s $\delta=0.684$, Mann–Whitney $p\approx 10^{-73}$; §C.3); E4 L_2 self/cross ratios of $60\times$ – $411\times$ across k . The within/cross IQRs meet near 15.1. Medians separate by roughly $30\times$; tails overlap.

Temporal adaptation does not. Ordering is neutral (§C.4.1, Cohen’s $d=0.202$). Recency does not beat cumulative memory at any of three half-lives (§C.4.2). The cold-start curve is reference dilution rather than convergence, replicated under two builders and two metrics (§C.4.3); a session-order shuffle leaves the Stage 2 non-fall of held-out error intact. Within-session windows are exchangeable (§A.6.8.4). Those nulls are controls that rule out temporal explanations for a signal that is real. One half of the named design is not needed on this corpus.

A description-built reference at $k=0$ does not. Sections §C.3–C.4.3 cannot reach $k=0$: there is no session to average. A store description exists before any session. Under llama3.2 zero-shot on fifty-three F-Droid apps, that construction fails self-versus-cross (AUC 0.501 against a ceiling of 0.979) and underperforms a population prevalence prior (median within-app AUC 0.725 against 0.952; §C.5). Of the two definitions of self tested here, past behaviour carries signal and stated purpose does not, at this observation depth.

The corpus asymmetry is structural, not a shortcoming. AndroCT (Chapter A) has malware labels and one session per app. The v2_extended corpus (Chapter B) has repeated ContextDroid sessions and no malware. Neither corpus can answer both readings. The empty cell of that 2×2 — self-reference evaluated as detection under malware labels—is the boundary of this chapter (§C.7).

No AUC is reported here except the description-seed diagnostics in §C.5, which are not detection scores. Every other finding is a corpus-level statement about within-app behavioural structure on benign Frida sessions. Wherever a v2 number sits beside an AndroCT number, the sensor differs: ContextDroid Frida hooks versus DroidFax / Soot mapping (§B.7.1; inventory sensor note).

Table A.1 marks which ABRG design commitments each corpus can exercise. Three are measured here at session granularity. The δ temporal edge filter’s corpus cost is Chapter B (§B.5). The exercised commitments are the recency channel w_{rec} , per-app cross-session refinement, and the ordering of reference sessions.

C.2 Corpus and method

Chapter B carries the corpus characterisation (§B.7.1, §B.7.2, §B.3.3). What follows is only what is needed to read §C.3–§C.4.3.

Eligible apps. Phase 0 inventory: 342 usable sessions, 59 apps, 40 graph-eligible (§B.3.1: δ -filtered export rule; 46 apps under the unfiltered Run 2 sequence builder). Design runs (E4) use the $n=26$ apps with at least eight usable sessions (maximum nine)—the Stage 2 eligible apps ($n=35$) that meet the ≥ 8 -session rule a 6+2 reference/test split requires; the remaining nine Stage 2 apps have five to seven usable sessions and are excluded from E4 on that rule alone ($|A \cap B|=26$; §C.3).

Reference / test split. For each eligible app, six sessions form the reference and two the test. PREFIX takes the earliest six (chronological); apps with nine sessions discard the middle index under PREFIX so that exactly six remain. SCATTERED draws six at random (seed $42 + \text{hash}(\text{app_id})$); the remaining two are the test. The same 6+2 rule applies at every k in the cold-start curve by taking the first k of the reference pool.

Statistic. Deviation d is the per-node aggregated absolute difference between a test session tensor X_s and a reference R_i built from the app’s other sessions—the same object used at window granularity in Chapter A (§A.6.8). Stage 2 scores `frobenius_combined` on equal-mean normalised session tensors (§C.3); E4 scores pooled L_2 combined deviation under sequence graphs. They are not the same number.

Builder. E4 uses `update_graph_sequence` after `graph_from_events`—the same Run 2 stage as Chapter B’s sequence density figures (§B.4; provenance pin in `e4_ordering/provenance.json`). Stage 2 uses the Chapter C timed session builder, then equal-weight tensor means via `equal_mean_normalised_session_tensors`. Builder identity is why the two results must not be merged.

Effective universe. v2’s effective PRODUCIBLE universe is 15/22; seven categories never fire on reference-tier sessions. Buckets and the three UNDETERMINED categories are in Chapter B (§B.7.2).

C.3 The premise: per-app references are strongly predictive

Per-entity behavioural baselining is not new: anomaly detection has long scored current activity against a profile learned from the entity’s own past rather than a population aggregate (Lane and Brodley [17]; Forrest et al. [9]). This section tests that pattern on per-app Frida category profiles.

Two constructions—different statistic, different builder, different normalisation; nested app sets ($E4 \subset \text{Stage 2}$)—find that a reference built from an app’s own sessions is far nearer to a held-out session of that app than a reference built from any other app. They are reported side by side and never merged (Table C.1). The reported claim is construction-independence on a nested population, not independent sampling.

Table C.1. Within-app versus cross-app reference error (two constructions). Stage 2: $n=35$ apps; E4: $n=26$ apps ($E4 \subset \text{Stage 2}$, $|A \cap B|=26$). No AUC.

arm	statistic	builder	within	cross	ratio	test
Stage 2 ($n=35$) $p=1.02 \times 10^{-73}$	<code>frobenius_combined</code>	equal-mean tensors	1.28314	38.3165	$\sim 30\times$	MW
E4 $k=1$ ($n=26$)	L_2 combined	<code>update_graph_sequence</code>	0.008143	3.3444	$411\times$	—
E4 $k=7$ ($n=26$)	L_2 combined	same builder	0.042989	3.1122	$72\times$	—

C.3.1 Distributions, effect size, and the Mann–Whitney

Stage 2 (Frobenius). Held-out session tensors against equal-weight means of earlier sessions of the same app, versus full-app means of every other app. Primary metric `frobenius_combined`; channel `both`; reference combine `equal_mean_normalised_session_tensors`. Within-app median 1.2831416802452953 ($n=240$ pairs); cross-app median 38.31648589584154 ($n=9350$ pairs). The medians separate by roughly $30\times$.

The interquartile ranges are within $[0.14115182493519912, 15.173454849282518]$ and cross $[15.114807007915823, 86.23015984316189]$. The IQRs touch at ≈ 15.1 (Figure C.1): the boxes do not nest, but the tails overlap. About a quarter of within pairs sit above the cross first quartile, and about a quarter of cross pairs sit below the within third quartile. The claim the data supports is therefore: medians separate by an order of magnitude, the rank test is decisive, and the distributions overlap in the tails.

Mann–Whitney $U=354425.0$, alternative within<cross, $p=1.0195951462519772\times 10^{-73}$. Cliff’s δ (equivalently the rank-biserial correlation) on the persisted `_within` and `_cross` lists is $\delta=0.6841131907308378$: among all 240×9350 pairs, 84.2% have within<cross and 15.8% have within>cross (no ties). A p -value at 10^{-73} on thousands of dependent pairs needs that effect size beside it.

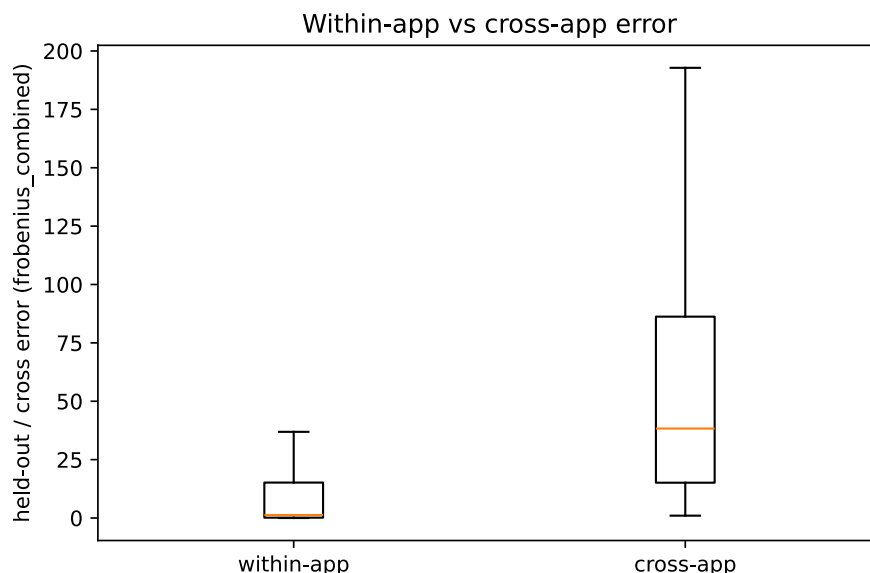


Figure C.1. Within-app versus cross-app `frobenius_combined` error ($n=240$ within, $n=9350$ cross; outliers suppressed). Medians 1.28 vs. 38.3; IQRs meet near 15.1. Source: `abrg/output/v2_chapter_c/figures/F3_within_vs_cross.svg`.

C.3.2 Independence caveat

The Mann–Whitney test compares 240 within pairs to 9350 cross pairs. Cross pairs are not mutually independent: each app appears in many comparisons. A per-app paired alternative (median within versus median cross per app, then a paired test across 35 apps) is not computable from the persisted artifact: Stage 2 stores `_cross` as an unlabeled flat list of 9350 floats without app attribution (`artifacts/stage2.json`). Recomputing it would require re-running `converge.py`. The unpaired test is what exists; the p -value should be read as directional evidence of a large separation, not as a calibrated Type I rate under dependence. Cliff’s δ is subject to the same dependence; it is reported as a magnitude on the stored lists, not as an independent-sample effect size.

C.3.3 Session-order shuffle (session-order control)

`stage2.shuffle` permutes the *chronological order of sessions within each app* (seeds 0–4, channel `both`) and recomputes the per-app drift and held-out curves. It does not recompute the within-versus-cross control: it is a null for whether the k -curve depends on calendar order, not a permutation null for the $30\times$ median ratio.

Under every seed, the Wilcoxon test that held-out error at the first k exceeds error at the last k fails (p from 0.1127 to 0.5611; chronological $p=0.9947$). Apps that never stabilise remain 19–22 of 35 (chronological: 24). Shuffling session order does not turn a rising held-out curve into a falling one. That matters for §C.4.3; it does not replace a null for the cross-app gap itself.

C.3.4 Stabilisation and per-app structure

Among the 35 Stage 2 apps, only 11 ever meet the stabilisation criterion; 24 never stabilise within the available sessions (`SUMMARY.md` lists them). Among the 11, median stabilisation k is 4 (IQR [3, 5.5]). Per-app references are strongly separated from cross-app references, and most apps do not reach a stable reference under mean aggregation within eight sessions. Both statements are true; the second is the Stage 2 face of the dilution result in §C.4.3.

C.3.5 E4 cold-start (L_2) and construction independence

Pooled median L_2 combined deviation under `update_graph_sequence` graphs. Self-deviation rises from 0.008143210928523527 at $k=1$ to 0.04298860433014455 at $k=7$; cross-app median stays near 3.11–3.34 (e.g. 3.1122056741914212 at $k=7$). Self/cross ratio is $411\times$ at $k=1$ and $72\times$ at $k=7$ ($n=26$). Full k -curve and dilution mechanism: §C.4.3.

What differs. Different statistic (Frobenius on equal-mean tensors versus L_2 on sequence graphs); different builder (Chapter C timed graphs versus Run 2 `update_graph_sequence`); different normalisation; different app counts (35 versus 26). App-set overlap: $|A \cap B|=26$, and the E4 set is a *subset* of the Stage 2 set—nine Stage 2 apps are absent from E4. The measurements are therefore not independent samples of apps; they are convergent evidence under independent *constructions* on a nested population.

What they share. Both find that per-app references dominate cross-app references by one to two orders of magnitude, in the same direction, under those distinct constructions on the nested app set. That is the chapter’s positive premise.

Window-granularity counterpart (AndroCT; Frida $\neq$ DroidFax). On labelled AndroCT at *window* granularity, a supervised hygiene control finds d_{self} 0.934043 versus d_{cross} 0.846975 on PREFIX node HGB ($\Delta=+0.087$, DeLong $p \approx 3 \times 10^{-6}$); one-class d_{cross} collapses to ≈ 0.50 – 0.53 while d_{self} holds ≈ 0.70 (§A.6.8.2; `results/d_self_vs_d_cross.md`). That metric is HGB floor AUC (and one-class floor AUC), not Frobenius or L_2 session distance. The three results—Stage 2, E4, and AndroCT windows—are not interchangeable numbers; they agree that self-reference carries identity beyond cross-reference.

C.4 What does not explain the per-app signal

The positive premise does not imply that behaviour drifts, that recency helps, or that more sessions sharpen the reference. Three negative results test the adaptive design commitments. Each carries a mechanism. They are controls that rule out temporal explanations for a signal that is real (§C.3), not failures of the chapter.

C.4.1 Ordering is neutral

PREFIX (earliest six sessions) versus SCATTERED (random six). Median $\Delta=0.000$ (PREFIX minus SCATTERED pooled L_2 deviation). Wilcoxon $W=94.0$, $p=0.45488884468908175$, $N_{\text{effective}}=21$ after dropping five exact-zero ties. Sign counts: 12 PREFIX<SCATTERED, 9 PREFIX>SCATTERED, 5 equal. Scattered-draw audit: 0/26 reference sets identical to PREFIX; 26/26 seed replay. Median $\|d\|$: PREFIX 0.0484530993594062; SCATTERED 0.05807944159277588. Paired Cohen’s $d=0.20184204360481664$ —a small effect the design could have detected and did not find as a systematic PREFIX advantage.

Rank-biserial correction. The originally stored rank-biserial used $n=26$ in the denominator while the Wilcoxon p -value uses $N_{\text{effective}}$ after dropping zero-difference pairs: stored $r=0.46438746438746437$; corrected $r=0.18614718614718617$ on $N_{\text{effective}}=21$. Both values are reported; the verdict is unchanged (ORDERING_NEUTRAL; Table C.2).

Table C.2. Session-ordering test ($n=26$ apps, ≥ 8 sessions each). Builder: `graph_from_events` $\rightarrow$ `update_graph_sequence`.

statistic	value
median Δ (PREFIX – SCATTERED)	0.000
Wilcoxon W	94.0
p	0.4549
$N_{\text{effective}}$	21
rank-biserial r (corrected, N_{eff})	0.1861
rank-biserial r (stored, $n=26$ denominator)	0.4644
Cohen’s d_{paired}	0.2018
signs ($<$ / $>$ / $=$)	12 / 9 / 5
verdict	ORDERING_NEUTRAL

Chapter A’s window-granularity exchangeability ($|\text{PREFIX} - \text{SCATTERED}|$ mean $\Delta=0.005545$; §A.6.8.4) was measured on eight slices of one 600 s Monkey run and is necessarily near-stationary. This section is the independent test at *session* granularity with real elapsed time between observations (median inter-session gap 17509.528 s; §B.7.1). Sensor: Frida (v2) versus DroidFax (AndroCT).

C.4.2 Recency does not beat cumulative memory

All 26/26 eligible apps span multiple calendar days. Recency decay half-lives: fast 3600 s, medium 17509.528 s (corpus median gap), slow 86400 s. Apps where W_{REC} lowers mean L_2 deviation: 12/26, 7/26, and 9/26 respectively. Pooled median deviation under W_{CUM} is 0.048453 at every decay; under W_{REC} the medians are 0.041192 (fast), 0.050305 (medium), and 0.047834 (slow). Paired Cohen’s d : -0.295 (fast; recency slightly lower), 0.172 (medium), 0.338 (slow). Verdict: the recency channel was specified, implemented, and measured as not helping on this corpus (`w_rec_beats_w_cum` = false; Table C.3). That closes the Table A.1 row that marks recency as available but unexercised on AndroCT (§A.10.2).

Table C.3. Recency versus cumulative reference ($n=26$ apps, all multi-day). Wins = apps with lower mean L_2 under W_{REC} . Reported r used $n=26$; corrected r uses $N_{\text{effective}}$ after dropping exact-zero paired diffs. Cohen’s d is paired ($W_{\text{REC}} - W_{\text{CUM}}$).

decay (s)	wins	p	r (rep.)	N_{eff}	r (corr.)	d
3600	12/26	0.3914	0.3162	24	0.2000	-0.295
17509.528	7/26	0.1036	0.4188	25	0.3723	0.172
86400	9/26	0.2109	0.3390	25	0.2862	0.338

C.4.3 The k -curve is reference dilution, not non-convergence

The claim that the per-app reference “has not converged within eight sessions” is withdrawn. It was supported by rising d_{self} medians from $k=1$ (0.008143) to $k=7$ (0.042989) and by the flag `has_not_converged_within_8_sessions` = true in `phase4_coldstart.json`. Diagnostics show the rise is `REFERENCE_DILUTION`, not behavioural non-convergence.

$\|R\|$ (combined L_2 of the reference) falls only 3.42%: 2.3858367534630704 at $k=1$ and 2.3042525427086877 at $k=7$. $d_{\text{self}}/\|R\|$ rises $\approx 5.33\times$, from 0.0035221012566672653 to 0.018769835871254168. Both chronological and permuted session orderings show the same rise (chrono medians 0.008143 to 0.042989). Scattered-order medians run 0.020773407401611058 to 0.04664009221384615; the effect is not monotonic drift.

Mechanism: averaging k sparse session graphs yields a smooth reference while the test session stays spiky; the magnitude mismatch grows with k . d_{cross} is stable (median ≈ 3.11 – 3.34); self/cross ratios across $k=1\dots 7$ run $411\times$, $184\times$, $60\times$, $124\times$, $78\times$, $89\times$, $72\times$ —non-monotonic, always large (Table C.4). Cold-start conclusion: $k=1$ is the minimum—one session suffices for a usable per-app reference; additional sessions dilute rather than sharpen under mean aggregation. `plateau_k` (median within 10% of the $k=7$ self median) is 5; Spearman of k against pooled self median is $\rho=0.607$ ($p=0.148$) on seven points—directionally rising, not a calibrated trend test.

The $k=1$ asymmetry (chronological 0.008143 versus scattered 0.020773) has a plain explanation: chronological $k=1$ is always session 1; scattered $k=1$ is a random session. It is reported, not over-interpreted.

Table C.4. Cold-start k -curve ($n=26$ apps). d_{self} and d_{cross} are pooled median L_2 combined deviations; IQR in parentheses. Builder `update_graph_sequence`.

k	d_{self}	med. (IQR)	d_{cross}	med. (IQR)	ratio
1	0.008143	(0.045)	3.344	(0.778)	411×
2	0.016692	(0.098)	3.075	(1.331)	184×
3	0.050298	(0.146)	3.020	(2.163)	60×
4	0.025720	(0.151)	3.183	(1.712)	124×
5	0.040403	(0.128)	3.168	(1.009)	78×
6	0.035837	(0.305)	3.197	(0.785)	89×
7	0.042989	(0.369)	3.112	(1.404)	72×

Independent replication (Stage 2 Frobenius). The same dilution pattern appears under a different builder and a different metric. On the $n=35$ Stage 2 apps, pooled median held-out `frobenius_combined` error rises from 0.803758 at $k=1$ to 10.5722 at $k=8$ (non-monotonic in between: 2.379 at $k=4$, 1.468 at $k=6$). Spearman of held-out error against k : $\rho=0.15136094900580135$, $p=0.018967715589492416$ ($n=240$ pairs). Wilcoxon testing first- k held-out greater than last- k fails ($W=162.0$, $p=0.9946693338570185$, $n=35$). The session-order shuffle (§C.3.3) leaves that failure intact. Two independent measurement paths—E4 L_2 under `update_graph_sequence`, and Stage 2 Frobenius under Chapter C timed equal-mean tensors—both show held-out error rising rather than falling with more reference sessions under mean aggregation (Figure C.2).

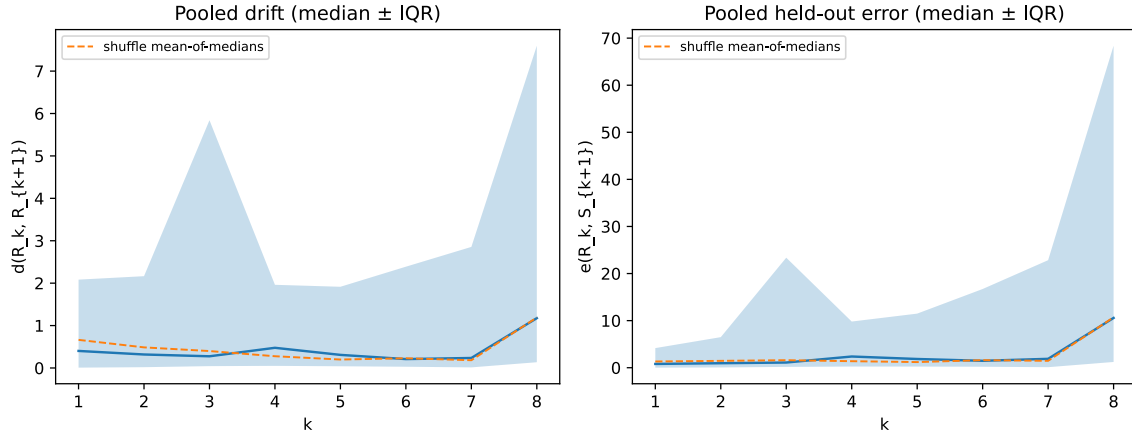


Figure C.2. Stage 2 pooled drift and held-out error (median ± IQR) versus reference size k , with shuffle mean-of-medians overlay (dashed). Held-out error does not fall with k . Source: `abrg/output/v2_chapter_c/figures/F2_pooled_median_iqr.svg`.

C.5 Description-seeded reference at $k=0$

The thesis asks whether an app behaves like itself. Chapter C answers with one definition: *itself* means the app’s own past sessions, and §C.3 shows those references are strongly separated from cross-app references. This section tests a second definition: *itself* means what the app says it is. Those are different constructions of the same identity; the chapter tests both.

The per-app reference in §C.3 requires sessions to exist. The k -curve in §C.4.3 measures how that reference behaves as sessions accumulate, and finds dilution rather than convergence. Neither measurement reaches $k=0$: with no prior session there is nothing to average. A store description exists before any session. This section is therefore the only measurement in the chapter that addresses the $k=0$ boundary of the central mechanism.

v2 is benign-only. The measurement asks whether descriptions carry app-specific behavioural information, not whether a description-seeded reference detects malware; that requires a labelled corpus under the same harness, which §C.7 identifies as the natural next collection.

Prior work. CHABADA [12] clusters apps by description topic and flags *static* API outliers within a cluster; it succeeded there and is why this test was worth running. WHYPER [27] and AutoCog [31] map descriptions to permissions. None compare descriptions to *runtime* profiles in the Frida category universe used here.

Population and metadata. Fifty-nine v2 apps with at least one usable session; six excluded—five absent from F-Droid (`ai.susi`; `at.mikenet.serbianlatintocyrillic`; `br.odb.knights`; `buet.rafi.dictionary`; `cat.jordiherandez.cinecat`) and one empty description (`app.fedilab.nitterizeme`). Fifty-three apps analysed. Four excluded apps are graph-eligible with five to nine sessions. `ai.susi` was the only `location` fire; that category is absent on the 53-app set and `ipc_intents` drops from eighteen to seventeen. F-Droid `index-v2.json`, fetched 2026-08-31 (timestamp 2026-08-31T08:25:43Z, AGPL-3.0): name, summary, description only (median length 738 characters; IQR 589.5; range 42–4000).

Prediction and decoding. llama3.2 via Ollama, JSON-schema constrained output, twenty-two category probabilities; package name withheld from the prompt. Definitions from `abrg/api_category_map.py` (Chapter A taxonomy). Temperature 0.0 primary (deterministic; seeds 42–46 identical: cross-seed SD 0.0, pairwise correlation 1.0). Stability arm at 0.7: cross-seed SD 0.162, mean pairwise correlation 0.503, displacement 0.148 from temp. 0.0 (correlation 0.786). Parse success 100% throughout. Observed profiles: a 53×22 binary matrix from `run2_comparison/v2_units.json` (`pooled[].active_categories`), with column sums validated against `category_fire.csv` and stored as `abrg/output/desc_seed/observed/observed_59x22.csv`.

C.5.1 Three framings, one conclusion

Each framing closes a different objection to the others.

Self versus cross. Does description i match app i ’s observed profile better than app j ’s? Fifty-three self pairs against 2756 cross pairs; cosine similarity between predicted and observed twenty-two-vectors. This parallels Chapter C’s $d_{\text{self}}/d_{\text{cross}}$ with the description standing in for prior sessions.

Table C.5. Self-versus-cross AUC ($n=53$ apps). Population-mean profile baseline uses the mean predicted vector for every app.

scorer	AUC
cosine (temp. 0.0)	0.501
cosine (temp. 0.7, five-seed mean)	0.515
predicted mass on observed-active / mass on all 22	0.516
Jaccard, predicted thresholded at 0.5	0.507
mean predicted on active minus mean on inactive	0.538
population-mean profile baseline	0.500

Mean self cosine 0.395 against mean cross 0.386 at temperature 0.0. Alternative scorers do not depart from chance (Table C.5).

Ceiling. Thirty-four distinct observed profiles among fifty-three apps; twenty-five apps share an exact profile with at least one other. Duplicate group sizes: 8, 7, 4, 2, 2, 2. The largest groups fire only `{storage, file_io}` (eight apps) and `{storage, file_io, ipc_intents}` (seven apps). Treating ties on identical profiles as 0.5, the maximum attainable AUC for a perfect predictor is 0.979. The observed 0.501 is 51% of that ceiling.

Within-app ranking. For app i , does the description rank its active categories above its inactive ones? Twenty-two categories per app; observed fire as label; predicted probability as score.

Table C.6. Within-app ranking AUC ($n=53$ apps).

	descriptions	population prevalence prior
median	0.725	0.952
mean	0.706	0.943
IQR	0.222	0.094
min / max	0.386 / 0.958	0.725 / 1.000
apps above 0.5	48/53	53/53

Median delta -0.24 ; the prior wins on fifty of fifty-three apps (Table C.6). Five apps fall below chance (0.386–0.475). Against 0.5, 0.725 looks informative; against a prior that ranks categories by corpus fire rate, descriptions are worse. That comparison is the point of this framing.

Per-category across apps. For category c , does the description rank apps that fire c above apps that do not? A constant predictor scores 0.5 here, so the prevalence prior has no advantage. Eight categories never fire on the 53-app set (undefined); five have fewer than five fires (unstable).

Table C.7. Per-category AUC across apps (temp. 0.0). Undefined: zero fires. Unstable: fewer than five fires.

category	fires	mean pred.	AUC	status
accounts	0	0.436	—	undefined
audio	1	0.345	1.000	unstable
camera	1	0.468	0.635	unstable
clipboard	0	0.160	—	undefined
content_access	3	0.625	0.487	unstable
crypto	12	0.294	0.456	stable
database	8	0.482	0.517	stable
device_info	0	0.369	—	undefined
dynamic_code_loading	0	0.196	—	undefined
file_io	39	0.483	0.420	stable
ipc_intents	17	0.528	0.538	stable
location	0	0.286	—	undefined
media	0	0.454	—	undefined
native_code	8	0.205	0.469	stable
network	10	0.564	0.500	stable
notifications	2	0.288	0.230	unstable
package_manager	1	0.394	0.760	unstable
process	2	0.302	0.510	unstable
sms	0	0.155	—	undefined
storage	40	0.547	0.508	stable
telephony	0	0.330	—	undefined
webview	6	0.549	0.544	stable

Descriptions do not discriminate across apps in the framing where a trivial prior cannot help (Table C.7). Stable AUCs sit near chance; `file_io` (39 fires) is below chance at 0.420.

Atypical-app hypothesis. Spearman correlation between per-app delta (model minus prior) and the prior’s own within-app AUC: $\rho = -0.17$, $p = 0.23$. The model beats the prior on three of fifty-three apps overall, and on three of the ten where the prior is weakest, including a failure on `ch.abertschi.waterme.water_me` (lowest baseline; active set `{camera, storage}` with `camera` firing in only one app). App-specific description signal is isolated, not systematic.

C.5.2 Mechanism and what it licenses

The failure is systematic. `file_io`, the second most common category (39/53 apps), scores AUC 0.420, below chance. `app.hypostats` (offline hypoglycemia logger) and `app.notesr` (offline encrypted notes) both fire `file_io` alone; the model assigns `file_io` 0.10 while giving `webview`, `storage`, and `telephony` 0.18–0.22. `app.alextran.immich` fires `{ipc_intents, native_code}`; the model pushes `webview`, `content_access`, `storage`, and `media` to 0.9–1.0. Descriptions state purpose; the twenty-two-category universe records system-call surfaces. The model reads a generic Android app from the prose and over-scores ubiquitous categories. The prevalence prior avoids that by construction because `storage` and `file_io` are usually the categories that fire.

Two conditions would have to change before the line is worth re-testing. (1) Observation depth. Twenty-five of fifty-three apps share an exact observed profile; the median app fires two of twenty-two categories. No per-app prior, description-derived or otherwise, can discriminate between apps whose observed profiles are identical. The constraint is the corpus, not the description. (2) An intermediate mapping. CHABADA succeeded against static API usage by going description $\rightarrow$ topic $\rightarrow$ API; this experiment went description $\rightarrow$ syscall category directly. Mapping stated

purpose through a permission or API layer before reaching behavioural categories is the untested variant, and the `file_io` anti-correlation is evidence that the direct jump is where it breaks.

What was tested: llama3.2 zero-shot on fifty-three F-Droid apps with median two active categories. That does not establish descriptions are uninformative in general. CHABADA found signal against static API usage on a much larger corpus.

Of the two definitions of app identity this chapter tests, the app’s own past behaviour carries strong signal (§C.3) and its stated purpose does not, at this observation depth.

Artefacts. Persisted under `abrg/output/desc_seed/`: configuration and reports (`reproduce_config.json`, `reproduce.ipynb`, `SUMMARY.md`, `stage2b_report.json`); prediction JSON under `predictions/`; `scores/self_cross_temp07.json`; `observed/observed_59x22.csv`. The config records llama3.2 predictions as frozen inputs not regenerated on validate, because Ollama output is not guaranteed reproducible across versions.

C.6 Interpretation: stability under session replication

Applications on this corpus are stable and identity-bearing rather than drifting. The self/cross separation (§C.3) holds across k and under PREFIX and SCATTERED reference draws (§C.4.1, §C.4.3). Recency weighting does not beat cumulative memory (§C.4.2). What varies with k is reference smoothness, not app identity.

Cold start is $k=1$. One session suffices to build a usable per-app reference; additional sessions dilute rather than sharpen under mean aggregation (§C.4.3). Pooled self median rises with k while cross-app medians stay near 3.1–3.3 (REFERENCE_DILUTION). The minimum is sharp at $k=1$ and the mechanism is named.

The adaptive update commitment has no empirical support at either granularity tested—within-session ordering on AndroCT (§A.6.8.4) and cross-session ordering here (§C.4.1). The ordering nulls are consistent with a stable session stream rather than with behavioural drift. If behaviour does not drift, a reference does not go stale: there is no update rule to run, no reference-poisoning surface from stale history, and no cold-start ordering dependency.

If benign apps drifted session to session, a changed app would be unremarkable. They do not: a per-app reference remains near the next session and far from every other app.

Per-app references on v2 encode app identity; within-session windows on a single Monkey run encode neither, and one-class detection fails there (Chapter A E0/E2; §A.6.8). The cross-corpus reading of when a reference is informative is developed in §A.8.5.

C.7 Threats and boundary

Sample size and scope. E4 uses $n=26$ apps; Stage 2 uses $n=35$, of which 26 overlap ($E4 \subset \text{Stage 2}$). Every finding below is corpus-level and provisional at this scale. Nothing reported is a detection result: the only AUCs are description-seed ranking diagnostics in §C.5, not malware scores.

Effective universe. v2’s effective PRODUCIBLE set is 15/22, with three categories left UNDETERMINED (§B.7.2).

Session is the unit. Session-level windowing is not viable on v2: at $W=120$ s the median zero-edge fraction under $k=5$ is 0.5; median session wall duration is 460.882s; median inter-mapped-event gap 7.0ms; median idle fraction 0.959 (§B.7.1; `e4_phase0/inventory.json`). The session is the unit of analysis.

Sensor. Statistics here are Frida-harness observations. Chapter A’s window results are DroidFax. Forms are comparable; scales are not.

Statistical caveats. (1) Stage 2’s Mann–Whitney p -value treats 9350 cross pairs as independent; they are not (§C.3). (2) Stage 2 and E4 use different builders and nested app sets; they corroborate a direction under independent constructions, not a single calibrated effect size. (3) Sessions are not independent draws: two collection batches differ on five of six structural metrics (§B.3.2), and 112/342 usable sessions are identical-mode near-replicates at median graph distance 0 (§B.8.2). Chapter C pools sessions regardless; session-level statistics inherit that nesting.

The boundary. Converting self-reference into detection requires cross-session observation *with* malware labels. No public corpus in this thesis provides both.

	malware labels	no labels
population reference	Chapter A (AndroCT)	—
self reference	empty cell	Chapter C (v2_extended)

Such a corpus would need: repeated sessions per app, both classes, one sensor held fixed across two stimulus regimes on a shared app set, and an effective node universe stated up front. ContextDroid was designed to produce that collection; extending it to labelled malware under the same harness is the natural next collection, not a re-analysis of AndroCT.

C.8 Conclusions

Four questions were posed at the opening of the chapter: what a self-reference is, whether it carries signal, whether it must be temporal, and whether it can be built before any session exists. The measurements below close each on v2_extended (benign Frida sessions; no malware labels).

Reference and signal. A per-app reference built from prior sessions lay one to two orders of magnitude closer to a held-out session than any cross-app reference. Stage 2 Frobenius medians were 1.28 versus 38.3 (Cliff’s $\delta=0.684$; §C.3). E4 pooled L_2 self/cross ratios ran $60\times$ – $411\times$ across k under a different builder and statistic that was never merged with Stage 2. The pattern is identity structure in the session stream, not a detector.

Temporal adaptation. Session ordering was exchangeable (Cohen’s $d=0.202$; §C.4.1). Recency weighting did not beat cumulative memory at any of three half-lives (§C.4.2). Adding sessions to a mean reference diluted rather than converged under both E4 L_2 and Stage 2 Frobenius (§C.4.3; REFERENCE_DILUTION). Behaviour was stable enough that a $k=1$ reference already captured what the next session looked like; the adaptive update rule had no empirical job on this corpus.

Description at $k=0$. Stated purpose failed where past behaviour succeeded: self-versus-cross AUC 0.501 against a ceiling of 0.979, and within-app median AUC 0.725 against a prevalence prior of 0.952 (§C.5). That marks the $k=0$ boundary of the mechanism, not a rescue of the session-built reference.

The design was tested component by component. Per-app reference from past sessions is strongly supported. Temporal adaptation of that reference is not needed to explain the signal. Description-as-identity is a null on this corpus. The nulls are controls, not chapter failures.

What remains empty is evaluation. Self-reference as a detection claim needs repeated sessions *and* malware labels under one sensor. Until that cell is filled, Chapter A’s population reading and the self-reference reading developed here stay complementary halves of the title question, not a single pipeline.

References

- [1] Giuseppina Andresini, Annalisa Appice, Vincenzo Belvedere, Giuseppe Fiameni, and Donato Malerba. “Anakin: explainable android malware detection with graph neural networks”. In: *Cybersecurity* 9 (2026), p. 116. DOI: [10.1186/s42400-026-00552-z](https://doi.org/10.1186/s42400-026-00552-z).
- [2] Federico Barbero, Feargus Pendlebury, Fabio Pierazzi, and Lorenzo Cavallaro. “Transcending TRANSCEND: Revisiting Malware Classification in the Presence of Concept Drift”. In: *IEEE Symposium on Security and Privacy (S&P)*. 2022, pp. 805–823. DOI: [10.1109/SP46214.2022.9833659](https://doi.org/10.1109/SP46214.2022.9833659).
- [3] Cosmin I. Bercea, Daniel Rueckert, and Julia A. Schnabel. “What Do AEs Learn? Challenging Common Assumptions in Unsupervised Anomaly Detection”. In: *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. Vol. 14224. Lecture Notes in Computer Science. Does not use the term “identical shortcut”. 2023. DOI: [10.1007/978-3-031-43904-9_30](https://doi.org/10.1007/978-3-031-43904-9_30).
- [4] Roel Bouman and Tom Heskes. *Autoencoders for Anomaly Detection are Unreliable*. Preprint; withdrawn ICLR 2025 submission. 2025. arXiv: [2501.13864](https://arxiv.org/abs/2501.13864) [cs.LG].
- [5] Yu Cai, Hao Chen, and Kwang-Ting Cheng. “Rethinking Autoencoders for Medical Anomaly Detection from a Theoretical Perspective”. In: *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2024. arXiv: [2403.09303](https://arxiv.org/abs/2403.09303).
- [6] Han Chen, Hanchen Wang, Hongmei Chen, Ying Zhang, Lu Qin, and Wenjie Zhang. *HiGraph: A Large-Scale Hierarchical Graph Dataset for Malware Analysis*. Preprint; cite v2 (499K apps); v1 reported 595K. 2025. arXiv: [2509.02113](https://arxiv.org/abs/2509.02113) [cs.CR].
- [7] Gabriele Corso, Luca Cavalleri, Dominique Beaini, Pietro Liò, and Petar Veličković. “Principal Neighbourhood Aggregation for Graph Nets”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020. arXiv: [2004.05718](https://arxiv.org/abs/2004.05718).
- [8] Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. “Attention is Not All You Need: Pure Attention Loses Rank Doubly Exponentially with Depth”. In: *International Conference on Machine Learning (ICML)*. Venue: ICML 2021 (not ICLR). 2021. arXiv: [2103.03404](https://arxiv.org/abs/2103.03404).
- [9] Stephanie Forrest, Steven A. Hofmeyr, Anil Somayaji, and Thomas A. Longstaff. “A Sense of Self for Unix Processes”. In: *IEEE Symposium on Security and Privacy (S&P)*. 1996, pp. 120–128. DOI: [10.1109/SECPRI.1996.502675](https://doi.org/10.1109/SECPRI.1996.502675).
- [10] Han Gao, Shaoyin Cheng, and Weiming Zhang. “GDroid: Android malware detection and classification with graph convolutional networks”. In: *Computers & Security* 106 (2021), p. 102264. DOI: [10.1016/j.cose.2021.102264](https://doi.org/10.1016/j.cose.2021.102264).
- [11] Dong Gong, Lingqiao Liu, Vuong Le, Budhaditya Saha, Moussa Reda Mansour, Svetha Venkatesh, and Anton van den Hengel. “Memorizing Normality to Detect Anomaly: Memory-Augmented Deep Autoencoder for Unsupervised Anomaly Detection”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2019. DOI: [10.1109/ICCV.2019.00179](https://doi.org/10.1109/ICCV.2019.00179). arXiv: [1904.02639](https://arxiv.org/abs/1904.02639).
- [12] Alessandra Gorla, Ilaria Tavecchia, Florian Gross, and Andreas Zeller. “Checking App Behavior Against App Descriptions”. In: *Proceedings of the 36th International Conference on Software Engineering (ICSE)*. CHABADA; authors are Gorla, Tavecchia, Gross, Zeller (not Tonella). 2014. DOI: [10.1145/2568225.2568276](https://doi.org/10.1145/2568225.2568276).
- [13] Mostafa Haghir Chehreghani. *Persistent Gaussian Perturbations Prevent Oversmoothing in Recurrent Graph Neural Networks*. Secondary preprint. 2026. arXiv: [2607.28185](https://arxiv.org/abs/2607.28185).

- [14] Md Ahsanul Haque, Ismail Hossain, Md Mahmuduzzaman Kamol, Md Jahangir Alam, Suresh Kumar Amalapuram, Sajedul Talukder, and Mohammad Saidur Rahman. *LAMDA: A Longitudinal Android Malware Benchmark for Concept Drift Analysis*. Preprint; dataset / benchmark, not a drift detector; ICLR 2026 forthcoming per citing text. 2025. arXiv: [2505.18551 \[cs.CR\]](#).
- [15] Yiling He, Yubing Liu, Lei Wu, Ziqi Yang, Kui Ren, and Zhan Qin. “MsDroid: Identifying Malicious Snippets for Android Malware Detection”. In: *IEEE Transactions on Dependable and Secure Computing* 20.3 (2023). DOI dated 2022; journal issue 20(3) 2023. DOI: [10.1109/TDSC.2022.3168285](#).
- [16] Roberto Jordaney, Kumar Sharad, Santanu K. Dash, Zhi Wang, Davide Papini, Ilia Nouretdinov, and Lorenzo Cavallaro. “Transcend: Detecting Concept Drift in Malware Classification Models”. In: *26th USENIX Security Symposium (USENIX Security 17)*. Venue is USENIX Security 2017 (not NDSS). USENIX Association, 2017, pp. 625–642. URL: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/jordaney>.
- [17] Terran Lane and Carla E. Brodley. “Temporal Sequence Learning and Data Reduction for Anomaly Detection”. In: *Proceedings of the 5th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1998, pp. 150–158. DOI: [10.1145/288090.288122](#).
- [18] Qimai Li, Zhichao Han, and Xiao-Ming Wu. “Deeper Insights into Graph Convolutional Networks for Semi-Supervised Learning”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2018. DOI: [10.1609/aaai.v32i1.11604](#).
- [19] Wen Li, Xiaoqin Fu, and Haipeng Cai. “AndroCT: Ten Years of App Call Traces in Android”. In: *IEEE/ACM 18th International Conference on Mining Software Repositories (MSR)*. AndroCT 2017 release; Zenodo record 4470320. 2021, pp. 570–574. DOI: [10.1109/MSR52588.2021.00076](#).
- [20] Zhen Liu, Ruoyu Wang, Nathalie Japkowicz, Heitor Murilo Gomes, Bitao Peng, and Wenbin Zhang. “SeGDroid: An Android malware detection method based on sensitive function call graph learning”. In: *Expert Systems with Applications* 235 (2024), p. 121125. DOI: [10.1016/j.eswa.2023.121125](#).
- [21] Ruiying Lu, YuJie Wu, Long Tian, Dongsheng Wang, Bo Chen, Xiyang Liu, and Ruimin Hu. “Hierarchical Vector Quantized Transformer for Multi-class Unsupervised Anomaly Detection”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2310.14228](#).
- [22] Rongrong Ma, Guansong Pang, Ling Chen, and Anton van den Hengel. “Deep Graph-level Anomaly Detection by Glocal Knowledge Distillation”. In: *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM)*. 2022, pp. 704–714. DOI: [10.1145/3488560.3498473](#). arXiv: [2112.10063](#).
- [23] Enrico Mariconti, Lucky Onwuzurike, Panagiotis Andriotis, Emiliano De Cristofaro, Gordon Ross, and Gianluca Stringhini. “MaMaDroid: Detecting Android Malware by Building Markov Chains of Behavioral Models”. In: *Network and Distributed System Security Symposium (NDSS)*. 2017. DOI: [10.14722/ndss.2017.23353](#).
- [24] Xinyi Meng and Daofeng Li. “HertDroid: Android Malware Detection Method with Influential Node Filter and Heterogeneous Graph Transformer”. In: *Applied Sciences* 14.8 (2024), p. 3150. DOI: [10.3390/app14083150](#).

- [25] Lucky Onwuzurike, Mario Almeida, Enrico Mariconti, Jeremy Blackburn, Gianluca Stringhini, and Emiliano De Cristofaro. “A Family of Droids—Android Malware Detection via Behavioral Modeling: Static vs Dynamic Analysis”. In: *16th Annual Conference on Privacy, Security and Trust (PST)*. 2018, pp. 1–10. DOI: [10.1109/pst.2018.8514191](https://doi.org/10.1109/pst.2018.8514191).
- [26] Kenta Oono and Taiji Suzuki. “Graph Neural Networks Exponentially Lose Expressive Power for Node Classification”. In: *International Conference on Learning Representations (ICLR)*. 2020. arXiv: [1905.10947](https://arxiv.org/abs/1905.10947).
- [27] Rahul Pandita, Xusheng Xiao, Wei Yang, William Enck, and Tao Xie. “WHYPER: Towards Automating Risk Assessment of Mobile Applications”. In: *22nd USENIX Security Symposium (USENIX Security 13)*. USENIX Association, 2013, pp. 527–542. URL: <https://www.usenix.org/conference/usenixsecurity13/technical-sessions/presentation/pandita>.
- [28] Feargus Pendlebury, Fabio Pierazzi, Roberto Jordaney, Johannes Kinder, and Lorenzo Cavallaro. “TESSERACT: Eliminating Experimental Bias in Malware Classification across Space and Time”. In: *28th USENIX Security Symposium (USENIX Security 19)*. USENIX Association, 2019, pp. 729–746. URL: <https://www.usenix.org/conference/usenixsecurity19/presentation/pendlebury>.
- [29] Hezhe Qiao, Hanghang Tong, Bo An, Irwin King, Charu Aggarwal, and Guansong Pang. “Deep Graph Anomaly Detection: A Survey and New Perspectives”. In: *IEEE Transactions on Knowledge and Data Engineering* 37.9 (2025). Distinct from Ma et al. TKDE 2021, pp. 5106–5126. DOI: [10.1109/TKDE.2025.3581578](https://doi.org/10.1109/TKDE.2025.3581578). arXiv: [2409.09957](https://arxiv.org/abs/2409.09957).
- [30] Chen Qiu, Marius Kloft, Stephan Mandt, and Maja Rudolph. “Raising the Bar in Graph-level Anomaly Detection”. In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI)*. 2022, pp. 2196–2203. DOI: [10.24963/ijcai.2022/305](https://doi.org/10.24963/ijcai.2022/305). URL: <https://www.ijcai.org/proceedings/2022/0305.pdf>.
- [31] Zhengyang Qu, Vaibhav Rastogi, Xinyu Zhang, Yan Chen, Tiantian Zhu, and Zhong Chen. “AutoCog: Measuring the Description-to-permission Fidelity in Android Applications”. In: *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2015. DOI: [10.1145/2660267.2660287](https://doi.org/10.1145/2660267.2660287).
- [32] Ladislav Rampásek, Mikhail Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. “Recipe for a General, Powerful, Scalable Graph Transformer”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022. arXiv: [2205.12454](https://arxiv.org/abs/2205.12454).
- [33] Jeanine Romano, Jeffrey D. Kromrey, Jesse Coraggio, and Jeff Skowronek. “Appropriate statistics for ordinal level data: Should we really be using *t*-test and Cohen’s *d* for evaluating group differences on the NSSE and other surveys?” In: *Annual Meeting of the Florida Association of Institutional Research*. Conference presentation (not a journal article); source of Cliff’s δ thresholds 0.147 / 0.33 / 0.474. 2006.
- [34] Lukas Ruff, Robert Vandermeulen, Nico Görnitz, Lucas Deecke, Shoaib Ahmed Siddiqui, Alexander Binder, Emmanuel Müller, and Marius Kloft. “Deep One-Class Classification”. In: *Proceedings of the 35th International Conference on Machine Learning*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 2018, pp. 4393–4402. URL: <https://proceedings.mlr.press/v80/ruff18a.html>.
- [35] Nino Shervashidze, Pascal Schweitzer, Erik Jan van Leeuwen, Kurt Mehlhorn, and Karsten M. Borgwardt. “Weisfeiler–Lehman Graph Kernels”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2539–2561.

- [36] Floriano Tori, Vincent Holst, and Vincent Ginis. “The Effectiveness of Curvature-Based Rewiring and the Role of Hyperparameters in GNNs Revisited”. In: *International Conference on Learning Representations (ICLR)*. Paper (not a survey). 2025. arXiv: [2407.09381](#).
- [37] Ngoc N. Tran, Anwar Said, Waseem Abbas, Tyler Derr, and Xenofon D. Koutsoukos. *Quantifying the Generalization Gap: A New Benchmark for Out-of-Distribution Graph-Based Android Malware Detection*. Preprint. 2025. arXiv: [2508.06734 \[cs.CR\]](#).
- [38] Yili Wang, Yixin Liu, Xu Shen, Chenyu Li, Rui Miao, Kaize Ding, Ying Wang, Shirui Pan, and Xin Wang. “Unifying Unsupervised Graph-Level Anomaly Detection and Out-of-Distribution Detection: A Benchmark”. In: *International Conference on Learning Representations (ICLR)*. Proceedings title order: anomaly detection then OOD; authors verified against OpenReview / ICLR 2025 poster. 2025.
- [39] Haojie Wu, Nurbol Luktarhan, Gaoqi Tian, and Yangyang Song. “An Android Malware Detection Approach to Enhance Node Feature Differences in a Function Call Graph Based on GCNs”. In: *Sensors* 23.10 (2023). S-FCSG is a subgraph extracted in the paper, not the paper title, p. 4729. DOI: [10.3390/s23104729](#).
- [40] Yueming Wu, Xiaodi Li, Deqing Zou, Wei Yang, Xin Zhang, and Hai Jin. “MalScan: Fast Market-Wide Mobile Malware Scanning by Social-Network Centrality Analysis”. In: *34th IEEE/ACM International Conference on Automated Software Engineering (ASE)*. 2019. DOI: [10.1109/ASE.2019.00023](#).
- [41] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. “How Powerful are Graph Neural Networks?” In: *International Conference on Learning Representations (ICLR)*. 2019. arXiv: [1810.00826](#).
- [42] Limin Yang, Wenbo Guo, Qingying Hao, Arridhana Ciptadi, Ali Ahmadzadeh, Xinyu Xing, and Gang Wang. “CADE: Detecting and Explaining Concept Drift Samples for Security Applications”. In: *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 2021, pp. 2327–2344. URL: <https://www.usenix.org/conference/usenixsecurity21/presentation/yang-limin>.
- [43] Zhenyu Yang, Ge Zhang, Shan Xue, Xiaoxiao Ma, Jian Yang, Hao Peng, Amin Beheshti, and Jia Wu. “Revisiting Graph-Level Anomaly Detection: From Partially to Fully Unsupervised Learning”. In: *Proceedings of the ACM Web Conference (WWW)*. 2026, pp. 901–912. DOI: [10.1145/3774904.3792307](#).
- [44] Chengxuan Ying, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. “Do Transformers Really Perform Badly for Graph Representation?” In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2021. arXiv: [2106.05234](#).
- [45] Zhiyuan You, Lei Cui, Yujun Shen, Kai Yang, Xin Lu, Yu Zheng, and Xinyi Le. “A Unified Model for Multi-class Anomaly Detection”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Origin of the “identical shortcut” term in this literature. 2022. arXiv: [2206.03687](#).
- [46] Lingxiao Zhao and Leman Akoglu. “On Using Classification Datasets to Evaluate Graph Outlier Detection: Peculiar Observations and New Insights”. In: *Big Data* 11.3 (2021). Journal article (not a conference), pp. 151–180. DOI: [10.1089/big.2021.0069](#).